MATH SURVEY

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2025-2026

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Stuyvesant Math Survey

2025-2026 Winter Edition

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Editor's Note

We are delighted to present to you the 2025-2026 Edition of *Math Survey*! This edition is filled with articles exploring various aspects of mathematics, from arithmetic functions to graph theory and origami, as well as engaging computer science articles and puzzles. Thank you so much to our writers for their hard work and dedication to research in math and computer science as well as to our publication.

We would like to thank our faculty advisor Mr. Rubinstein for his guidance and support as well as Mr. Scheiman for agreeing to do an interview for the feature article. We would also like to extend our appreciation to the Student Union, for without their generous support we would not be able to print hard copies.

Tanvir Ahmed, Vincent Chen
2025-2026

Guess Who I Am

Aulinda Wei

A nonogram (also called a monogram or picross) is a logic puzzle where you shade cells in the grid to reveal a hidden picture. The numbers beside each row and column show how many consecutive filled cells appear in that line. For example, a clue "4 1" means a group of 4 filled cells, then at least one blank space, then a single filled cell. Use logic to determine which squares are filled and which are empty—no guessing needed! Mark cells you're sure are blank with an "×" or dot to keep track of your reasoning.

	3 3	11 1	7 3	3	5	7	8 5	16	15	16	8 5	7	5	3	7 3	11 1	3 3
1																	
5																	
7																	
13																	
15																	
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2 9 2																	
2 9 2																	
2 7 2																	
3 3 3																	
2 5 2																	
3 5 3																	
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3 5 3																	
2 5 2																	
1 1 1 1																	
1 1 1 1																	

The solution is on page 3.

Sequences Puzzle

Liz Burnell

Each of the sequences below follows a pattern which can be numeric, linguistic, or encoded. For example, sequence $1, 4, 9, 16, 25, ____$ would continue with 36 as the rule is squares of natural numbers $1, 2, 3, 4, 5$, and so on. Below, you need to recognize the pattern and fill in the blank appropriately.

Some of the sequences below are straightforward, while others may require you to look for less obvious patterns, so use your imagination!

1. 1, 2, 4, 8, _____
2. 3, 4, 6, 8, 12, 14, 18, 20, 24, 30, 32, 38, 42, 44, _____
3. M, T, W, T, F, _____
4. J, F, M, A, M, J, J, A, S, O, N, _____
5. 2, 1, 3, 4, 7, 11, _____
6. $--$, $.-$, $-$, _____
7. 3, 1, 4, 1, 5, 9, 2, 6, 5, 3, 5, _____
8. $P, r, _$ $S, c, _, _ n, t, _, _$ $_, t, q, _, _$ $S, _, p, _, _, n, t, _, _$
9. 2, 7, 1, 8, 2, 8, 1, 8, 2, _____
10. -1, 2, 7, 14, 23, 34, 47, _____
11. 0, 1, 2, 3, 4, 5, 6, 0, 1, 2, 3, 4, 5, 6, _____
12. 1, 1, 1, 2, 1, 1, 3, 3, 1, 1, 4, 6, 4, 1, 1, 5, _____
13. 1, i, -1, _____
14. 1, 11, 21, 1211, 111221, _____
15. 4, 3, 3, 5, 4, 4, 3, 5, 5, _____
16. 1, 7, 19, 37, 61, 91, 127, 169, 217, 271, 331, _____
17. 0, 1, 9, 36, 100, 225, 441, 784, _____
18. 1, 2, 2, 3, 3, 4, 4, 4, 5, 5, 5, 6, 6, 6, 6, 7, 7, 7, 7, 8, 8, 8, 8, 9, 9, 9, 9, 9, 10, 10, 10, 10, 10, 11, 11, 11, 11, 11, 12, 12, 12, 12, 12, _____

The solution is on page 4.

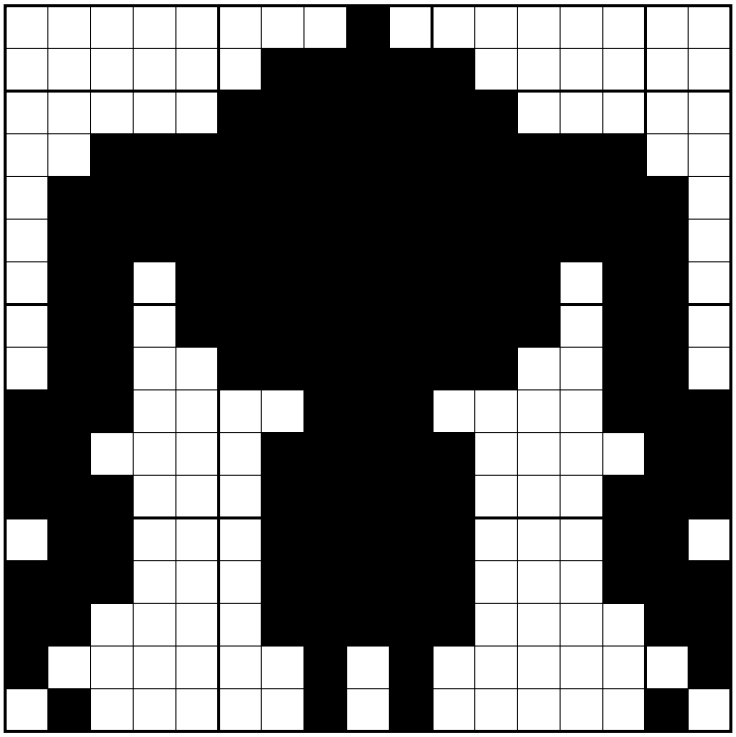
Solution to Guess Who I Am

Aulinda Wei

Here are some general strategies that are useful to tackle these sorts of puzzles:

1. Start with rows or columns that have very large numbers (like 15 or 16) — these cover most of the row and are easiest to place.
2. Fill the guaranteed middle section first. For a 17-wide grid with clue 16, all but one cell are filled, so only one edge cell stays blank.
3. Notice the puzzle's left-right symmetry and use it to deduce mirrored fills.
4. Use intersections: where filled columns overlap filled rows, confirm those cells.
5. Continue eliminating impossible blanks until the entire shape is revealed.

Completed Solution



Solution to Sequences Puzzle

Liz Burnell

1. 16 (Multiply by 2)
2. 48 (Primes $2, 3, 5, 7, 11, 13$ plus 1)
3. S (Saturday)
4. D (December)
5. 18 (Lucas sequence)
6. .... (Morse code: $M =$ `--`, $A =$ `.-`, $T =$ `-`, so $H =$ ....)
7. 8 (Digits of π)
8. `Pro Scientia Atque Sapientia` (Stuyvesant motto)
9. 8 (Digits of Euler's number e)
10. 62 ($x^2 - 2$)
11. 0 (Arithmetic modulo 7)
12. 10 (Pascal's triangle read by rows)
13. $-i$ (Powers of i)
14. 312211 (Look-and-say sequence)
15. 4 (Number of letters in month names)
16. 397 (Centered hexagonal numbers)
17. 235416 (Sum of first n cubes; $(1 + \cdots + n)^2$)
18. 12 (Golomb's sequence)

Comments

Question 16 (OEIS A003215). The sequence

$$1, 7, 19, 37, 61, 91, 127, 169, 217, 271, 331, \ldots$$

is given by

$$a_n = 3n(n+1) + 1.$$

These numbers count lattice points inside a regular hexagon and are sometimes called the crystal ball sequence for the hexagonal lattice.

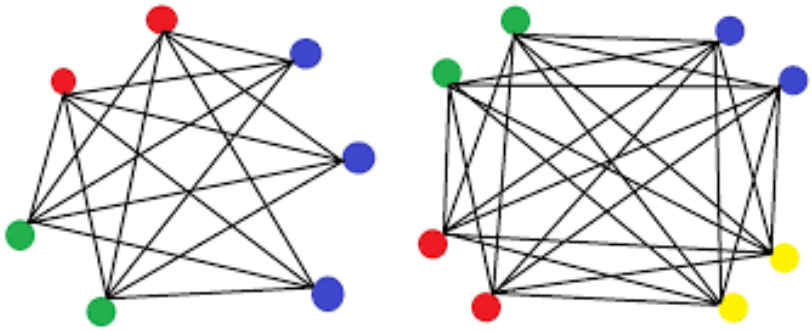
Question 18. The Golomb sequence is an integer sequence that grows slowly. For more information, see:

"The golden ratio from a calculus point of view," Mathematics Magazine, 2023.

`https://mat.uab.cat/web/matmat/wp-content/uploads/sites/23/2023/11/v2023n04.pdf`

Forbidden Subgraphs in Extremal Graph Theory

Ryan Peng



1 Introduction

Graph theory is an area of combinatorics that studies graphs, which are structures made up of vertices and edges.

Extremal graph theory is a branch of graph theory that studies with how global properties of a graph (such as how many vertices or edges a graph has) are connected with local substructure.

The most fundamental problem in extremal graph theory is the **forbidden subgraph problem**, which asks: How dense can you make a graph before the existence of a specific subgraph is guaranteed? For example, how many edges can a graph have before it must contain a triangle? What about a pentagon? In this article, we will explore these questions.

2 Definitions

2.1 Basics of Graph Theory

To begin, we will define some of the basic terminology of graph theory.

Definition 2.1

A **graph** is an ordered pair $G = (V, E)$, where:

- V is a set of vertices (points)
- E is a set of edges, where an edge is defined as an unordered pair of vertices (each edge connects two distinct vertices).

Definition 2.2

Two graphs G_1 and G_2 are **isomorphic** if there exists a bijection (one-to one mapping) $f : V_1 \to V_2$ between the vertices of the two graphs such that if a,b is an edge in G_1 then $f(a)$, $f(b)$ is an edge in G_2.

Definition 2.3

For an edge x, y, the vertices x and y are said to be the **endpoints** of the edge.

Definition 2.4

Two vertices x and y are **adjacent** if they are connected by an edge.

Definition 2.5

The **degree** of a vertex x (denoted $d(x)$) is the number of edges of which x is an endpoint.

2.2 Named Graphs

Definition 2.6

A graph $G_1 = (V_1, E_1)$ is a **subgraph** of $G_2 = (V_2, E_2)$ if $V_1 \subseteq V_2$ and $E_1 \subseteq E_2$.

Definition 2.7

A **path** (denoted P_n) is a subset of n distinct vertices and $n - 1$ edges, where any two consecutive vertices are adjacent.

Definition 2.8

An n**-cycle** (denoted C_n) can be obtained from a path P_n by connecting the first and last vertex of the path with an edge.

Definition 2.9

An n**-clique** (denoted K_n) is a subset of n vertices where any two vertices in the clique are adjacent.

Definition 2.10

A graph is **bipartite** if it is possible to divide the vertices of the graph into two disjoint sets such that no two vertices in the same set are adjacent.

Definition 2.11

The **complete bipartite graph** $K_{m,n}$ is a bipartite graph whose vertices can be divided into two disjoint sets of size m and n such that every vertex in the first set is adjacent to every vertex in the second set.

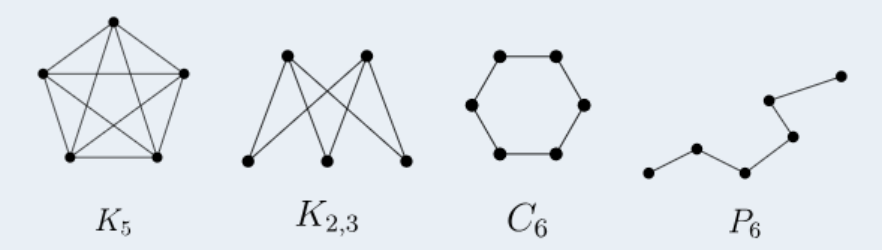


Figure 1: Examples of named subgraphs

2.3 Forbidden Subgraphs

We now define the central focus of this article: the extremal number of a forbidden subgraph.

Definition 2.12

The **extremal number** (denoted $\text{ex}(n, G)$) or **Turán number** (denoted $t(n, G)$) of a graph G is the maximum possible number of edges of a graph on n vertices that does not contain a subgraph isomorphic to G.

3 Results

Now that the definitions are out of the way, we can start proving some results!

3.1 Handshaking Lemma

We will start with a fairly obvious assertion: the sum of the degrees of all of the vertices in a graph is equal to twice the number of edges.

Lemma 1: Handshaking Lemma

For any graph G,

$$\sum_{v \in V(G)} d(v) = 2|E(G)|$$

This makes sense, since each edge is comprised of two vertices, so each edge contributes 2 to the sum of degrees.
This observation is very useful because it allows us to relate the degrees of individual vertices of a graph to the total number of edges in the graph. The idea of relating properties of individual components of a graph to properties of the whole graph is central to extremal graph theory.

(Note: This statement is called the handshaking lemma because of the following analogy:
In a party where people shake hands with each other, if you count the number of handshakes each person received and sum them, the result is twice the total number of handshakes.)

3.2 Mantel's Theorem

For our first forbidden subgraph, we will consider one of the simplest graphs: the triangle. The triangle (or K_3 or C_3) consists of three vertices and 3 edges.

Mantel's theorem was published in 1907 by dutch mathematician Willem Mantel.

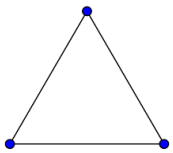


Figure 2: A triangle.

Theorem 1: Mantel's Theorem

The maximum number of edges that a graph on n vertices which does not contain any triangles can have is $\left\lfloor \frac{n^2}{4} \right\rfloor$. In other words, $\text{ex}(n, K_3) = \left\lfloor \frac{n^2}{4} \right\rfloor$.

Proof: Let G be a triangle-free graph on n vertices. Since we are looking to prove an upper bound on the number of edges G can have, assume G has at least one edge.

First, consider an arbitrary edge of G. Call the two vertices that make up this edge x and y.
Now consider a third vertex z. If z is adjacent to both x and y, then x, y, z would form a triangle!

Therefore, no vertex of G can be adjacent to both x and y. In other words, the total number of vertices adjacent to x plus the total number of vertices adjacent to y cannot exceed the total number of vertices in G.
This leads us to our first key observation:

$$d(x) + d(y) \leq n$$

Now, how can we relate this to the total number of edges in G? In order to use the handshaking lemma, we need the sum of degrees over all vertices of G, not just two vertices.

So, let's consider taking the sum of $d(x) + d(y)$ over all edges x,y in G. Call this sum S.

$$S = \sum_{\{x,y\} \in E(G)} d(x) + d(y)$$

Since each $d(x) + d(y)$ is less than or equal to n, we have $S \leq n|E(G)|$.

Now, let's look at this sum another way. Consider an arbitrary vertex z of G. Since there are $d(z)$ edges containing z, the term $d(z)$ will be added to S $d(z)$ times. Therefore,

$$S = \sum_{z \in V(G)} d(z)^2$$

Substituting this into our inequality, we have

$$\sum_{z \in V(G)} d(z)^2 \leq n|E(G)|.$$

(Note: this technique of expressing the same sum in two different ways and then combining the two expressions is called **double counting**. It is a useful proof technique in combinatorics.)

Now, we just need to get rid of the power of 2, and then we can use the handshaking lemma. To do this, we can use the Cauchy-Schwarz inequality:

Lemma 2: Cauchy-Schwarz Inequality

Let $x_1, x_2, ..., x_n$, $y_1, y_2, ..., y_n$ be real numbers. Then

$$\left(\sum_{i=1}^{n} x_i y_i\right)^2 \leq \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right).$$

Setting $x_1, x_2, ..., x_n$ as the degrees of the vertices in G and $y_1, y_2, ..., y_n = 1$, we obtain

$$n \cdot \sum_{z \in V(G)} d(z)^2 \geq \left(\sum_{z \in V(G)} d(z)\right)^2$$

Therefore

$$\left(\sum_{z \in V(G)} d(z)\right)^2 \leq n^2 |E(G)|$$

To finish, we apply the handshaking lemma:

$$(2|E(G)|)^2 \leq n^2 |E(G)|$$

$$4|E(G)|^2 \leq n^2 |E(G)|$$

$$4|E(G)| \leq n^2$$

$$|E(G)| \leq \frac{n^2}{4}$$

□

The graph that maximizes the number of edges without containing any triangles is the complete bipartite graph $K_{\frac{n}{2},\frac{n}{2}}$.

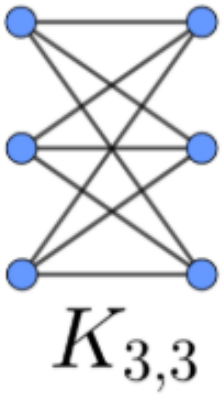


Figure 3: A triangle-free graph on 6 vertices with $\frac{6^2}{4} = 9$ edges.

In this graph, each vertex has degree $\frac{n}{2}$, so the total number of edges is $\frac{1}{2}(n \cdot \frac{n}{2}) = \frac{n^2}{4}$. The floor function is only necessary when n is odd, in which case the number of edges is maximized in $K_{\frac{n+1}{2},\frac{n-1}{2}}$.

3.3 Turán's Theorem

Mantel's theorem was generalized for cliques of all sizes by Hungarian mathematician Pál Turán in 1941.

Theorem 2: Turán's Theorem

$$\mathrm{ex}(n, K_r) = \left(1 - \frac{1}{r-1} + o(1)\right)\left(\frac{n^2}{2}\right).$$

$o(1)$ (pronounced "little-o of 1") denotes a function that approaches 0 as n approaches infinity. Adding the term $o(1)$ accounts for error when n is not divisible by $r-1$, similar to the use of the floor function in Mantel's theorem. This error term does not affect the proof, so we will ignore it.

Proof: We will prove the theorem by showing that for any graph G on n vertices, $|E(G)|$ is less than or equal to our claimed value of $\mathrm{ex}(n, G)$. We will show this by strong induction on n.

For the base case, consider a graph G with n vertices, where $n < r$.

A graph with fewer than r vertices cannot possibly contain K_r, so the maximum number of edges is simply $\binom{n}{2}$. Therefore we have

$$|E(G)| \leq \binom{n}{2}$$

$$|E(G)| \leq \frac{n(n-1)}{2}$$

$$|E(G)| \leq \left(1 - \frac{1}{n}\right)\left(\frac{n^2}{2}\right)$$

Since $n < r$, $n \leq r-1$, so $\frac{1}{n} \geq \frac{1}{r-1}$. Therefore

$$|E(G)| \leq \left(1 - \frac{1}{r-1}\right)\left(\frac{n^2}{2}\right).$$

Now for our induction hypothesis, assume that the theorem holds for all $n \leq k$ for some positive integer k. We want to show that the theorem then holds for $n = k+1$.

Consider a graph G on $k+1$ vertices that does not contain K_r. We want to show that

$$|E(G)| \leq \left(1 - \frac{1}{r-1}\right)\frac{(k+1)^2}{2}.$$

Suppose G has a maximal number of edges. Then G must contain K_{r-1}, or else we would be able to add more edges.
Now, split the vertices in G into two sets A and B, where A contains $r-1$ vertices that form a K_{r-1} and B contains the remaining $k-r+2$ vertices in G.

Then the edges in G can be split into three categories: edges in A, edges in B, and edges between A and B. We will consider each of these categories separately, and then take the sum to get the total number of edges.

Since A is a clique, the number of edges in A is simply

$$\binom{r-1}{2} = \frac{(r-1)(r-2)}{2}.$$

Since B contains no K_r, we can apply the inductive hypothesis. Therefore, the number of edges in B is at most

$$\left(1 - \frac{1}{r-1}\right)\frac{(k-r+2)^2}{2}$$

To find the number of edges between A and B, consider an arbitrary vertex x in B. If x was adjacent to every vertex in A, then $x \cup A$ would be a clique of size r, which is not allowed. Therefore, every vertex in B is adjacent to at most $r-2$ vertices in A, so the number of edges between A and B is at most

$$(k-r+2)(r-2).$$

Adding these three expressions, we get that the number of edges in G is at most

$$\frac{(r-1)(r-2)}{2} + \left(1 - \frac{1}{r-1}\right)\frac{(k-r+2)^2}{2} + (k-r+2)(r-2)$$

$$= \frac{1}{2}\left(1 - \frac{1}{r-1}\right)((r-1)^2 + (k-r+2)^2 + 2(r-1)(k-r+2))$$

$$= \frac{1}{2}\left(1 - \frac{1}{r-1}\right)((r-1) + (k-r+2))^2$$

$$= \frac{1}{2}\left(1 - \frac{1}{r-1}\right)(k+1)^2$$

$$= \left(1 - \frac{1}{r-1}\right)\frac{(k+1)^2}{2}.$$

Thus, $|E(G)| \leq \left(1 - \frac{1}{r-1}\right)\frac{(k+1)^2}{2}$, completing the induction. □

The graph that maximizes the number of edges without containing K_r is the **Turán graph** $T_{n,r-1}$, a complete $(r-1)$-partite graph on n vertices.

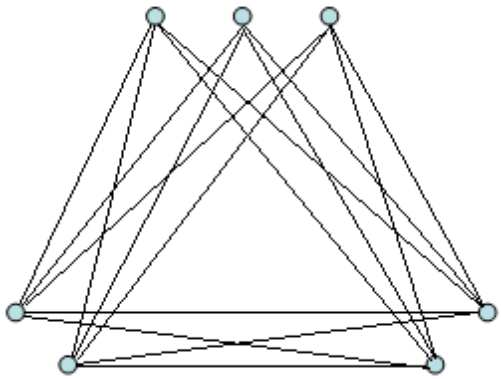


Figure 4: $T_{7,3}$

When n is divisible by $r-1$, each vertex in this graph has degree $n \cdot \left(1 - \frac{1}{r-1}\right)$, so the total number of edges is $\left(1 - \frac{1}{r-1}\right)\frac{n^2}{2}$, the exact value of our upper bound.

(Note: The two diagrams on the first page are of $T_{7,3}$ and $T_{8,4}$.)

3.4 Further Generalization

Now that we've found the extremal number for all cliques, the natural question to ask is: can we generalize this to find the extremal number for all graphs? Well, it turns out there is a theorem that does this, often called the fundamental theorem of extremal graph theory:

Theorem 3: Erdös–Stone–Simonovits Theorem

For any graph G, we have

$$\mathrm{ex}(n, G) = \left(1 - \frac{1}{\chi(G) - 1} + o(1)\right)\left(\frac{n^2}{2}\right)$$

(The proof of this theorem is significantly more involved than the proof of Turán's theorem, so we will not reproduce it in this article.)

To understand this theorem, we need to define $\chi(G)$:

Definition 3.1

The **chromatic number** of a graph G (denoted $\chi(G)$) is the minimum number of unique colors needed to color every vertex of G in such a way that no two adjacent vertices are the same color.

While this definition might seem random at first, it turns out that the chromatic number is the fundamental property of a graph that determines its extremal number!

For example, in a clique, any two vertices are adjacent, so no two vertices can be the same color. Therefore, $\chi(K_r) = r$. Then the Erdös–Stone–Simonovits theorem states

$$\mathrm{ex}(n, K_r) = \left(1 - \frac{1}{r - 1} + o(1)\right)\left(\frac{n^2}{2}\right)$$

This recovers Turán's theorem.

The Erdös–Stone–Simonovits theorem does not tell you the structure of the extremal graph that maximizes the number of edges. In general, this is an open problem.

4 Problems

By constructing graphs to represent pairwise interactions between objects, we can apply results from graph theory to problems that do not explicitly involve graphs. This is a useful technique for solving problems.

For example, consider the following problem:

Problem 1

Let S be a set of n irrational numbers. Find the maximum possible number of unordered pairs (a, b) such that $a, b \in S$ and $a + b$ is rational.

Solution: We can construct a graph G where each number in S corresponds to a vertex in $V(G)$, and two vertices of G are adjacent if and only if their corresponding numbers in S sum to a rational number. Then the problem becomes finding the maximum number of edges that G can have.

Claim: G is triangle-free.

Proof: Assume there exist three irrational numbers x, y, z such that $x+y, y+z, x+z$ are all rational. Since the sum of two rational numbers is also rational, then $(x+y)+(y+z)+(x+z) = 2(x+y+z)$ must be rational, so $x+y+z$ is rational.
Since the difference of two rational numbers must be rational, $(x+y+z)-(x+y), (x+y+z)-(y+z), (x+y+z)-(x+z)$ must all be rational, so x, y, z must all be rational, which is a contradiction.
Therefore there cannot exist three irrational numbers whose pairwise sums are all rational.

Since we know G is triangle-free, we can apply Mantel's theorem, so the maximum number of edges in G is $\boxed{\left\lfloor \frac{n^2}{4} \right\rfloor}$.

Here's another example, this time utilizing Turán's theorem:

Problem 2

In a volleyball tournament with 99 teams, each team plays against each other team exactly once. Every game results in either a win for one of the two teams, or a draw.

Show that if there does not exist a team that draws against more than 8 other teams, then there must exist a set of 11 teams such that no two teams in the set draw against each other.

Solution: We can construct a graph G where each vertex corresponds to a volleyball team, and two vertices are adjacent if their corresponding teams did not draw against each other.
Then we want to prove that G must contain a K_{11}.

Since each team plays 98 other teams, and each team draws against at most 8 other teams, $d(v) \geq 90$ for all vertices $v \in V(G)$.

Then by the handshaking lemma, $\sum_{v \in V(G)} d(v) = 2|E(G)| \geq (99)(90)$, so

$$|E(G)| \geq \frac{(99)(90)}{2}.$$

By Turán's theorem, the maximum number of vertices that a graph on 99 vertices that does not contain K_{11} can have is $\frac{99^2}{2} \cdot \frac{9}{10}$.

For the sake of contradiction, assume that G does not contain K_{11}. Then

$$|E(G)| \leq \frac{99^2}{2} \cdot \frac{9}{10} < \frac{99^2}{2} \cdot \frac{10}{11} = \frac{(99)(90)}{2}.$$

This is a contradiction, so G must contain K_{11}, and we are done.

Lastly, here are some more problems to try on your own:

Problem 3

Let $v_1, v_2, ..., v_n$ be complex numbers such that $|v_i| \geq 1$ for all i, where $|z|$ denotes the magnitude of a complex number z. Prove that there are at most $\lfloor \frac{n^2}{4} \rfloor$ pairs v_i, v_j such that $|v_i + v_j| < 1$.

Problem 4

A graph has 15 vertices. Each edge is colored either red or blue so that there no monochromatic triangles. Find the largest possible number of edges of this graph.

Problem 5

Let n be a positive integer. For a set S of $2n$ real numbers, find the maximum possible number of pairwise (positive) differences between two elements in S, that are in the range $(1, 2)$.

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The Wave That Should Not Exist

Liz Burnell

1 Introduction

It is easy to observe a wave in water—almost everyone has thrown a stone into a lake or river and watched the circular ripples spread outward and weaken quickly. This is the normal experience most of us have with waves. If the energy of waves on an ocean beach is sustained, it is due to the wind we can feel. What about sitting on a beach and observing a single wave without visible force? Our experience says that it should not be possible. Yet such a wave, called a *soliton*, does exist.

The best known first record of a soliton in a mathematical journal was made by the Scottish engineer John Scott Russell:[1]

> "I was observing the motion of a boat which was rapidly drawn along a narrow channel by a pair of horses, when the boat suddenly stopped—not so the mass of water in the channel which it had put in motion; it accumulated round the prow of the vessel in a state of violent agitation, then suddenly leaving it behind, rolled forward with great velocity, assuming the form of a large solitary elevation, a rounded, smooth and well-defined heap of water, which continued its course along the channel apparently without change of form or diminution of speed. ... I followed it on horseback, and overtook it still rolling on at a rate of some eight or nine miles an hour, preserving its original figure some thirty feet long and a foot to a foot and a half in height. ... Such, in the month of August 1834, was my first chance interview with that singular and beautiful phenomenon which I have called the Wave of Translation."

Indeed, a wave traveling for such a long time with little change is remarkable; see Figure 1 for the commemorative plaque. Assuming it took John several minutes to get on the horse, ride, and catch up to the wave, it covered roughly a mile and a half at 9 mph for guesstimated about 10 minutes. The wave was travelling in the same shape, surviving interaction with canal banks.

2 Describing and Solving a Soliton Mathematically

Common water waves break down due to dispersion and nonlinearity. Dispersion means that different parts of the wave move at different speeds thus the wave flattens and stretches.

[1] John Scott Russell, "Report on Waves," in *Report of the Fourteenth Meeting of the British Association for the Advancement of Science* (London: John Murray, 1845), 311–390.

Figure 1: Commemorative plaque reads "Near this spot JOHN SCOTT RUSSELL discovered the SOLITARY WAVE in August 1834."



Source: Graeme, "Discovering Mathematical Tourism," February 25, 2010, `https://maths.straylight.co.uk/archives/256`.

Nonlinearity means that the wave steepens and distorts, as the taller parts of the wave move faster, shorter parts lag behind, the wave steepens, forming shocks or breaking. In a soliton wave, dispersion perfectly cancels nonlinearity, therefore the wave keeps its form, moves at a constant speed, and survives collisions [2] or interactions with canal walls or banks.

In 1895, the Dutch mathematician Diederik Korteweg, a professor of mathematics, mechanics, and astronomy at the University of Amsterdam, together with his doctoral student Gustav de Vries, derived a partial differential equation whose solutions include solitary waves.

The logic was as follows. We consider the simplest case in only one space variable x and look for a function u (height, or deviation from equilibrium), depending on the variables x and t, i.e. $u(x,t)$ describes the one-dimensional wave's height at the place x at time t. Using the typical short notations [3] ,

$$
\begin{aligned}
u_t(x,t) &= \frac{\partial u(x,t)}{\partial t} && \text{(speed of height change at one location)},\\
u_x(x,t) &= \frac{\partial u(x,t)}{\partial x} && \text{(slope of the wave)},\\
u_{xx}(x,t) &= \frac{\partial^2 u(x,t)}{\partial x^2} && \text{(curvature)},\\
u_{xxx}(x,t) &= \frac{\partial^3 u(x,t)}{\partial x^3} && \text{(change of curvature)}.
\end{aligned}
$$

The resulting equation, now known as the Korteweg–de Vries (KdV) equation, can be

[2] Even in a collision of two solitons, each of them emerges with the same shape and speed. Such behavior is more typical of particles than waves.

[3] Klaus Brauer (University of Osnabrück, Department of Mathematics & Computer Science), "The Korteweg-de Vries Equation: History, exact Solutions, and graphical Representation," May 2000, accessed January 2026, `https://math.arizona.edu/~gabitov/teaching/141/math_485/KDV.pdf`.

written as

$$u_t(x,t) + 6\,u(x,t)\,u_x(x,t) + u_{xxx}(x,t) = 0. \tag{1}$$

In this equation, term $6uu_x$ is responsible for the fact that higher parts of the wave move faster, which steepens the wave; mathematically, value of the function affects its rate of change with respect to time represented by the first term, u_t. The term u_{xxx} represents dispersion which acts to spread the wave out. Most often, either the force of steepening or the force of dispersion dominates. The wave tends to break if steepening dominates, or fades if dispersion dominates. The KdV equation captures the mathematically perfect balance between the two forces, cancelling them exactly, and allowing the wave to become a solitary one.

A note on the coefficient 6 in the term uu_x. Mathematically, the coefficient 6 is introduced without loss of generality, meaning that with another coefficient derived from related physics constants the equation $u_t + auu_x + u_{xxx} = 0$ would describe the same physics, on a less convenient scale mathematically, and the *soliton* solution will still exist. The KdV's coefficient of 6 is a convenient rescaling of variables that simplifies the mathematics, making it elegant.

Since the wave keeps its form while travelling, this suggests seeking solutions in the form $u(x,t) = f(x-ct)$ where c is the constant speed of travel to the right of the whole profile defined by the same f. Based on this representation, we will convert the equation into single-variable calculus denoting substitution $y = x - ct$ (further, f' is understood as a derivative with respect to y):

$$u_t(x,t) = \frac{\partial f(x-ct)}{\partial t} = -cf'(y), u_x = f'(y), \text{ and } u_{xxx} = f'''(y)$$

Then KdV equation becomes $-cf' + 6ff' + f''' = 0$, and we can integrate both sides, collecting constants on the right (denoted A):

$$-cf + 3f^2 + f'' = A$$

We note that $f(y) \to 0, \quad f'(y) \to 0$, and $\quad f''(y) \to 0 \quad$ as $|y| \to \infty$, forcing $A = 0$ and the equation to become $f'' = cf - 3f^2$.

Omitting certain justifications, we multiply both sides by f' and then integrate both sides, denoting the constant B:

$$f''f' = cff' - 3f^2f'$$

$$\frac{1}{2}(f')^2 = \frac{c}{2}f^2 - f^3 + B$$

Again, since $f(y) \to 0$ and $\quad f'(y) \to 0 \quad$ as $|y| \to \infty$, it forces $B = 0$. The equation becomes

$$(f')^2 = cf^2 - 2f^3 = f^2(c - 2f)$$

This equation is known from physics because it is similar to a conservation-of-energy equation with the left side representing kinetic energy and the right side representing potential energy. It follows that $0 \le f \le \frac{c}{2}$, the right inequality due to the fact that permitting

f above $\frac{c}{2}$ would make the equation right-hand side negative while the left-hand side is a square.

tThe solution to such equation is also known and it is hyperbolic secant sech (which is the reciprocal of hyperbolic cosine cosh) $sech(z) = \dfrac{2}{e^z + e^{-z}}$. The equation solution is:

$$f(y) = \frac{c}{2} sech^2\left(\frac{\sqrt{c}}{2} y\right) \text{ and } u(x,t) = \frac{c}{2} sech^2\left(\frac{\sqrt{c}}{2}(x - ct)\right).$$

3 Modern View

After almost 70 years of little progress since the re-discovery of the KdV equation, in 1965 the successful application of computer-based numerical methods by Norman Zabusky at Bell Labs and Martin Kruskal at Princeton University revived interest in the soliton phenomenon.[4] This work also pointed to hierarchies of solitons, structures in nonlinear soliton waves, and possible solutions to more complex mathematical models.

One of the best applications of soliton theory was found in transmitting signals via fiber-optic cables.[5] The ability of soliton pulses to travel along optical fiber while maintaining their original wave shape makes solitons an attractive choice for transmitting data signals over very long distances.

[4]N. J. Zabusky and M. D. Kruskal, "Interaction of 'Solitons' in a Collisionless Plasma and the Recurrence of Initial States," *Physical Review Letters* 15, no. 6 (August 9, 1965): 240–243, `https://journals.aps.org/prl/pdf/10.1103/PhysRevLett.15.240`.

[5]Alex Kasman, "A Brief History of Solitons and the KdV Equation," *Current Science* 115, no. 8 (October 25, 2018): 1486–1496, `https://www.jstor.org/stable/10.2307/26978440`.

Apollonian Circles

Zhuo Zhan Chen

1 Second Definition of Circle

1.1 Introduction

Since the sixth century B.C.E., Greek mathematicians have formalized and explored the study of geometry. It is well known that Euclid formally defined circles as the set of points equidistant from a given point at a given distance. However, people often overlook the second definition of circle by another excellent geometrician, Apollonius, with the definition being "the set of points that have a specific ratio of distance to two fixed points". The circle defined with Apollonius' definition is called "Apollonius Circle." A synthetic proof of that definition goes like the following:

1.2 Proof

Here, we will prove the statement that "the set of points with a specific ratio of distance to two fixed points is a circle." However, before we prove this statement, we will go over a few definitions and theorems.

1.2.1 Definitions

Definition 1.2.1 (Locus): A locus is a set of all points that satisfy a specific geometric rule, which often is a shape, curve, or line. A well-known example of a locus is a circle, which consists of all points equidistant from its center.

Definition 1.2.2 (Angle Bisector): An angle bisector is a line that bisects an angle into two congruent angles.

Definition 1.2.3 (Exterior Angle): An exterior angle is an angle formed by extending one of the lines that enclose the original angle.

Definition 1.2.4 (Exterior Angle Bisector): An exterior angle bisector is the line that bisects an exterior angle.

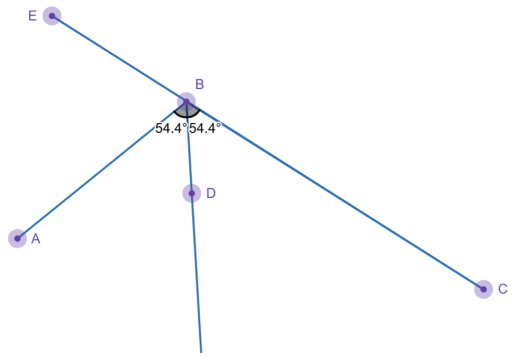


Figure 1: Angle ABE is the exterior angle of angle ABC, BD is the angle bisector of angle ABC

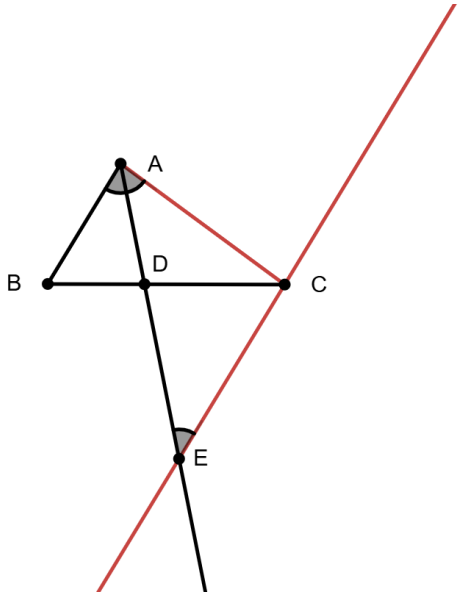


Figure 2: Angle Bisector

1.2.2 Theorems

Angle Bisector Theorem: The angle bisector theorem states that in $\triangle ABC$, if the angle bisector of $\angle BAC$ intersects BC at point D, then $\frac{AB}{BD} = \frac{AC}{CD}$

There are several ways to prove this theorem; here, we will do an entirely synthetic proof that goes as follows:

We will first construct a line intersecting point C that is parallel to line AB. We then extend AD until it intersects the line through point E that we constructed. $\angle BAD \cong \angle AEC$ because of $AB \parallel CE$, which means that $AC = CE$, since $\angle DAC = \angle DEC$. It is clear that $\triangle DAB \sim \triangle DEC$, therefore it is true that $\frac{AB}{BD} = \frac{CE}{DC}$. Substituting $AC = CE$ yields the desired result.

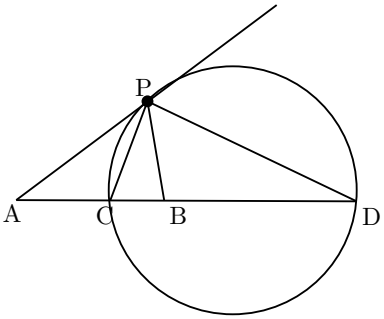
The Exterior Angle Bisector Theorem: The exterior angle bisector of a triangle divides the opposite side externally in the ratio of the adjacent sides. As the proof is similar to the proof of the angle bisector theorem, the proof will be left as an exercise for the reader.

1.2.3 Proof of the Statement

Let the two fixed points be point A and point B, and point P a point satisfying $\frac{PA}{PB} = k$, with k as a positive constant that is not 1

We first construct the angle bisector of the $\angle APB$ such that it intersects the line AB at point C. By the angle bisector theorem, $\frac{PA}{PB} = \frac{CA}{CB}$, which means that point C is a fixed point in the set. Then we will construct the exterior angle bisector of the $\angle APB$, which intersects the line AB at the point D. By the exterior angle bisector theorem, $\frac{PA}{PB} = \frac{AD}{BD}$, which means that point D is another point within the set. By the definitions of an angle bisector and an exterior angle bisector, $\angle CPD$ is always a right angle. By Thales' Theorem, the point P can be any point that lies on the circle with diameter CD.

The converse of the statement that we have proven: "all points on the circle satisfy the ratio $\frac{PA}{PB} = k$ for two fixed points A and B" is left as an exercise for the reader. (Hint: Angle Bisector Theorem)



1.2.4 Degenerate Case

We know that the definition of the locus is all points that satisfy $\frac{PA}{PB} = k$, with k as a constant. However, a degenerate circle will be constructed if $k = 1$, which is pretty intuitive. Since we want $PA = PB$, that is, the point P is on the perpendicular bisector of segment AB.

2 Applications

2.1 But why?

You may be wondering why we need a definition like this or how this is useful. This property of a circle allows us to solve the Apollonius Pursuit problem, which concerns minimizing the time of the chase.

The problem goes as follows: Given a ship moving in a constant direction and speed v, in what direction should the chase ship with speed V go, to intercept the other boat with minimal time spent?

The solution to this problem is also pretty simple. We want to find all points reachable by both ships simultaneously. Every t amount of time passes, the chasing ship will displace by Vt units of distance, while the evading ship will displace by vt units of distance. That means that the ratio of their displacement is a constant $\frac{V}{v}$. The set of locations that the two ships can simultaneously reach is a circle centered on the evading ship's location. Therefore, when the direction of the evading ship is given, the chasing boat has to move towards the area where the circle and the evading ship's direction intersect.

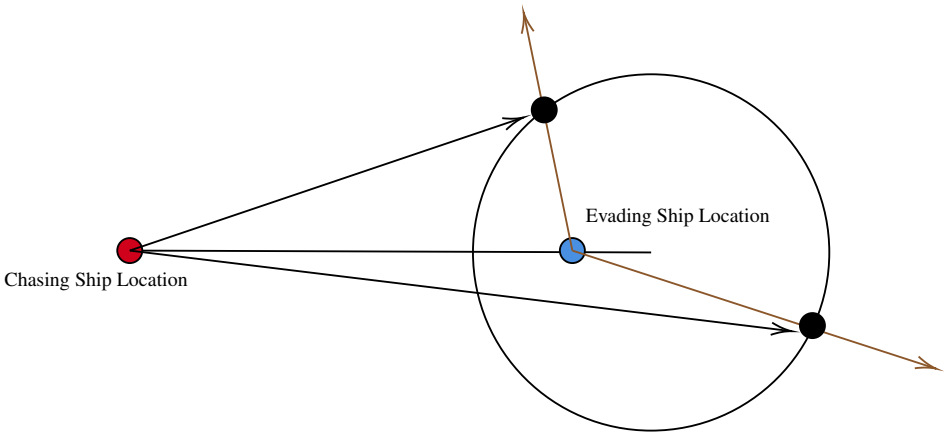


Figure 3 (The brown arrow is the direction the evading ship is heading, and the intersection is where the chasing ship should head.)

2.2 Exercise

Here is a problem that applies the Apollonius circle to real-life issues.

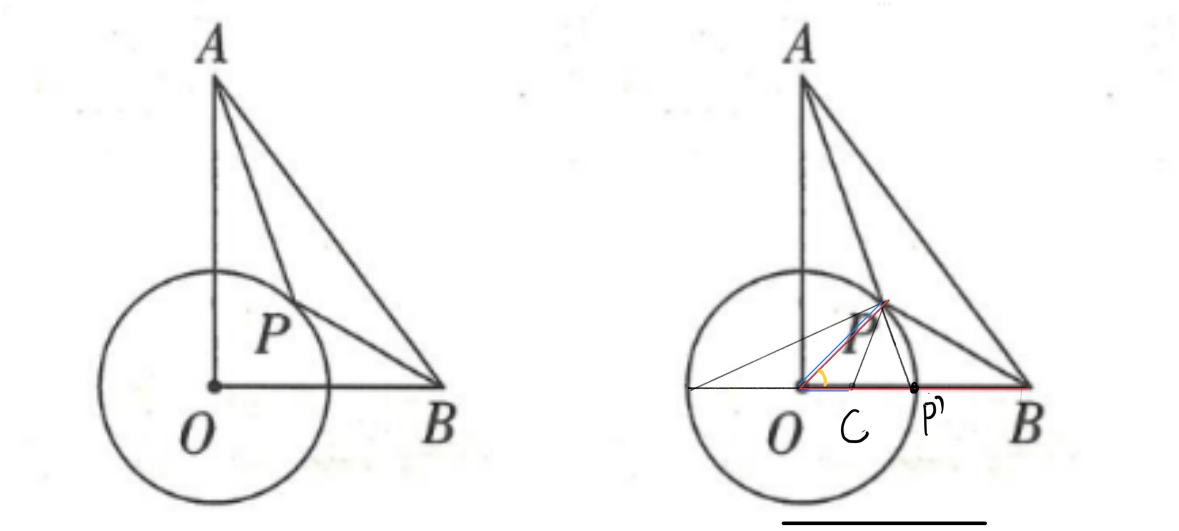


Figure 3: diagram

A city has three logistics hubs: one at point A, one at point B, and one at point O. Your company wants to build a transfer station at point P, 2 miles from the logistics hub at point O. However, the shipping from hub A to station P costs 100 dollars per mile while the shipping from point P to hub B costs 50 dollars per mile. Given that the $\angle AOB$ is a right angle, the distance from hub A to hub O is 4 miles, and the distance from hub B to hub O is 6 miles. How would we determine the optimal location for the transfer station to minimize shipping costs? What is the shipping cost to deliver from hub A to station P and then deliver from station P to hub B? (Hint: We want the minimized value of $100(PA + \frac{PB}{2})$, so we would minimize $PA + \frac{PB}{2}$. How can we convert $\frac{PB}{2}$ into some other segment using Apollonius' circle?)

2.3 Solution!!!

Solution: It is clear that point P can be any point that is 2 units away from point O, which makes all possible location of point P a circle.

Key Observation: There exist a point C on line OB such that $PC = \frac{PB}{2}$.

Reasoning: Although this may seems intuitive to be true for some people, it is actually not easy to prove. Let's first locate point C and review the givens we have. We can first take an extreme case where P is on segment OB, which is also the intersection of circle O and segment OB. To not cause any confusions, we'll name this point P'. We want $PC = \frac{PB}{2}$, so $CP' = \frac{P'B}{2}$. We know that $P'B = BO - P'O$, which yields $P'B = 4 - 2 = 2$. And so point C would be one unit away from point P', such that it is in the interior of the circle.

Now we can review the side lengths, $OC = 1, PO = 2, OB = 4$. We want to create a ratio relationship between PC and PB, and a common way to do it is to prove similarity. Notice that $\angle POB$ is on the opposite of side PC in $\triangle POC$ and is also on the opposite of side PB in $\triangle POB$. Notice that $\frac{OP}{OC} = \frac{OB}{OP} = 2$. Through the SAS similarity Theorem, we can prove that $\triangle POC \sim \triangle BOP$. This implies that $\frac{PB}{PC} = 2$, and we have proven the claim.

Since the side ratio holds true, the circle formed by all possible point P is certainly the Apollonius circle of with fixed point B and C. This means that $PA + \frac{PB}{2} = PA + PC$. This expression is clearly minimized when point P is on line AC. The existence of such point P also have to be taken into consideration. Still, we can make the argument that "all lines that contains points in the interior of a circle intersects the circle", the proof of this will be skipped since this is trivial. If point P is on line AC, $PA + PC = AC$. By Pythagorean

Theorem, we can find the length of AC to be $\sqrt{37}$. The minimized value of $100(PA + \frac{PB}{2})$ will be $100\sqrt{37}$ and we have determined the position of point P.

3 Conclusion

The Apollonius circle provides an elegant geometric configuration that explores ratio-based distance loci. Beyond the theoretical significance, it also offer solutions to pursuit and optimization problems that involves weighted distance. These applications can be extended to robotics path planning, physics, warehouse placements, and more. As a concept that existed for more than 2000 years, it remains significant for its applications and theoretical values.

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Spherical Geometry

Narnia Poddar

1 Introduction

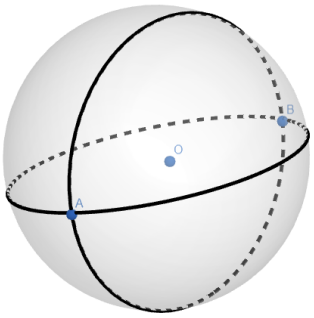
We're all familiar with Euclidean geometry – geometry on a plane – but not so much with geometry on a sphere like the Earth. Though spherical geometry still follows the first four axioms that Euclidean geometry is governed by, it doesn't abide by the fifth one: the existence of parallel lines (which Euclid initially thought could be proved by the other four, but later discovered couldn't)! This may not seem like a great change, but it leads a geometry on the surface of a sphere to have very different properties than those we are used to seeing on a plane.

2 Lines and Distances

On a sphere, we define a line to be a great circle. A great circle is a circle that lies on the surface of the sphere and whose center is at the center of the sphere (and thus has the same radius as the sphere). We chose great circles to be our lines rather than any other circle on the surface because the distances between two points on the surface of the sphere are minimized by traveling on a great circle.

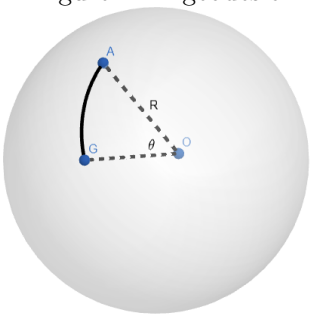
From our definition of a line, we can also find that there are no parallel lines on a sphere, as every pair of great circles intersects at two antipodal points.

Figure 1: A pair of great circles intersecting at two antipodal points (diagram made with Geogebra)



Finally, we define the distance between two points (also known as a geodesic) to be shortest arc of a great circle connecting the points. Since all great circles have same radius as the sphere and are centered at the sphere's center, the angle subtended by the geodesic uniquely determines the length of the geodesic. The length of a geodesic with central angle θ can therefore be computed as $\frac{\theta}{2\pi} \cdot 2\pi R = R\theta$.

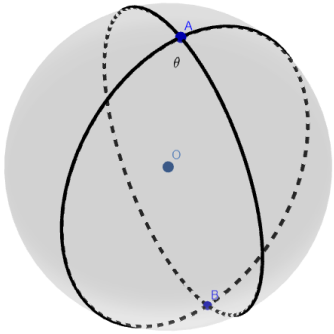
Figure 2: A geodesic



3 Bigons

We've already observed that any two different lines intersect in exactly two antipodal points of the sphere, and they split the sphere into several closed regions. Each of these regions only have two edges, meaning we can have two sided polygons – bigons – in spherical geometry.

Figure 3: A bigon with antipodal points A and B



To find the area of a bigon, we want to know what fraction of the surface area of the sphere it takes up. Since the two points of intersection of the bigons are antipodal, we can rotate the sphere so that the two endpoints of the bigon are the northmost and southmost points (the poles) of the sphere. The planes of the great circle that define the bigon intersect along the line joining the north and south pole. Suppose the angle between them is θ. Then the bigon makes up $\frac{\theta}{2\pi}$ of the surface area of the sphere, meaning it has area $\frac{\theta}{2\pi} \cdot 4\pi R^2 = 2R^2\theta$.

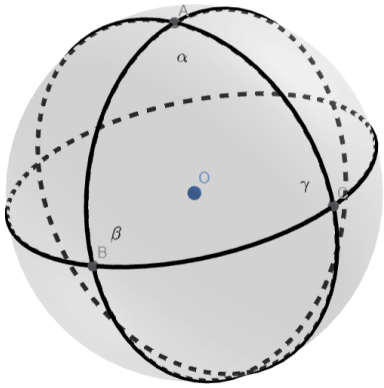
A slightly easier way to find the area is to measure the angle where the two great circles intersect on the surface of the sphere. We know this angle is the same as θ because the great circles intersect at the same angle as the planes they determine intersect at. From here, we use a similar logic to determine that the area is $\frac{\theta}{2\pi} \cdot 4\pi R^2 = 2R^2\theta$.

4 Triangles

4.1 Computing areas

Suppose we have a triangle ABC on the sphere, with $\angle BAC = \alpha$, $\angle ABC = \beta$, and $\angle BCA = \gamma$, as shown in figure 4 below.

Figure 4: A triangle



Let the area of the triangle be A. If we take the union of the three bigons determined by each pair of lines, we get half the area of the sphere (since for every point in a bigon, its antipodal point is not included in a bigon). This means the union of the areas is half the area of the sphere, or $\frac{1}{2} \cdot 4\pi r^2 = 2\pi r^2$.

The union of the areas of the bigons can also be written as the sum of the areas of each bigon, minus two times the area of the triangle as it is overcounted twice (it is included once in each bigon). Equating the two expressions for the union of the areas of the bigons:

$$\begin{aligned} 2\alpha R^2 + 2\beta R^2 + 2\gamma R^2 - 2A &= 2\pi R^2 \\ 2A &= 2\alpha R^2 + 2\beta R^2 + 2\gamma R^2 - 2\pi R^2 \\ A &= R^2(\alpha + \beta + \gamma - \pi) \end{aligned}$$

4.2 More about angles

We know that area is always positive, so applying this to our area formula for triangles:

$$\begin{aligned} R^2(\alpha + \beta + \gamma - \pi) &> 0 \\ \alpha + \beta + \gamma - \pi &> 0 \\ \alpha + \beta + \gamma &> \pi \end{aligned}$$

Therefore, the sum of the angles in a triangle is always greater than π.

5 A new congruence theorem

Like in Euclidean geometry, SSS, SAS, and ASA congruence hold true on a sphere. An interesting result of this is a new spherical congruence theorem: AAA (which had only guaranteed similarity in Euclidean geometry)! For simplicity, we assume we are working with a unit sphere ($R = 1$) in this section.

5.1 Poles

First, we define a pole of a line l to be one of the two points where the line going through the center and perpendicular to the plane of the great circle of l intersects the sphere. We observe the following properties of poles:

1. The length PA of the geodesic connecting pole P of line to any point A on line l is $\frac{1}{4} \cdot 2\pi = \frac{\pi}{2}$ since the line connecting the pole to the center is perpendicular to the plane of the line (and because we are assuming $R = 1$ for this section).

2. Given a line segment AB and a point P such that $PA = \frac{\pi}{2}$ and $PB = \frac{\pi}{2}$, then P is a pole of AB because $PA = PB = \frac{\pi}{2}$ implies that PO is perpendicular to the great circle containing AB.

3. The length of the geodesic between two points A and B is $\angle APB$, because $\angle APB \cong \angle AOB$.

5.2 Polar triangles

We define the polar triangle of $\triangle ABC$ to be $\triangle A'B'C'$ such that A' is a pole of BC on the same side of BC as A, B' is a pole of AC on the same side of AC as B, and C' is a pole of AB on the same side of AB as C. Using the three properties above, we can prove the following theorems:

1. Given the polar triangle of $\triangle ABC$, $\triangle A'B'C'$, $\angle A + B'C' = \pi$.

   To prove this, we extend AB and AC to meet $B'C'$ at points X and Y. We then have that $B'Y = C'X = \frac{\pi}{2}$ by property 1 of section 5.1. $B'Y + C'X = B'C' + XY$, so $B'C' + XY = \frac{\pi}{2} + \frac{\pi}{2} = \pi$. By property 3 of section 5.1, we know that $\angle A = XY$. Plugging this into the equation, we get $\angle A + B'C' = \pi$, as desired.

2. If $\triangle A'B'C'$ is the polar triangle to $\triangle ABC$, then $\triangle ABC$ is also the polar triangle of $\triangle A'B'C'$.

   We know B' is a pole of AC, so by property 1 of section 5.1, $AB' = \frac{\pi}{2}$. By a similar logic, $AC' = \frac{\pi}{2}$, which implies that A is a pole of $B'C'$. Similarly, we can show that B is a pole of $A'C'$ and C is a pole of $A'B'$.

5.3 AAA congruence

Let $\triangle ABC$ and $\triangle DEF$ have $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$. Consider their polar triangles, $\triangle A'B'C'$ and $\triangle D'E'F'$. By theorem 1 of section 5.2:

$$B'C' + \angle A = \pi$$
$$E'F' + \angle D = \pi$$

Since $\angle A \cong \angle D$ by assumption, we know $B'C' = E'F'$ and can similarly find that $A'B' = D'E'$ and $A'C' = D'F'$. Therefore, by SSS congruence, we know that $\triangle A'B'C' \cong \triangle D'E'F'$.

We know that by theorem 2 of section 5.2, $\triangle ABC$ is the polar triangle of $\triangle A'B'C'$ and $\triangle DEF$ is the polar triangle of $\triangle D'E'F'$. Applying theorem 1 of section 5.2 again:

$$BC + \angle A' = \pi$$
$$EF + \angle D' = \pi$$

Since we've shown that $\triangle A'B'C' \cong \triangle D'E'F'$, we know that $\angle A' \cong \angle D'$, so $BC = EF$. Similarly, we find that $AB = DE$ and $AC = DF$, which tells us that $\triangle ABC \cong \triangle DEF$ by SSS.

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Dirichlet Convolutions

Steven Breger

1 Arithmetic Functions

In all that follows, $\mathbb{N}$ denotes the set of positive integers.

Definition 1.1. *A function f is said to be arithmetic if f maps $\mathbb{N} \to \mathbb{C}$.*

Remark 1.2. We will mainly deal with functions that map $\mathbb{N} \to \mathbb{Z}$ in these notes.

Definition 1.3. *An arithmetic function f is said to be multiplicative if $f(mn) = f(m)f(n)$ for all coprime $m, n \in \mathbb{N}$ and if $f(1) \neq 0$.*

Definition 1.4. *An arithmetic function f is said to be completely multiplicative if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$ and if $f(1) \neq 0$.*

Note that for multiplicative or completely multiplicative f, $f(1) = f(1 \cdot 1) = f(1)^2$. Since $f(1) \neq 0$, $f(1) = 1$ (if we were to allow $f(1) = 0$, we would introduce the function $f(n) = 0$, which we will come to see does not play nicely with what follows). Consider σ, the sum of divisors function. We can write σ as

$$\sum_{d|n} d.$$

Let $D(n)$ be the set of divisors of n. It is not too hard to see that σ is multiplicative since there is a bijection between $D(mn)$ and $D(m) \times D(n)$ for coprime $m, n \in \mathbb{N}$. For instance, if we take $12 = 3 \cdot 2^2$, any divisor of 12 must have a component that comes from 3 and a component that comes from 2^2 (for example 6 can be split into 3 and 2 while 2^2 can be split into 1 and 2^2).

So, we can notice that if we write the sum of the divisors of 3: $\sigma(3) = 1 + 3$, the sum of the divisors of 2^2: $\sigma(2^2) = 1 + 2 + 2^2$, and the sum of the divisors of $3 \cdot 2^2$: $\sigma(3 \cdot 2^2) = 1 + 2 + 2^2 + 3 + 3 \cdot 2 + 3 \cdot 2^2$, the third is the product of the first, as when multiplying $\sigma(3)$ and $\sigma(2^2)$, each resulting term is exactly a divisor of $\sigma(3 \cdot 2^2)$ by our bijection. In general, we have that σ is multiplicative.

Because σ is multiplicative, if we can evaluate $\sigma(p^k)$ for all primes p and $k \geq 0$, we can evaluate $\sigma(n)$ for any n by prime factorizing n. If $p_1^{a_1} \cdots p_i^{a_i}$ is the canonical prime factorization of n, then $\sigma(n) = \sigma(p_1^{a_1} \cdots p_i^{a_i}) = \sigma(p_1^{a_1}) \cdots \sigma(p_i^{a_i})$.

2 The Dirichlet Convolution

Note the form we found for σ. We have that

$$\sum_{d|n} d.$$

We can generalize this by consider arithmetic functions F and f such that

$$F(n) = \sum_{d|n} f(d).$$

If you play around a little bit more, you will come to find that the following sum for 2 arithmetic functions f and g is very useful:

Definition 2.1. *(Dirichlet Convolution) Given arithmetic functions f and g, we define their Dirichlet Convolution to be $f * g$ where*

$$(f * g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right)$$

Example 2.2. $1 * \mathrm{id} = \sigma$. Note that here 1 refers to function that sends every natural to 1 and id refers to the identity function that sends every n to n. In other words, for all n, $1(n) = 1$ and $\mathrm{id}(n) = n$. Whether or not 1 refers to a number or a function should be clear based on context. The same is true for n.

Theorem 2.3. *Dirichlet convolution is commutative, associative, and distributes over addition.*

Proof. Commutativity and the distributive property are trivial. Note that to prove associativity, that instead of summing over $d \mid n$, we can some over all $ij = n$ and i will sum range over the divisors of n. So,

$$\begin{aligned}
&((f * g) * h)(n) \\
&= \sum_{kc=n} (f * g)(k)h(c) \\
&= \sum_{kc=n} \left(\sum_{ab=k} f(a)f(b) \right) h(c) \\
&= \sum_{kc=n} \sum_{ab=k} f(a)f(b)f(c) = \sum_{abc=n} f(a)f(b)f(c)
\end{aligned}$$

Analogously, the last quantity above is also equal to $f * (g * h)(n)$. This finishes. □

Theorem 2.4. *If any two of f, g, $f * g$ are multiplicative, then so is the third.*

Proof. Suppose that f and g are multiplicative. Let $m, n \in \mathbb{N}$ and coprime. Then, by the bijection between $D(mn)$ and $D(m) \times D(n)$ that we mentioned before,

$$\begin{aligned}
&(f * g)(mn) \\
&= \sum_{d|mn} f(d)g\left(\frac{mn}{d}\right) \\
&= \sum_{d_1|m} \sum_{d_2|n} f(d_1 d_2)g\left(\frac{mn}{d_1 d_2}\right) \\
&= \sum_{d_1|m} \sum_{d_2|n} f(d_1)f(d_2)g\left(\frac{m}{d_1}\right) g\left(\frac{n}{d_2}\right) \\
&= \left(\sum_{d_1|m} f(d_1)g\left(\frac{m}{d_1}\right) \right) \left(\sum_{d_2|n} f(d_2)g\left(\frac{n}{d_2}\right) \right) \\
&= (f * g)(m) \cdot (f * g)(n)
\end{aligned}$$

We now need to show that if g and $f * g$ are multiplicative, then so is f. Suppose the contrary. Then there exist $m, n \in \mathbb{N}$ such that $f(mn) \neq f(m)f(n)$. By the well ordering principle, we can choose m, n such

that mn is minimal. Consider that

$$\begin{aligned}
(f*g)(mn) &= \sum_{d|mn} f(d)g\left(\frac{mn}{d}\right) \\
&= f(mn) + \sum_{d|mn, d<mn} f(d)g\left(\frac{mn}{d}\right) \\
&= f(mn) + \sum_{d_1|m, d_1<m} \sum_{d_2|n, d_2<n} f(d_1d_2)g\left(\frac{mn}{d_1d_2}\right) \\
&= f(mn) + \sum_{d_1|m, d_1<m} \sum_{d_2|n, d_2<n} f(d_1)f(d_2)g\left(\frac{m}{d_1}\right)g\left(\frac{n}{d_2}\right) \quad \text{(by the minimality of } mn\text{)} \\
&= f(mn) + \sum_{d_1|m, d_1} \sum_{d_2|n, d_2} f(d_1)f(d_2)g\left(\frac{m}{d_1}\right)g\left(\frac{n}{d_2}\right) - f(m)f(n) \\
&= f(mn) + \sum_{d_1|m} f(d_1)g\left(\frac{m}{d_1}\right) \sum_{d_2|n} f(d_2)g\left(\frac{n}{d_2}\right) - f(m)f(n) \\
&= f(mn) + (f*g)(m)\cdot(f*g)(n) - f(m)f(n)
\end{aligned}$$

Since $f*g$ is multiplicative, the above implies that $f(mn) = f(m)f(n)$, a contradiction. □

Remark 2.5. As we will see, this theorem becomes false if we replace "multiplicative" with "completely multiplicative."

3 More about $*$

One natural question to ask about convolution is whether or not there is an arithmetic function ε that acts as an identity under convolution. That is for any arithmetic function f, $f*\varepsilon = f$. It is not too hard to see that

$$\varepsilon(n) = \begin{cases} 1, n = 1 \\ 0, \text{otherwise} \end{cases}$$

satisfies this property. It is up to the reader to verify this.

Remark 3.1. This is not the identity function on the naturals (the one denoted as id or n that sends every natural to itself)! id is the identity function on the naturals, which is not an identity under convolution as $f*\text{id}$ will not give you f most of the time! When we refer to an identity, the context will make it clear which function is being referenced. As far as math notation though, we will use ε if and only if we are referring to the identity under convolution.

Another natural question to ask is whether every arithmetic function f has a Dirichlet inverse. Well, suppose we have a function f^{-1} satisfying $f*f^{-1} = \varepsilon$. By plugging in 1 on both sides, we get that $f^{-1}(1) = \frac{1}{f(1)}$. Thus, $f(1) \neq 0$. Now, for any $n > 1$ we have that

$$\begin{aligned}
&(f*f^{-1})(n) = \varepsilon(n) \\
\implies &\sum_{d|n} f^{-1}(d)f\left(\frac{n}{d}\right) = 0 \\
\implies &f^{-1}(n)f(1) + \sum_{d|n, d<n} f^{-1}(d)f\left(\frac{n}{d}\right) = 0 \\
\implies &f^{-1}(n) = -\frac{1}{f(1)} \sum_{d|n, d<n} f^{-1}(d)f\left(\frac{n}{d}\right)
\end{aligned}$$

This gives us a recursive way of computing f^{-1} as long as $f(1) \neq 0$. So to summarize, there is an identity under $*$ and every function with $f(1) \neq 0$ has an inverse. Note that by definition, we get that each multiplicative (and completely multiplicative) function has an inverse.

What's cool is that with all of these properties, we get that arithmetic functions form a ring under $+$ and $*$ (note that they don't form a field because not every nonzero element has an inverse under $*$ by the above). Essentially what this means is that multiplication and addition are nice, but not everything has an inverse under $*$. This is analogous to how $\mathbb{Z}$ functions under $+$ and $\cdot$.

Also, the set of arithmetic functions with $f(1) \neq 0$ forms the group of units of this ring. This is another way of saying "the set of elements with an inverse" (think about $\{1, -1\}$ for $\mathbb{Z}$). The reason we say group is because this set forms a group under $*$, meaning its closed under $*$ and satisfies a few nice properties. Furthermore, the set of multiplicative arithmetic functions is a subgroup of this group of units (mainly because it satisfies closure under $*$ and inverses). The set of completely multiplicative functions is not though, since it is not closed under $*$, simply consider $1 * 1$.

It's not to important exactly what the necessary properties are (you can look them up if you like) but they are mostly the things proven above.

4 Möbius Inversion

Although we now know that we can theoretically compute the inverse of any multiplicative function, let us try to explicitly compute the inverse of the function 1.

Let $1^{-1} = \mu$. For starters, μ must be multiplicative by theorem 2.4. Thus, $\mu(1) = 1$. Furthermore, if we can determine the value of μ on each prime power we would be done since if we decompose any integer n into its prime canonical factorization $p_1^{a_1} \cdots p_i^{a_i}$, then we can compute $\mu(n)$ by using

$$\mu(n) = \mu(p_1^{a_1} \cdots p_i^{a_i}) = \mu(p_1^{a_1}) \cdots \mu(p_i^{a_i}).$$

Well, for any prime p and integer $a > 0$ we must have that

$$\begin{aligned} &(\mu * 1)(p^a) = \varepsilon(p^a) \\ \implies &\sum_{d|p^a} \mu(d) = 0. \end{aligned}$$

We can start by setting $a = 1$. We get that $\mu(p) = -1$. Then, by setting $a = 2$ we get that $\mu(p^2) = 0$. It is not too hard to see that in fact $\mu(p^a) = 0$ for all $a > 1$. In other words, μ zeros out when there is a prime of multiplicity more than 1 and gives -1 for a prime otherwise.

By using the prime factorization above, we can come up with the following

Definition 4.1. *We define the arithmetic function μ as follows.*

$$\mu(n) = \begin{cases} 1, n = 1 \\ (-1)^i, n = p_1 \cdots p_i \\ 0, \text{otherwise} \end{cases}$$

Of course, p_i are distinct primes in the above, and $\mu(n) = 0$ is equivalent to $p^2 \mid n$ for some prime p.

We have determined that $\mu = 1^{-1}$. This lets us do a cool trick.

Theorem 4.2. *(Möbius Inversion) Suppose that we have an arithmetic function f. Then,*

$$F(n) = \sum_{d|n} f(d).$$

if and only if

$$f(n) = \sum_{d|n} F(d)\mu\left(\frac{n}{d}\right).$$

In other words, $F = f * 1$ if and only if $f = F * \mu$.

Proof. Of course, $F = f * 1$. Because convolution is associative, convoluting both sides with μ gives us

$$\begin{aligned} & F * \mu = f * 1 * \mu \\ \iff & F * \mu = f * \varepsilon \\ \iff & F * \mu = f \end{aligned}$$

This finishes. □

Example 4.3. For a prime p and $a > 0$, $\varphi(p^a) = p^a - p^{a-1}$. Observe that this formula is simply $(\mathrm{id} * \mu)(p^a)$. Since φ is multiplicative, or because id and μ are multiplicative, we have that $\varphi = \mathrm{id} * \mu$. Thus, $\mathrm{id} = \varphi * 1$ and we get that

$$n = \sum_{d|n} \varphi(d).$$

5 Sample (Easy) Problems

- Find $\mu(30)$.
- True or False: A completely multiplicative function has an inverse.
- True or False: Completely multiplicative functions are closed under convolution.
- True or False: The convolution of two multiplicative functions is multiplicative.
- Compute $(\varphi * \sigma)(6^5)$.

6 Sample (Hard) Problems

- Find

$$f(n) = \sum_{k=1}^{n} \mu(k) \left\lfloor \frac{n}{k} \right\rfloor.$$

- Show that

$$\sum_{k=1}^{n} \varphi(k) = \frac{1}{2}\left(\sum_{k=1}^{n} \mu(k) \left\lfloor \frac{n}{k} \right\rfloor^2 + 1\right).$$

- Show that

$$\sum_{1 \le k \le n, (k,n)=1} e^{2\pi i \cdot \frac{k}{n}} = \mu(n)$$

  (Hint: Consider the summing over all $k \le n$ and inserting a function that is 0 unless $(k, n) = 1$). Generalize this to find

$$\sum_{1 \le k \le n, (k,n)=1} f\left(\frac{k}{n}\right)$$

  in terms of

$$\sum_{1 \le k \le n} f\left(\frac{k}{n}\right).$$

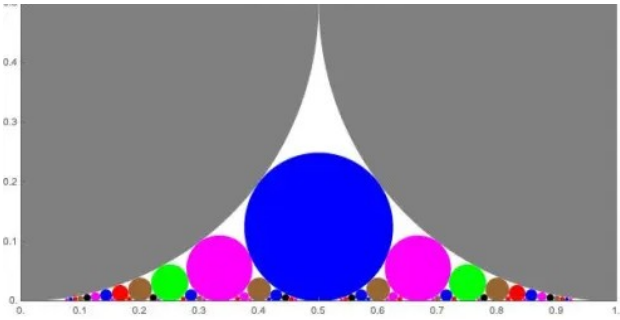
- Do there exist zero divisors in the Dirichlet Ring? If yes, give and justify an example. If not, provide a proof of the fact. Note that a zero divisor in the Dirichlet Ring is an arithmetic function f different from $\mathbf{0}$ such that there exists an arithmetic g different from $\mathbf{0}$ satisfying $f * g = \mathbf{0}$.

Ford Circles And Farey Sequences

Jacob Levin

1 What are Ford Circles?

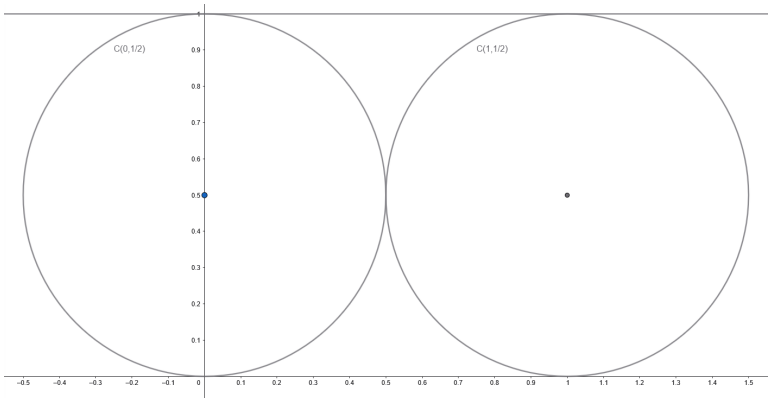
1.1 Introduction



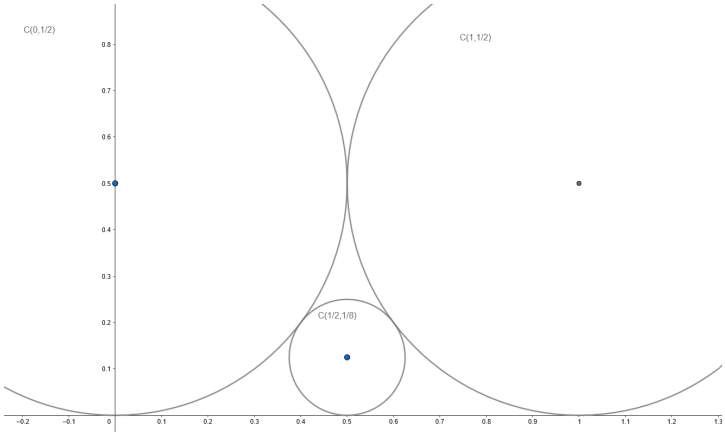
Don't these colorful circles look cool? Everything looks very symmetrical and perfect. What are these mystical, oddly satisfying-to-look-at circles, what rules do they obey, and how do they relate to a seemingly irrelevant sequence of fractions?

1.2 Definition

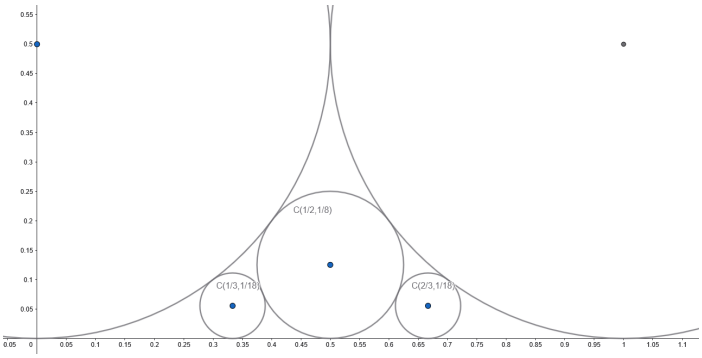
We can say that a circle $C(i, j)$ tangent to the x-axis has center (i, j). Next, we realize that the radius of $C(i, j)$ is the y-coordinate, or just j. Now, consider a set of circles C tangent to the x-axis. We are given that $C(0, \frac{1}{2})$ and $C(1, \frac{1}{2})$ make up the ordered subset C_1 (notice that $C(0, \frac{1}{2})$ and $C(1, \frac{1}{2})$ are tangent to each other, since both have radius $\frac{1}{2}$ and are 1 apart). For $k \geq 1$ C_{k+1} is constructed by starting with C_k. Then, for any pair of tangent circles in C_k, C_{k+1} contains the $C(i, j)$ tangent to both circles in that pair. For instance, C_1 will look like this:



C_2 will look like this:



And C_3 would look like this:

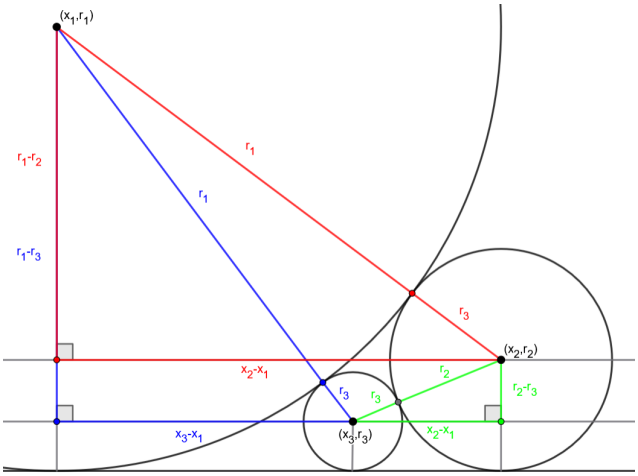


We say that C is the set of all of these circles, or in other words just the union of all C_k.

1.3 Properties

1.3.1 Radius

First, we wish to find the radius of a circle between two others. Suppose that $C(x_1, r_1)$ and $C(x_2, r_2)$ are two tangent Ford Circles. If we say that the Ford circle tangent to them both is $C(x_3, r_3)$ where $r_3 < r_1, r_2$, we want to express r_3 in terms of r_1 and r_2.



As shown in the diagram, we can drop a perpendicular to the lines connecting $C(x_1, r_1)$ and $C(x_2, r_2)$ to their respective tangency points to the x-axis. As a result, we get two right triangles, one with hypotenuse $r_1 + r_3$ and legs $r_1 - r_3$ and $x_3 - x_1$, and another with hypotenuse $r_2 + r_3$ and legs $r_2 - r_3$ and $x_2 - x_3$. Applying the Pythagorean Theorem on them both, we get that $(r_1 + r_3)^2 = (r_1 - r_3)^2 + (x_3 - x_1)^2 \implies 4r_1r_3 = (x_3 - x_1)^2 \implies 2\sqrt{r_1r_3} = x_3 - x_1$, and $(r_2 + r_3)^2 = (r_2 - r_3)^2 + (x_2 - x_3)^2 \implies 4r_2r_3 = (x_2 - x_3)^2 \implies 2\sqrt{r_2r_3} = x_2 - x_3$. Adding these gives us $2\sqrt{r_1r_3} + 2\sqrt{r_2r_3} = x_2 - x_1$.

By the same logic, if we construct a similar right triangle with $C_(x_1, r_1)$ and $C_(x_2, r_2)$ and do the Pythagorean Theorem again, we get that $2\sqrt{r_1r_2} = x_2 - x_1$. So since both $2\sqrt{r_1r_3} + 2\sqrt{r_2r_3}$ and $2\sqrt{r_1r_2}$ are equal to $x_2 - x_1$, we have that $2\sqrt{r_1r_3} + 2\sqrt{r_2r_3} = 2\sqrt{r_1r_2} \implies \sqrt{r_3}(\sqrt{r_1} + \sqrt{r_2}) = \sqrt{r_1r_2} \implies \frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$. (Interestingly, this is a degenerate case of Descartes' Circle Theorem, where the x-axis actually a circle with infinite radius). So, we have our first property:

Property 1 For some ford circle $C(x_3, r_3)$ tangent to two other ford circles $C(x_1, r_1), C(x_2, r_2)$ such that $r_1, r_2 > r_3$, $\frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$.

1.3.2 Tangency Point

We will make the claim that for some ford circle $C(x_3, r_3)$ tangent to two other ford circles $C(\frac{p}{q}, r_1), C(\frac{r}{s}, r_2)$ such that $r_1, r_2 > r_3$ and $\gcd(p, q) = \gcd(r, s) = 1$, $x_3 = \frac{p+r}{q+s}$, and additionally, for some ford circle $C(\frac{a}{b}, r)$ for $\gcd(a, b) = 1$, $r = \frac{1}{2b^2}$. These claims can be proven by induction.

Base Case: $C(0, \frac{1}{2})$ and $C(1, \frac{1}{2})$

The circle tangent to them both will lie directly in the middle by symmetry, since the circles have equal radii and their radial axis is parallel to the x-axis. Thus, the circle tangent to them both touches the x-axis at $x = \frac{0+1}{1+1} = \frac{1}{2}$. By Property 1.1, we also get that $\frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{\frac{1}{2}}} + \frac{1}{\sqrt{\frac{1}{2}}} \implies \frac{1}{\sqrt{r_3}} = 2\sqrt{2} \implies r_3 = \frac{1}{8} = \frac{1}{2(2^2)}$, so our second condition also holds.

Induction Hypothesis: The radii of $C(\frac{p}{q}, r_1)$ and $C(\frac{r}{s}, r_2)$ are $\frac{1}{2q^2}, \frac{1}{2s^2}$, respectively.

Induction Step: From our previous right triangles, we have that $2\sqrt{r_1r_3} = x_3 - x_1$ and $2\sqrt{r_2r_3} = x_2 - x_3$. Plugging in the values from the induction hypothesis, we have that $2\sqrt{r_1r_3} = x_3 - \frac{p}{q}$ and $2\sqrt{r_3r_2} = \frac{r}{s} - x_3$. Adding the equations, we get $\frac{r}{s} - \frac{p}{q} = 2\sqrt{r_3}(\sqrt{r_1} + \sqrt{r_2}) \implies \sqrt{r_3} = \frac{\frac{r}{s} - \frac{p}{q}}{2(\sqrt{r_1} + \sqrt{r_2})}$.

Plugging this into $x_3 = \frac{p}{q} + 2\sqrt{r_1r_3}$, we get $x_3 = \frac{p}{q} + 2\sqrt{r_1r_3} = \frac{p}{q} + (\frac{2\sqrt{r_1}(\frac{r}{s} - \frac{p}{q})}{2(\sqrt{r_1} + \sqrt{r_2})}) = \frac{p}{q} + \frac{\sqrt{r_1}(\frac{r}{s} - \frac{p}{q})}{\sqrt{r_1} + \sqrt{r_2}}$.

Using $\sqrt{r_1} = \frac{1}{\sqrt{2}q}$ and $\sqrt{r_2} = \frac{1}{\sqrt{2}q}$, we have that $\frac{\sqrt{r_1}}{\sqrt{r_1}+\sqrt{r_2}} = \frac{\frac{1}{\sqrt{2}q}}{\frac{1}{\sqrt{2}q}+\frac{1}{\sqrt{2}s}} = \frac{\frac{1}{q}}{\frac{1}{q}+\frac{1}{s}} = \frac{s}{q+s}$. This means that $x_3 = \frac{p}{q} + \frac{s(\frac{r}{s}-\frac{p}{q})}{q+s} = \frac{p}{q} + \frac{rq-sp}{q(q+s)} = \frac{pq+ps+rq-sp}{q(q+s)} = \frac{pq+rq}{q(q+s)} = \frac{p+r}{q+s}$, which is our desired x-coordinate.

Finally, we want to prove that our third circle has radius $\frac{1}{2x_3^2}$. Since $r_1 = \frac{1}{2q^2}, r_2 = \frac{1}{2s^2}$, and $\frac{1}{\sqrt{r_3}} = \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}}$, we have that $\frac{1}{\sqrt{r_3}} = \frac{1}{\frac{1}{\sqrt{2}q}} + \frac{1}{\frac{1}{\sqrt{2}s}} = \sqrt{2}q + \sqrt{2}s \implies r_3 = \frac{1}{(\sqrt{2}(q+s))^2} = \frac{1}{2(q+s)}$. And now, since $x_3 = \frac{p+r}{q+s}$, $r_3 = \frac{1}{2(q+s)^2}$ is our desired result. Thus, we now have two more properties:

Property 2 For some ford circle $C(x_3, r_3)$ tangent to two other ford circles $C(\frac{p}{q}, r_1), C(\frac{r}{s}, r_2)$ such that $r_1, r_2 > r_3$ and $\gcd(p, q) = \gcd(r, s) = 1$, $x_3 = \frac{p+r}{q+s}$.

Property 3 For some ford circle $C(\frac{a}{b}, r)$ such that $\gcd(a, b) = 1$, $r = \frac{1}{2b^2}$.

2 Farey Sequences

Consider the following sequence: for some positive integer n, the sequence F_n is the set of reduced fractions in increasing order between 0 and 1 inclusive such that their denominators do not exceed n. We can see that F_1 is the set $[\frac{0}{1}, \frac{1}{1}]$, F_2 is $[\frac{0}{1}, \frac{1}{2}, \frac{1}{1}]$, F_3 is $[\frac{0}{1}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{1}{1}]$, and so on.

2.1 Properties

2.1.1 $bc - ad = 1$

We will prove that for any two adjacent fractions $\frac{a}{b} < \frac{c}{d}$ in F_n, $bc - ad = 1$.

Proof: We can show the contrapositive: If two fractions in F_n form a matrix of determinant not -1, then they are not adjacent in F_n.

Take two Farey fractions $\frac{a}{b} < \frac{c}{d}$ in F_n. We have that $bc - ad > 0$. We will suppose that $bc - ad = k > 1$. By Bezout's lemma, there exist integers x_0, y_0 such that $bx_0 - ay_0 = 1$. Bezout's lemma also tells us that we can replace x_0 and y_0 with $x = x_0 + at$ and $y = y_0 + bt$ for any integer t to get the entire solution set to this equation. We take y to be the largest positive integer solution which will satisfy $y \leq n$. Notice that $n - b < y < n$, since otherwise you could find a larger y by adding b and adding a to x. Since we defined y to be less than n, $\frac{x}{y}$ is in F_n. And since $bx - ay = 1$, $bx > ay \implies \frac{x}{y} > \frac{a}{b}$.

Let us examine the difference $\frac{c}{d} - \frac{x}{y} = \frac{cy-dx}{dy}$. Plugging in for x, we find that $cy - dx = cy - d(\frac{1+ay}{b}) = \frac{cyb-db-dya}{b} = \frac{y(bc-ad)-d}{b} = \frac{yk-d}{b}$. We wish to prove that $yk - d \geq b$, which would imply that $cy - dx > 0$ and that $\frac{c}{d} > \frac{x}{y}$. Since $bx - ay = 1$, we can treat (x, y) like a vector, so we can write c and d as $c = kx + ma$ and $d = ky + mb$ (we can plug these values into $bc - ad = k$ to see that this is a valid representation). So, we can manipulate $mb = d - ky$. Let us plug in $ky > k(n - b)$ and $d \leq n$: $mb < n - k(n - b) < 2b - n < b$, so $m < 1$. If $m < 0$, then $yk - d = -mb \geq b > 0$, so $cy - dx > 0$. If $m = 0$, then $yk - d = -mb = 0$, however then $cy - dx = 0 \implies \frac{c}{d} = \frac{x}{y}$, which cannot be true (since $bc - ad$ cannot equal 1 and k at the same time). Thus, $yk - d > 0$, so $cy - dx > 0 \implies \frac{c}{d} > \frac{x}{y}$, so $\frac{a}{b} < \frac{x}{y} < \frac{c}{d}$, and we are done.

Property 4 For any two consecutive fractions $\frac{a}{b}, \frac{c}{d}$ in F_n, $bc - ad = 1$.

2.1.2 Mediant (Farey Property)

Next, let us look at a fraction in between two fractions: say $\frac{1}{2}$, which is in between $\frac{0}{1}$ and $\frac{1}{1}$. Well, we can see that the numerator of $\frac{1}{2}$ is equal to the sum of the numerators of $\frac{0}{1}$ and $\frac{1}{1}$, and similarly the denominator of $\frac{1}{2}$ is equal to the sum of the denominators of $\frac{0}{1}$ and $\frac{1}{1}$. And surprisingly, as we look at higher cases, we notice that this property also holds for any fraction in F_n (note: the fraction $\frac{a+c}{b+d}$ is known as the mediant of $\frac{a}{b}$ and $\frac{c}{d}$, and will be referred to as such from now on). Let us try to prove that this is the case for all fractions between two others in any Farey Sequence.

Proof: We want to show that in F_k all fractions of the form $\frac{x}{y}$ are right between some two fractions $\frac{a}{b} < \frac{c}{d}$ from F_k for whom $\frac{x}{y}$ is the mediant. We know that $bx - ay = 1, yc - xd = 1$ from our previous proof. Next, multiply the first equation by d and the second by b, and then sum them. We get $bxd - ayd + ycb - xdb = b + d \implies y(bc - ad) = b + d \implies y = b + d$.

We can conduct a similar process to prove that $x = a + c$, simply by multiplying the first equation by c, the second by a, and then summing them, to get $bxc - ayc + yca - xda = a + c \implies x(bc - ad) = a + c \implies x = a + c$. Thus, $\frac{x}{y} = \frac{a+c}{b+d}$ and the following property holds:

Property 5 For any three consecutive fractions $\frac{a}{b}, \frac{x}{y}, \frac{c}{d}, \frac{x}{y} = \frac{a+c}{b+d}$.

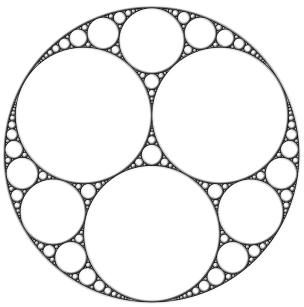
Before we move on, it is important to note that $b + d$ could be greater than k. However, our proven property only reveals information about the value of the fraction in between, and since we assumed that all three fractions are in F_k, it is implied that the middle fraction could be simplified.

2.2 Bringing Everything Together

By Property 2, two tangent circles $C(\frac{a}{b}, \frac{1}{2b^2}), C(\frac{c}{d}, \frac{1}{2d^2})$ are tangent to circle $C(\frac{a+b}{c+d}, \frac{1}{2(c+d)^2})$. Furthermore, by Property 5, for any two consecutive fractions $\frac{a}{b}, \frac{c}{d}$ in a Farey sequence, the fraction in a later sequence which will go between them will be $\frac{a+c}{b+d}$. These properties are very similar, so let us find a way to connect them.

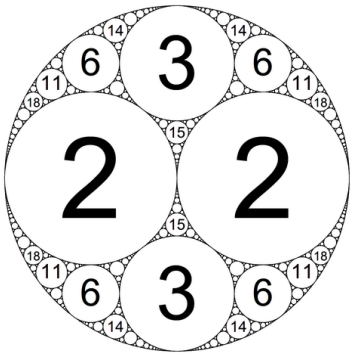
We can prove that the x-values of two tangent circles are adjacent in some Farey Sequence, and that the circle tangent to both of them has an x-value of the mediant of their x-values. Our base case is $C(\frac{0}{1}, \frac{1}{2})$. It is tangent to $C(\frac{1}{1}, \frac{1}{2}), C(\frac{1}{2}, \frac{1}{8}), C(\frac{1}{3}, \frac{1}{18}), ...$, all of which clearly have x-values in the Farey Sequence, and are all adjacent to $\frac{0}{1}$ in the Farey Sequence. Next, suppose that some circle $C(\frac{a}{b}, \frac{1}{b^2})$ is tangent to some circles and also adjacent to them in a Farey Sequence. We know that the fractions between $\frac{a}{b}$ and the fractions of those other circles must be their mediants. However, we also know that a circle's x-value is the mediant of the x-values of the circles tangent to each other and it. So, $C(\frac{a}{b}, \frac{1}{2b^2})$ must be adjacent to that circles as well. Thus, the x-values of two tangent circles are adjacent in some Farey Sequence, and that the circle tangent to both of them has an x-value of the mediant of their x-values.

3 Extra: Ford-eque Circles



The image above is of an Apollonian Gasket, a shape created from starting with three circles tangent to each other and indefinitely constructing circles tangent to three others. It is a fun challenge to try to figure out for yourself the radii, if say, the radius of the big circle is 1 and the three circles inside are congruent (here's a hint to start off with: Descartes's Theorem states that for 4 mutually tangent circles, $(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4})^2 = 2(\frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r_4^2})$, however the three circles are internally tangent to the larger one, so the radius of the larger one would have to be considered as -1). There exist other Apollonian Gaskets (this term covers a very wide variety of fractals): below is another gasket which has its own properties and

relationships (the numbers inside the circles represent the circle's curvature, or the inverse of the radius, importantly the curvature of the large circle is -1), feel free to play around with it as well!



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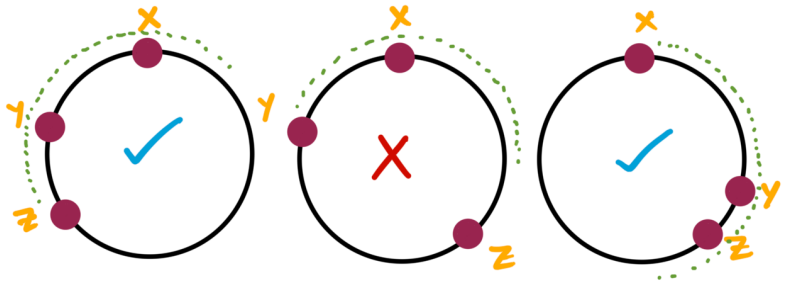
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Wendel's Theorem

Ryan Peng

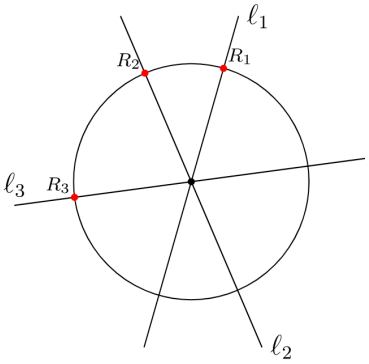
1 Three Points on a Circle

Suppose you have a circle, and you choose 3 random points on it. What is the probability that all 3 of these points lie on some half of the circle?

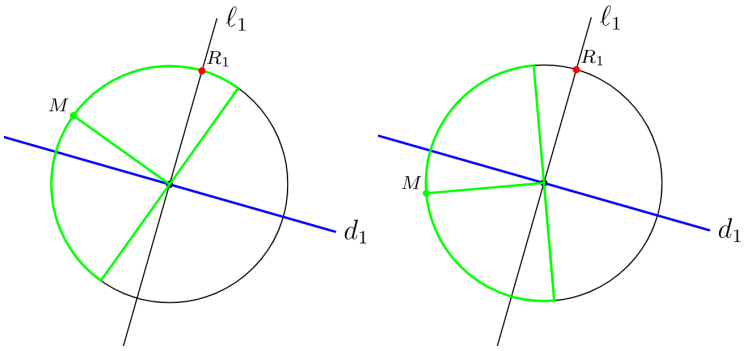


Here is one particular way to approach this problem:

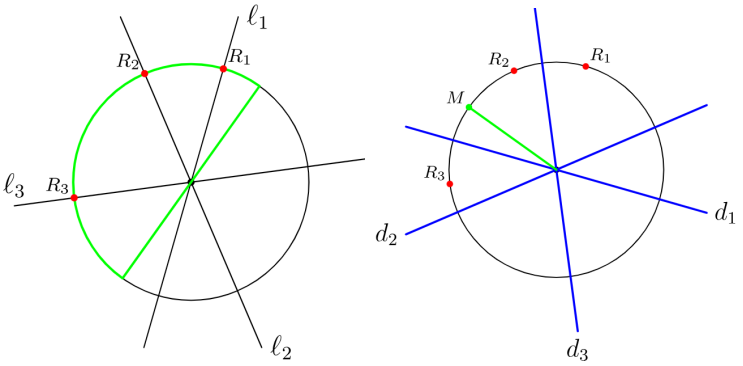
Label the three points R_1, R_2, R_3, and draw the lines connecting the points to the center of the circle. Call these lines ℓ_1, ℓ_2, ℓ_3, respectively. We can assume that ℓ_1, ℓ_2, ℓ_3 are distinct, since the probability that two randomly chosen points on a circle are diametrically opposite each other is 0. This is because there are infinitely many choices for the second point, but only one point that is diametrically opposite the first.



Consider a semicircle whose diameter does not coincide with any of the three lines. Let M be the midpoint of the semicircle.
Let d_1 be the line passing through the center of the circle that is perpendicular to ℓ_1.
Notice that the semicircle contains R_1 if and only if R_1 and M are on the same side of d_1.

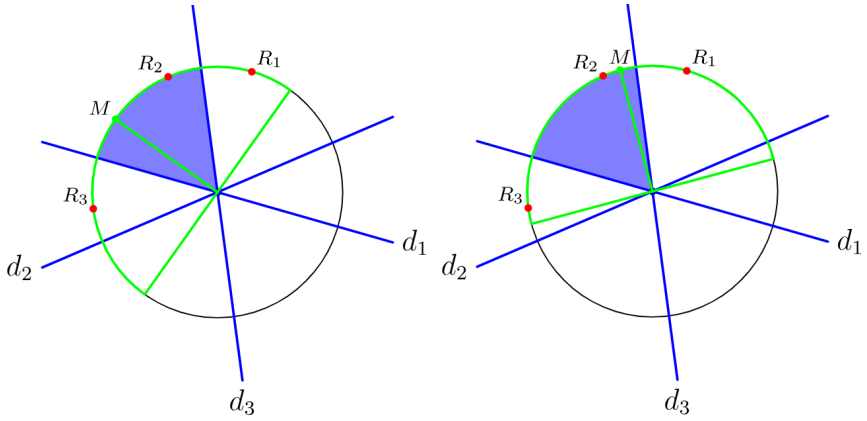


Define d_2 and d_3 analogously to d_1.



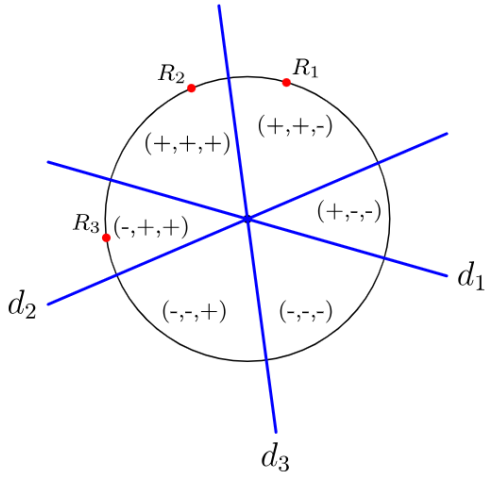
In this diagram, the semicircle with midpoint M contains all 3 of R_1, R_2, R_3. This is because M and R_1 are on the same side of d_1, M and R_2 are on the same side of d_2, and M and R_3 are on the same side of d_3.

Since this relies only on the position of M relative to d_1, d_2, d_3, the vector from M to the center of the circle can lie anywhere in the shaded region between d_1 and d_3, and the semicircle with midpoint M will still contain all 3 red points. However, if this vector lies outside of the shaded region, then the semicircle with midpoint M will not contain all 3 red points.

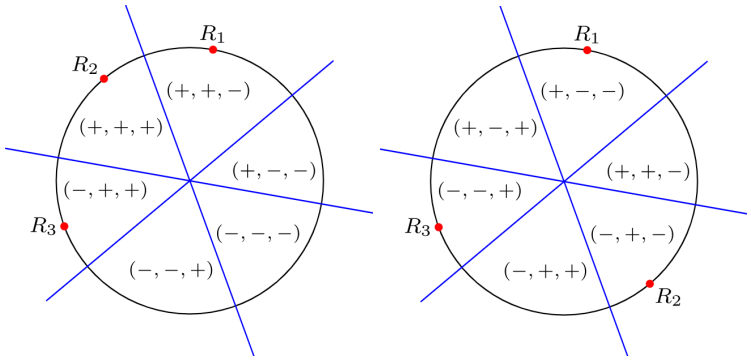


Extending this logic, each of the 6 regions that d_1, d_2, d_3 divide the plane into can be represented with a 3-tuple $s = (s_1, s_2, s_3)$, where $s_1, s_2, s_3 \in \{+, -\}$. For $1 \leq i \leq 3$, s_i is $+$ if and only if any semicircle with midpoint in the region contains R_i. Furthermore, we say that a 3-tuple (s_1, s_2, s_3) *occurs* if there exists some

region with $s = (s_1, s_2, s_3)$. For example, $(+,-,+)$ and $(-,+,-)$ do not occur in the diagram below. We want to find the probability that $(+,+,+)$ occurs.



Notice that reflecting R_i across the center of the circle will flip s_i in all 6 regions ($i = 2$ shown below).



Therefore, any sign is equally likely to be $+$ or $-$, so every 3-tuple is equally likely to occur.

Any 3 lines d_1, d_2, d_3 through the center of the circle divide the plane into 6 regions, and there are $2^3 = 8$ possible 3-tuples (s_1, s_2, s_3).

Therefore, the probability that $(+,+,+)$ occurs is $\frac{6}{8} = \boxed{\frac{3}{4}}$.

This might seem like an overly complicated solution for such a simple problem. And while it is true that simpler solutions exist, the strength in this approach is that it can be easily generalized, as we will see below.

2 More Points

What if instead of 3 points, we picked 4? 5? What about n points?

Well, notice that nowhere in the above argument did we rely on the number of points chosen being 3, except at the end when we calculated the numbers 6 and 8.

n lines will always divide the plane into $2n$ regions, and there are always 2^n possible $n-$tuples of $+$ and $-$

signs. Therefore, the general solution for n points is

$$P = \frac{\#\text{ of regions}}{\#\text{ of possible n-tuples}} = \frac{2n}{2^n} = \boxed{\frac{n}{2^{n-1}}}$$

3 Higher Dimensions

We can further generalize our result to higher dimensions. For example, instead of picking points on a circle, we can pick points on a sphere, and find the probability that they all lie on some hemisphere.

3.1 Definitions

To describe objects in higher dimensions, we must introduce some terms.

Definition 1

The **real coordinate space** of dimension d, denoted $\mathbb{R}^d$, is the set of all $d-$tuples of real numbers. For example, the real number line is $\mathbb{R}^1$ and the real coordinate plane is $\mathbb{R}^2$.

Definition 2

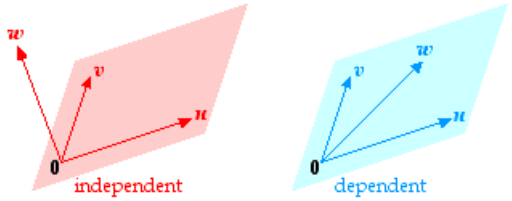
A **hyperplane** in $\mathbb{R}^d$ is a flat $(d-1)$-dimensional surface. It is represented by the linear equation

$$a_1x_1 + a_2x_2 + ... + a_dx_d = b$$

Where $a_1, a_2, ..., a_d$ are coefficients and b is a constant.
For example, a hyperplane in $\mathbb{R}^2$ is a line, and a hyperplane in $\mathbb{R}^3$ is a 2D plane.

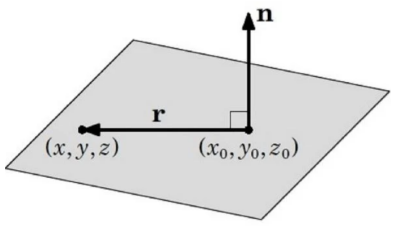
Definition 3

A set of vectors is said to be **linearly independent** if none of the vectors can be expressed as a linear combination of the others.



Definition 4

A **normal vector** to a hyperplane is a non-zero vector that is perpendicular to every vector lying within the hyperplane.



Definition 5

A set of hyperplanes is linearly independent if the set of their normal vectors is linearly independent.

Definitions 3, 4, and 5 are important because of the following fact:

Theorem 1

If a set of hyperplanes in $\mathbb{R}^d$ is linearly independent, then the intersection of any n of these hyperplanes has dimension $d - n$ if $n \leq d$, and is empty if $n > d$.

This is a consequence of the **rank-nullity theorem**, which states that the dimension of the solution set of a system of d independent linear equations in n variables is $d - n$.

Intuitively, $d - 1$ of the n variables can be eliminated by adding and subtracting multiples of the original equations. This leaves a linear equation in $d - n + 1$ variables, which describes a $(d - n)$-dimensional solution set.

$$4w + 7x + 3y + 5z = 12 \tag{1}$$
$$3w + 4x + 5y + 6z = 7 \tag{2}$$
$$6w + 10x + y + 3z = 4 \tag{3}$$

$$\text{Eliminiation: } 3 \cdot (1) - 2 \cdot (2) - 1 \cdot (3) \implies 3x - 2y = 18$$

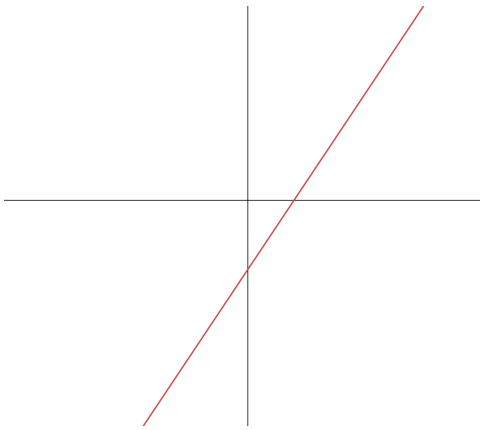


Figure 1: Solution set of 3 independent linear equations in 4 variables is a 1-dimensional line

3.2 Generalization

Now we can generalize our argument to higher dimensions.

Suppose we have a unit sphere centered at the origin in $\mathbb{R}^d$. Choose points $R_1, R_2, ..., R_n$ on the sphere at random.
For $1 \leq i \leq n$, let ℓ_1 be the line passing through R_i and the origin.
Then, let $d_1, d_2, ..., d_n$ be the hyperplanes through the origin perpendicular to $\ell_1, \ell_2, ..., \ell_n$, respectively. This means that $\overrightarrow{R_1}, \overrightarrow{R_2}, ..., \overrightarrow{R_n}$ are the normal vectors to $d_1, d_2, ..., d_n$.

We assume that any subset of size d of $\left\{\overrightarrow{R_1}, \overrightarrow{R_2}, ..., \overrightarrow{R_n}\right\}$ is linearly independent, therefore any subset of size d of $\{d_1, d_2, ..., d_n\}$ is linearly independent. We can do this because

- Since $d_1, d_2, ..., d_n$ all intersect at the origin, any subset of size greater than d of $\{d_1, d_2, ..., d_n\}$ cannot be linearly independent.
- We can make this assumption because the probability that d randomly chosen vectors in $\mathbb{R}^d$ are linearly independent is 1. Proving this rigorously requires measure theory, but intuitively this is true because any d linearly dependent vectors in $\mathbb{R}^d$ must all lie in some subspace of dimension less than d, and the probability that random vectors all lie in some lower-dimensional subspace is 0.

This is essentially a generalization of the assumption that no two points on the circle are diametrically opposite each other, from the original 2D problem.

Let $R(n, d)$ be the number of regions into which $d_1, d_2, ..., d_n$ divide $\mathbb{R}^d$. It turns out that $R(n, d)$ is constant for fixed n and d, given the above assumption. We will see why this is true in the next section.

Each region y can be assigned a $n-$tuple $s = (s_1, s_2, ..., s_n)$, where $s_i \in \{+, -\}$ for all $1 \leq i \leq n$. s_i is $+$ if a unit hemisphere centered at the origin with midpoint in y contains R_i, and $-$ otherwise. Here we define the midpoint of a hemisphere to be the intersection of the line through the center perpendicular to the base with the curved surface of the hemisphere.
Then $P(n, d)$ is equal to the probability that $(+, +, ..., +)$ occurs.
Since there are two choices for each s_i, there are 2^n possible $n-$tuples. Every $n-$tuple is equally likely to occur, and no two regions have the same $n-$tuple, so

$$\boxed{P(n, d) = \frac{R(n, d)}{2^n}}$$

4 Hyperplanes Through the Origin

In this section we will present and verify a general formula for $R(n, d)$.

4.1 Recurrence Relation

Suppose we have n hyperplanes through the origin in $\mathbb{R}^d$, whose normals are linearly independent in sets of d. These hyperplanes split $\mathbb{R}^d$ into $R(n, d)$ regions. Consider removing one of these hyperplanes, call it H_n. Then the number of regions that remain is $R(n-1, d)$. These regions can be split into two sets R_1 and R_2, such that for any region $r_i \in R_1$, $r_i \cap H_n$ is nonempty, and for any region $r_j \in R_2$, $r_j \cap H_n$ is empty.

$$R(n-1, d) = |R_1| + |R_2|$$

H_n splits each region in R_1 two and does not touch any region in R_2, so

$$R(n, d) = 2|R_1| + |R_2|$$

Substituting,

$$R(n, d) = R(n-1, d) + |R_1|$$

The intersection of each of the $n-1$ other hyperplanes with H_n is a hyperplane in the $(d-1)-$dimensional space H_n. Importantly, since we assumed that our original hyperplanes in $\mathbb{R}^d$ are linearly independent in sets of d, each of these intersections is distinct. Therefore, these intersections divide H_n into $R(n-1, d-1)$ regions. These regions are just the intersections of H_n with the regions in R_1, so

$$|R_1| = R(n, d-1)$$

Substituting, we obtain the recurrence relation

$$\boxed{R(n+1, d) = R(n, d) + R(n, d-1)}$$

4.2 Boundary Cases

Note that $R(n, d)$ must satisfy the following boundary cases:

- n lines through the origin divide the $2D$ plane into $2n$ regions, so $R(n, 2) = 2n$.
- Consider the case where $n \leq d$. We will show by induction that $R(n, d) = 2^n$.

  One hyperplane will always divide $\mathbb{R}^d$ in two, so $R(1, d) = 2$ for all d. This will be our base case. For the inductive step, assume that for some positive integer k, $k \leq d \implies R(k, d) = 2^k$ holds for all d. Then for all $d \geq k + 1$,

  $$R(k + 1, d) = R(k, d) + R(k, d - 1) = 2^k + 2^k = 2^{k+1}$$

  Since $k = 1$ satisfies $k \leq d \implies R(k, d) = 2^k$, by induction $n \leq d \implies R(n, d) = 2^n$ holds for all positive integers n.

4.3 The Formula

Consider the formula

$$R(n, d) = 2 \sum_{k=0}^{d-1} \binom{n-1}{k}$$

We can verify that this formula satisfies both boundary conditions:

$$\begin{aligned} R(n, 2) &= 2 \sum_{k=0}^{1} \binom{n-1}{k} \\ &= 2 \left(\binom{n-1}{0} + \binom{n-1}{1} \right) \\ &= 2(1 + (n - 1)) \\ &= 2n \end{aligned}$$

$\binom{n-1}{k} = 0$ if $k \geq n$, so if $n \leq d$,

$$\begin{aligned} R(n, d) &= 2 \sum_{k=0}^{d-1} \binom{n-1}{k} \\ &= 2 \sum_{k=0}^{n-1} \binom{n-1}{k} \\ &= 2 \left(2^{n-1} \right) \\ &= 2^n \end{aligned}$$

Lastly, verifying that this formula satisfies the recurrence relation:

$$\begin{aligned} R(n + 1, d) &= 2 \sum_{k=0}^{d-1} \binom{n}{k} \\ &= 2 \sum_{k=0}^{d-1} \left(\binom{n-1}{k} + \binom{n-1}{k-1} \right) \text{ (Pascal's identity)} \\ &= 2 \sum_{k=0}^{d-1} \binom{n-1}{k} + 2 \sum_{k=0}^{d-1} \binom{n-1}{k-1} \\ &= 2 \sum_{k=0}^{d-1} \binom{n-1}{k} + 2 \sum_{k=0}^{d-2} \binom{n-1}{k} \\ &= R(n, d) + R(n, d - 1) \end{aligned}$$

Theorem 2

n hyperplanes in $\mathbb{R}^d$ that each pass through the origin and are linearly independent in sets of d divide $\mathbb{R}^d$ into

$$2\sum_{k=0}^{d-1}\binom{n-1}{k}$$

disjoint regions.

This result is attributed to the Swiss mathematician Ludwig Schläfli.

5 Wendel's Theorem

Now that we have a formula for $R(n,d)$, we finally have the general solution

$$P(n,d) = \frac{R(n,d)}{2^n} = 2^{-n+1}\sum_{k=0}^{d-1}\binom{n-1}{k}$$

Theorem 3: Wendel's Theorem

The probability that n randomly chosen points on a sphere in $\mathbb{R}^d$ all lie on some hemisphere is

$$2^{-n+1}\sum_{k=0}^{d-1}\binom{n-1}{k}$$

This result was proven by J.G.Wendel in 1962.

6 Coin Tossing

Let's say you flip a fair coin $n-1$ times. What is the probability that you get less than d heads? Well, the probability of getting k heads in $n-1$ flips is

$$2^{-n+1}\binom{n-1}{k}$$

So the probability of getting less than d heads is

$$2^{-n+1}\sum_{k=0}^{d-1}\binom{n-1}{k}$$

Notice that this is exactly the same formula as in Wendel's Theorem!

In Wendel's original paper, he notes this connection, but remarks "it does not seem possible to find an isomorphism between coin-tossing and the given problem that would make the result immediate" [1].

While a direct mapping is elusive, we can draw parallels between coin tossing and the points-on-a-sphere problem.

First, we can convert the recurrence relation and boundary conditions on $R(n,d)$ into statements about $P(n,d)$.

Recurrence Relation:

$$R(n+1,d) = R(n,d) + R(n,d-1)$$

$$2^{n+1}P(n+1,d) = 2^nP(n,d) + 2^nP(n,d-1)$$

$$\boxed{P(n+1,d) = \frac{1}{2}P(n,d) + \frac{1}{2}P(n,d-1)}$$

Boundary Condition 1 ($d = 2$)

$$R(n,2) = 2n$$

$$2^nP(n,2) = 2n$$

$$\boxed{P(n,2) = \frac{n}{2^{n-1}}}$$

Boundary Condition 2 ($n \leq d$)

$$n \leq d \implies R(n,d) = 2^n$$

$$n \leq d \implies 2^nP(n,d) = 2^n$$

$$\boxed{n \leq d \implies P(n,d) = 1}$$

Let $C(n,d)$ be the probability of getting less than d heads in $n-1$ flips.

Suppose a coin is flipped n times and comes up heads less than d times. Then one of the following statements must be true:

- The coin came up heads less than d times in the first $n-1$ flips, then came up tails on the last flip.
- The coin came up heads less than $d-1$ times in the first $n-1$ flips, then came up heads on the last flip.

The last flip is equally likely to be heads or tails, so

$$C(n+1,d) = \frac{1}{2}C(n,d) + \frac{1}{2}C(n,d-1)$$

The probability of flipping zero heads in $n-1$ flips is $\frac{1}{2^{n-1}}$, and the probability of flipping heads exactly once is $\binom{n-1}{1} \cdot \frac{1}{2^{n-1}} = \frac{n-1}{2^{n-1}}$. Therefore, the probability of getting less than 2 heads in $n-1$ flips is $\frac{n}{2^{n-1}}$, so

$$C(n,2) = \frac{n}{2^{n-1}}$$

If $n \leq d$, then it is not possible to get more than $d-1$ heads in $n-1$ flips, so

$$n \leq d \implies C(n,d) = 1$$

Therefore, the boundary conditions and recurrence relation are the same, so

$$\boxed{P(n,d) = C(n,d)}$$

7 Conclusion

Wendel's Theorem is a fundamental result in geometric probability. While its statement might not seem practical, generalized versions of the theorem have surprising applications in various fields.

For example, forces on a particle can be modeled as points on a sphere, so Wendel's theorem can be applied to help calculate the local stability of particles.

Another example is in linear programming. Random linear programs are powerful optimization tools that take random vectors as parameters, and a generalized version of Wendel's Theorem can be used to calculate the probability that a random linear program is bounded.

Such applications commonly use the fact that if a set of points on a sphere do not all lie on a hemisphere, then their convex hull (the smallest possible convex shape that contains all of the points) contains the origin.

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Knight's Tour

Shanya Tissera

1. Knight's Tour

The knight is an unusual chess piece. It moves in an "L" shape, once in one direction and twice in a perpendicular direction:

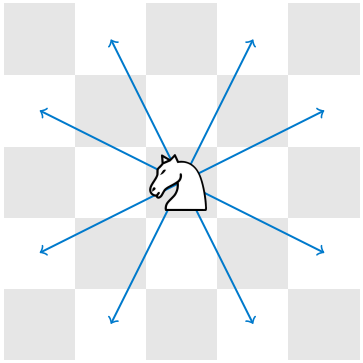


Figure 1: A knight and its possible moves.

The **Knight's Tour** is a sequence of moves where a knight visits each square of a chessboard exactly once. To analyze it, we can model the chessboard as a **graph**, $G = (V, E)$. In this graph, the **vertices** V represent the set of $m \times n$ squares on the board, and an **edge** E connects two vertices if and only if a knight can move between them.

In graph theory, we have a few terms that can help us talk about this tour: A **walk** is a series of vertices where each vertex is connected to the next by an edge. A knight travelling across a chessboard is a walk. A **path** is a walk that never visits the same vertex twice. The Knight's tour is a special kind of path: it visits every vertex in V exactly once. In graph theory, this is known as a **Hamiltonian path.** A "closed" tour, where a knight can return to its starting square, is a **Hamiltonian cycle.** All other tours are "open."

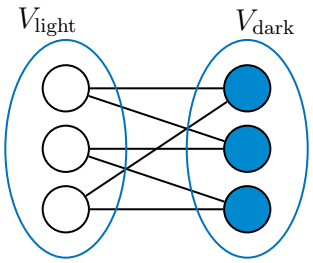


Figure 2: A bipartite graph with disjoint vertex sets V_{light} and V_{dark}.

Because the knight always moves from a light square to a dark square (or vise versa), our graph G is **bipartite**: We can partition the vertices into two disjoint sets V_{light} and V_{dark}, where every edge in E connects a vertex in V_{light} to one in V_{dark}, but no vertex connects to one in the same set.

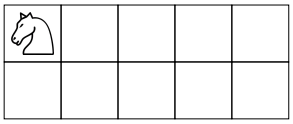
2. Restrictions

2.1. Disconnection

The Classical Knight's Tour considers an $m \times n$ chessboard. We can see very quickly that a Knight's tour cannot exist on a $1 \times n$ or $2 \times n$ chessboard: the Knight cannot possibly move enough to return to its starting square.



1×5 board, Knight just can't move



2×5 board, Knight can only backtrack

Figure 3: A Knight's Tour is impossible on these boards.

On a $1 \times n$ board, the graph G has n vertices but zero edges! Since the knight cannot move from one vertex to another, no path exists, and no tour is possible. (Except for the trivial 1×1 case.) On a $2 \times n$ board, the graph is **disconnected**! A knight on $(1, 1)$ can only reach squares in one **component** of the graph, while a knight starting on $(2, 2)$ can only travel the other component. Since the graph is disconnected, no path exists between all vertices, no Knight's Tour is possible.

2.2. Bipartite Coloring

Recall that the chessboard is a bipartite graph: The knight also alternates black, white, black...

Theorem 1

A closed knight's tour can only exist on a board with an even number of squares.

Proof. For a Knight's Tour to be closed, it must return to its starting square, meaning there must be an equal number of light and dark squares. When the dimensions of the chessboard are both odd, this is not possible, and therefore a closed knight's tour is impossible on any $m \times n$ board where m and n are odd. □

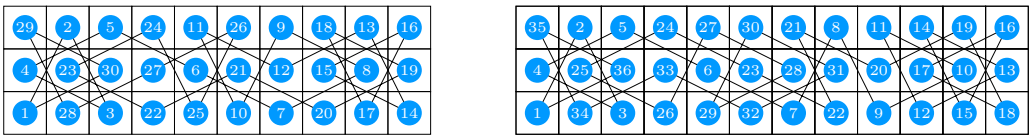


Figure 4: Euler thought closed tours were impossible for boards with sides < 5, but Ernest Bergholt found 3×10 and 3×12 tours in 1917.

2.3. Open Tours

Well, what about paths? An open tour on an odd board still needs to alternate colors, meaning an open tour is possible only if it starts and ends on the majority color.

If we consider a 3×3 board, we can see that there is no way for the knight to get to the middle square, and there is no way for the knight, once starting in the middle, to get to any other square. So, the square in the middle is disconnected from the graph, and no Hamiltonian path exists. This rules out $3 \times \{1, 2, 3\}$, but we can see that a Hamiltonian path is possible on a 3×4 grid!

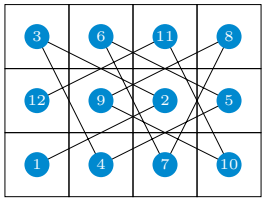


Figure 5: A Knight's Tour on a 3×4 chessboard.

A 3×4 grid has 12 squares: an even number. Our reasoning fails to rule out a closed tour, but a closed tour is still impossible. While bipartite coloring is a necessary condition for a closed tour, it is not sufficient to show that one exists.

2.4. 4 × n

Theorem 2

No closed Knight's Tour exists on a $4 \times n$ board.

Proof. We can justify this with a coloring argument. We can color the squares as follows, with A, B alternating on the top and bottom, and C, D alternating in the middle:

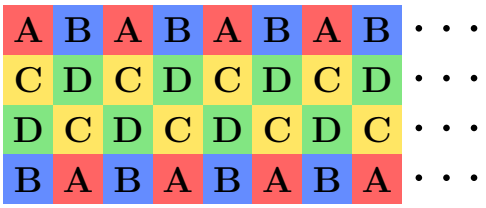


Figure 6: Each row has one of A, B, C, D.

There are an equal number of squares labeled A, B, C, D as this is true for each row. For a closed Knight's Tour to exist, we don't need to consider any specific starting point, since the knight goes to and comes from every square, so if it fails to go through one and come back to it, no Knight's Tour exists.

Thus, we assume the knight starts at square A. From here, it must go to a square C. However, it must have also *come* from a square C. This means that when it jumps from a C to a D cell, it *must* jump back to a C cell to get to another A cell. An example sequence may look like:

$$A \to C \to D \to \boxed{\text{C}} \to A \to C...$$

There are an equal number of A and C cells in the grid, and to form a closed Knight's Tour we must jump to all of them. However, this implies that we must travel to all of the A and C cells before any of the B and D cells, because otherwise we would count more C cells than A cells.

Then, we must finish the tour by traveling from D to B, back and forth. Since a B only leads to a D, this final D must connect back to an A. But a D cannot connect back to an A without travelling through a C. Thus, the tour can never be closed, so, no knight's tour exists on a $4 \times n$ board! □

2.5. Schwenk's Theorem

Schwenk's Theorem (1991)

For an $m \times n$ chessboard (with $m \leq n$), a closed Knight's Tour exists if and only if all of the following conditons are false:

- m and n are both odd.
- $m = \{1, 2, 4\}$.
- $m = 3$ and $n = \{4, 6, 8\}$.

Proof. We have already proven that a Knight's tour cannot exist if (a) or (b) are met. We

have also proven that a closed Knight's tour cannot exist on a 3×4 board.

An important feature of a graph with a Hamiltonian cycle is that removing k vertices from G can leave at most k connected components

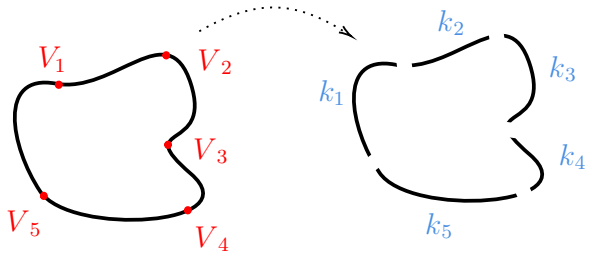


Figure 7: Removing $V_1 ... V_5$ leaves $k_1 ... k_5$ as parts.

Consider the board 3×6. Let's remove the vertices $(3, 1)$ and $(3, 3)$. If G contains a Hamiltonian cycle, the board should split into at most 2 components. However, we can see that the square $(1, 2)$ is disconnected, creating one component. Likewise $(5, 2)$ is disconnected. The other 14 squares remain connected to each other. Clearly, the board has split into 3 components, and thus a Hamiltonian cycle cannot exist!

We know now that a Knight's tour is impossible on the boards that satisfy the conditions of Schwenk's theorem. Additionally, we can construct every other board to prove they have Knight's tours! The full proof for this is too long to discuss here, but the construction is fairly similar for each case:

$(1,3)$	$(2,3)$	$(3,3)$	$(4,3)$	$(5,3)$	$(6,3)$
$(1,2)$	$(2,2)$	$(3,2)$	$(4,2)$	$(5,2)$	$(6,2)$
	$(2,1)$	$(3,1)$	$(4,1)$	$(5,1)$	$(6,1)$

$(1,3)$	$(2,3)$		$(4,3)$	$(5,3)$	$(6,3)$
$(1,2)$	$(2,2)$	$(3,2)$	$(4,2)$	$(5,2)$	$(6,2)$
$(1,1)$	$(2,1)$		$(4,1)$	$(5,1)$	$(6,1)$

Figure 8: A chessboard with two squares removed.

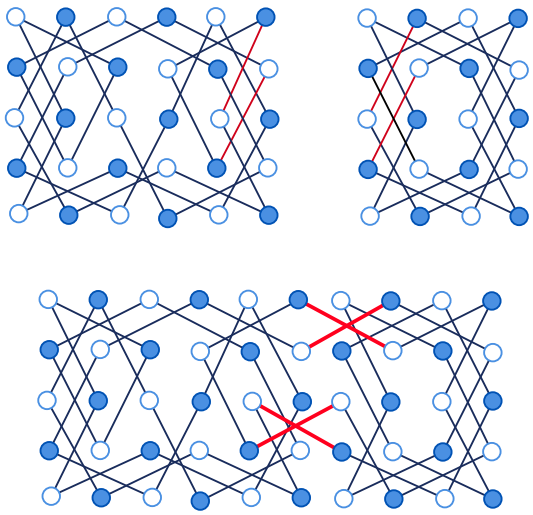


Figure 9: Two edges are cut, and then stitched together to create a larger Knight's Tour.

First, take a known tour, say, the 5×6 tour. Then, attach on a 5×4 block of squares. We can "cut" two specific edges on the 5×6 tour, right at the border, and do the same for the 5×4 block. We can then stitch them together to create a 5×10 tour. Note that the same process can be applied to the 5×10 board, and so on!

Note that the patterning on the completed board resembles the ending pattern of the original boards, meaning we can continuously stitch these boards together. The same argument can be applied to columns. Closed knight's tours can be constructed for the following pairs (m, n) although we will not explicitly show it here:

$$(3, 10), (3, 12), (5, 6), (5, 8), (6, 6), (6, 8), (7, 6), (7, 8), (8, 8)$$

From each of these pairs, by the stitching argument we described earlier, we can add 4 to any coordinate in the pairs above any number of times we want. By induction with these base cases, we get every pair that satisfies the conditions of Schwenk's Theorem, so closed Knight's tours are possible are on all of these chessboards. □

3. Variants

3.1. Holey Boards!

What if the chessboard isn't empty? In our proof that a knight's tour is impossible on a 3×6 board, we considered removing squares of the chessboard. This is the same as removing a vertex from a graph. If we place a knight on $(3, 3)$, we are no longer looking for a Hamiltonian cycle on G but on G', where the vertex $(3, 3)$ and all of its connected edges have been deleted.

While this can make tours impossible, it can make new tours possible! The smallest Knight's Tour on a 3×3 board only exists with 2 cells removed, while the smallest closed tour exists on a 3×3 board with just the center square removed.

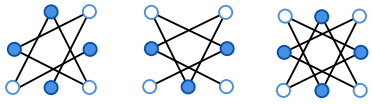


Figure 10: Two open Knight's Tours and one closed Knight's Tour.

Finding the second formation, the U-shape, appeared in one of the earliest Arabian chess manuscripts as a puzzle: the trick is to start a Knight's jump away from the empty non-central square!

3.2. ∞ Board

On an infinite board, there are no corners or edges to restrict the knight. Obviously, thinking about a closed knight's tour on an infinite $\mathbb{Z} \times \mathbb{Z}$ board is a little silly, but we can ask two other questions:

(a) Can our knight, starting at $(0, 0)$, reach any square (m, n)?

(b) Can we find a path that reaches every square exactly once?

For (a), we only need to prove that the Knight can move to $(0, 1)$ and $(1, 0)$. In other words, if we can find a sequence of moves that leaves us with a net $+1$ to the x and y coordinates of the knight, we can apply this sequence repeatedly to reach (m, n). We can easily find this sequence, after a few moves:

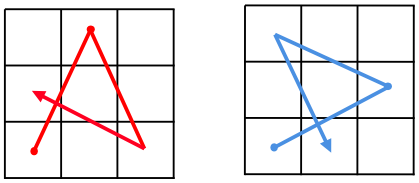


Figure 11: The path a Knight walks to reach $(0, 1)$ and $(1, 0)$.

...and so we can always reach (m, n). (We can do a similar procedure to achieve every negative ordered pair as well).

For (b), we can construct a tour that extends infinitely:

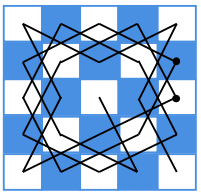


Figure 12: Infinite Knight's Tour!

This knight's tour, when repeated, will reach every square exactly once.

3.3. (a, b) Knight

Instead of changing the board, what if we change the *knight*? We can call our current knight a $(1, 2)$ knight, meaning it moves 1 step in one direction, and twice in a perpendicular direction. We can generalize this to an (a, b) knight, moving a in one direction, and b in a perpendicular direction. One question we can ask is: which (a, b) knights can reach every square on an infinite board?

The set of all possible positions a knight can reach forms a subgroup P of the lattice $\mathbb{Z} \times \mathbb{Z}$. This subgroup can be generated by 8 move vectors, representing the 8 moves of the knight:

$$P = \langle (\pm a, \pm b), (\pm b, \pm a) \rangle.$$

Can the knight reach every square (m, n)? is equivalent to asking: does $P = \mathbb{Z} \times \mathbb{Z}$? Similar to before, we can prove this by showing that the knight can make a net move of $(1, 0)$ and $(0, 1)$. It can then reach any point (m, n) by combining this sequence of moves.

First, $\gcd(a, b)$ must be 1 in order for the knight to reach every square. In other words, a and b must be coprime. Suppose otherwise. If $\gcd(a, b) = d > 1$, every coordinate a knight can reach from $(0, 0)$ is a multiple of d, which means not every coordinate can be reached.

There is another restriction on the value of a and b. If we color an infinite board in a checkerboard pattern, we need the knight to switch the color it's on every move. However, if $a+b$ is even, then the knight will continue to be on the same color. Therefore, it is necessary that a and b have different parity, or in other words, $a \not\equiv b \pmod 2$. In fact, this, along with a and b being coprime is sufficient for the (a, b) Knight to be able to traverse an infinite board.

To show that this condition is sufficient, suppose that $\gcd(a, b) = 1$ and a is odd while b is even. By Bezout's Lemma or the Extended Euclidean Algorithm, we can find integers c and d such that $ac + bd = 1$. In fact, we claim we can find a solution where c is odd and d is even. c is odd no matter what since if we reduce this equation mod 2, we see $ac \equiv 1 \pmod 2$, and a being odd implies c is odd. On the other hand, if d is odd, we can construct the new solution $a(c - b) + b(d + a)$. Here, $c - b$ is still odd, but since a and d are both odd, $d + a$ is even. Thus, we can just assume c is odd and d is even. The idea now is to construct a sequence of moves that will take the Knight to the square $(1, 0)$ based on the coefficients c and d. c will represent the net number of a-moves in the x-direction (# moves a to the right - # moves a to the left) while d will represent the net number of b-moves in the x-direction. If we want the net number of a-moves in the x-direction to be c, we can take the total number of a-moves to be any odd integer greater than or equal $|c|$; suppose we set the total to some large odd integer N. We have to choose the y-direction of each these moves to either be $+b$ or $-b$. We will choose a of them to be $-b$ and the rest (which come in pairs) to have an equal number of $+b$'s and $-b$'s. This makes the net movement in the y-direction due to these moves $-ab$.

Similarly, we can set the net number of b-moves in the x-direction to be any even integer greater than or equal to $|d|$, say M. We have to choose the y-movement for each of these moves to be either $+a$ or $-a$. We will choose b of them to be $+a$ and the rest to come in pairs of $+a$ and $-a$. Thus, overall, these moves contribute a total movement in the y-direction of

$+ab$.

Putting both of these types of moves together–ones where we move a in the x-direction and ones where we move b in the x-direction–we have a net movement in the x-direction of $ac + bd = 1$ and a total net movement in the y-direction of $-ab + ab = 0$. Thus, the Knight can take a series of moves to get to $(1, 0)$.

To summarize, since we can always construct a sequence of moves that results in a net move to $(1, 0)$ (and by symmetry, $(0, 1)$), the (a, b) knight can reach every square in $\mathbb{Z} \times \mathbb{Z}$ iff $\gcd(a, b) = 1$ and $a \not\equiv b \pmod 2$.

□

4. Boards all Glued Together

4.1. Cylinders and Tori

Imagine now that we take a chessboard and glue its sides together.

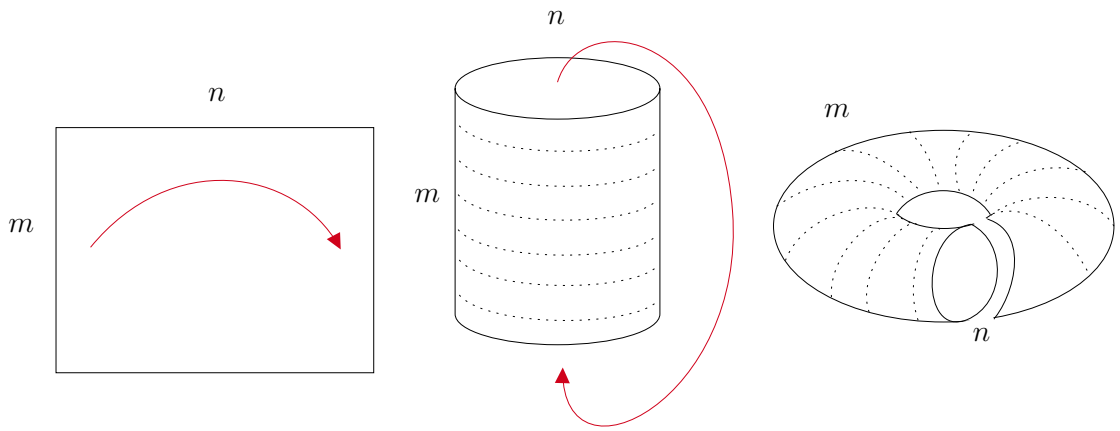


Figure 13: Construction of mxn cylinder and mxn torus from mxn rectangle chessboard.

We can see that an $m \times n$ rectangle and torus are identical to those that are $n \times m$. However, there is a distinction between an $m \times n$ cylinder and a $n \times m$ cylinder. If we take the left edge and "glue" it to the right edge, as in the diagram, we create an m-high cylinder with circumference n. Note that the "row" moves of the knight are the same as when the cylinder was unwrapped, but the "column" moves are repeated mod n.

Consider a knight on a rectangular board, on the far right edge, at square $(a, n-1)$. (Here, we are indexing at 0). The knight wants to move to $(a + 1, n + 1)$, but this is impossible due to the constraints of the board. However, if we wrap the rectangle to be a cylinder, the knight can wrap around the edge of the cylinder and end up at $(a + 1, 1)$. In general, when moving, we take the Knight's y-coordinate mod n. For an $n \times m$ cylinder, imagine rolling and sticking the top and bottom boundaries of the chessboard together. In this case, the column moves are unchanged, but the row moves wrap around.

For the $m \times n$ torus, we can imagine taking our cylinder and having *its* edges wrap

around. In this way, the torus chessboard has no boundary, and the knight can wrap around forever. We can take the new row mod m and the new column mod n to pinpoint a location. The graph G representing the torus (for every $m, n > 4$) is an 8-regular graph. A **regular** graph is one where each vertex has the same degree. This removes the edge restriction, and makes Knight's Tours easier to find.

4.2. Constructing Algorithms

Cylinder Construction

Consider an $m \times n$ cylinder (height m, circumference n). If there exists some Knight's Path of length $m+1$ that starts and ends at the bottom row, visiting every other row exactly once and has a net displacement of $(1, 0)$, then a closed Knight's Tour exists on the cylinder.

Proof. In essence, if we can start on the bottom row and can construct a Knight's Path that visits every row exactly once before returning to the bottom row, we can repeat this algorithm n times to cover every square of every row! Since the knight ends with a net movement $(1, 0)$, the row will be shifted exactly once over, so the knight will travel to every square. Also, since the knight ends by moving to square $(n+1, 0)$, it returns to its starting square, completing a closed Knight's Tour. □

4.3. Proving Existence

Let's try our coloring argument again. For a closed tour on a bipartite graph, the number of dark squares must equal the number of light squares. If mn is odd, $V_{light} \neq V_{dark}$. So, we may be tempted to say that no knight's tour exists on a torus where mn is odd, but this is false! Consider a square of a chessboard $(m-1, c)$, with m and c both odd. Let a square (a, b) with $a+b$ odd be dark and a square with $a+b$ even be light. Since the net movement of this knight is odd, the knight moves from odd to even net coordinates on its bipartite graph (and vice versa), thus moving from dark to light squares. A possible move for the knight is $(m+1, c+1)$. This seems to obey our principle: $m+1$ is even and $c+1$ is even, so the knight has moved from a net-odd square to a net-even square. However, as row $m-1$ is on the edge of the rectangle, $m+1$ is row 1. Thus the knight is on $(1, c+1)$, which is a net-odd/dark square. Because the knight can move from a dark square to another dark square, this checkerboard pattern isn't valid doesn't make the chessboard bipartite.

A similar argument can be used for the $m \times n$ cylinder. Consider square (m, c). Here, rows repeat mod m. The knight can move to $(1, c+1)$, creating a net move from an odd square to an odd square.

□

So, what restrictions *do* exist on bent boards?

Theorem 3

For any cylinder not of size $4 \times n$, $2 \times n$, or $1 \times n$, there exists a closed knight's tour.

Proof. We can prove the existence of a knight's tour on a $3 \times n$ board by construction, as demonstrated by Kelley Seibel in *The Knight's Tour on the Cylinder and Torus.*

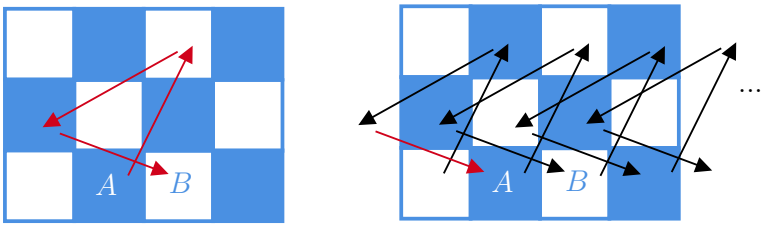


Figure 14: An algorithm for a closed Knight's Tour on a $3 \times n$ cylinder.

We start from A and we manuver to B in the above manner. In this way, we reach a square of each row exactly once per iteration, and we end at the square one to the right of A. If we continue this pattern around the cylinder, we loop around and end a knight's jump away from A, completing the Knight's tour. This closed Knight's Tour is constructed indistinguishably no matter the parity of the board, so there is no parity restriction. □

For boards $5 \times n$ and larger, (n here is ≥ 5), we know that a closed tour always exists on a cylinder where at least one of m, n are even, because they exist on an $m \times n$ board. So, we need only consider the case where both are odd. We can construct an algorithm by starting at a square and repeating the following sequence:

$$(+1, +2), (-1, +2)$$

until we reach the top of the board, at move $\frac{m-1}{2}$, because m is odd. From here, we can move $(-2, -1)$ and repeat the same sequence moving down, hitting the rows we missed previously. This sequence will always end up a knight's jump away from a square one unit to the right of our starting square, as the net move will be $(+1, 0)$.

To see this, we can consider when m is $1 \mod 4$ and when m is $3 \mod 4$. When m is $1 \mod 4$, the net move when going up the board is $(m, 0)$, because the arrows reach the top square, and return to the "color" of the starting square. (We can see this by observing $m = 5$, and seeing that the sequence repeats mod 4.) When going down, we will move net $(-m, 1)$. We can see that, in the x, the final arrow does not cancel, creating a net movement right. So, our total move vector is $(0, 1)$.

We can see a similar thing for when $m \equiv 3 \mod 4$. In this case, the net x move occurs when traveling up, creating a net $(m, 1)$ move. Then, when it goes down, it creates a $(-m, 0)$ move, totaling a net $(1, 0)$ move.

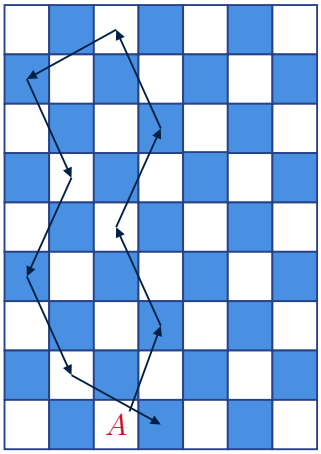


Figure 15: An example of the above algorithm on a 9×7 cylinder.

□

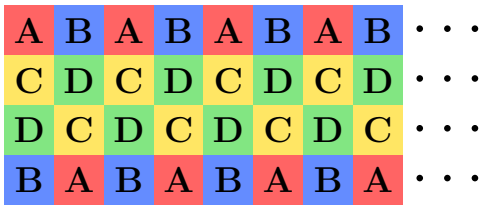
Theorem 4

No closed knight's tour exists on any $1 \times n$ cylinder, or $2 \times n$, $4 \times n$ cylinders with n even.

Proof. As with the $1 \times n$ board, the knight has no squares to travel to, so (besides the trivial 1×1 cylinder), there can exist no closed Knight's Tour on a $1 \times n$ cylinder.

For the $2 \times n$ board, with n even, a knight can move $(\pm 2, \pm 1)$ but not $(\pm 1, \pm 2)$. This means that the parity of the knight's x coordinate is unchanged after a knight's move. So, if the knight starts on an even numbered column, it can only reach other even numbered columns. When a knight reaches square $x = n + 1$ it goes to square $x = 1$, preserving parity under wrapping. Thus, the knight can only reach squares with columns numbered of its starting parity, and can never complete a full knight's tour.

Lastly, for the $m \times n$ cylinder, we can use an identical argument as we used for the board. To bring back our diagram from before:



We can split the chessboard into sections A, B, C, D, with A, B having degree 4 and opposite column parity, and C, D having degree 6 and opposite column parity. Each group A, B, C, D is equal in size, as there is one of each per row. Assume we start at square A. From here, even with wrapping, we must go to a square C, and we must have come from a square C. Thus, there are two $C's$ for one A. In order for the amount of C and A to be equal, we must then go to every C and every A before any B or D. But then, once we get

to B, D, there is no way to get back to A or C to complete the tour, as it would disrupt the balance of A and C. Thus, no closed knight's tour can exist on a $4 \times n$ cylinder, where n is even. □

For the $2 \times n$ cylinder, with n odd, we can construct a tour by considering the cylinder in 4 "parts". (Here, I quote "parts" because they really are all connected). We can start on the top left corner of the unwrapped cylinder, and number each column, indexed at 0. We can have our knight continuously move $(+2, \pm 1)$, starting with -1 and alternating each turn. Consider the net-xy move k, l. When k exceeds n the first time, k is set to 1 and our column parity is switched from even to odd. We can then continue to apply $(+2, \pm 1)$ until we reach $k = n$, and when this happens, k is set to 0.

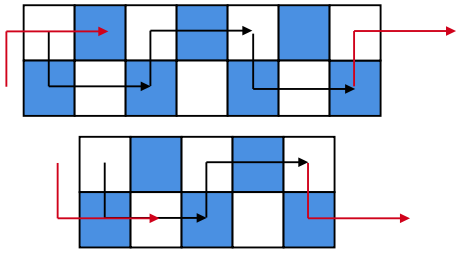


Figure 16: Knight's Tour on 2×7 and 2×5 cylinders.

When $n + 1 \equiv 2 \mod 4$, the row parity will switch during the first iteration, and will reach -1, not 0, as its net movement, before wrapping. There are say, j moves before the first wrapping, and there will be j moves before the second wrapping, so there will be $j + 1$ to complete the second wrapping. Because of this, the knight will have a net -1 row move by the end of the second iteration, and the knight will be on the second row, on an odd numbered column. It will continue to move forward, in the pattern $(+2, \pm 1)$, filling each square it hadn't before.

When $n + 1 \equiv 0 \mod 4$, this does NOT switch the row parity. However, before the end of the second wrapping, there will be exactly one column of space, meaning that for the next two iterations, the Knight will be on the 2nd row.

We move a total of $2n$ moves, and because there are an even number of $+1$s and -1s, we have a net 0 row move. Also, we shift our x-coordinate by $4n$, which is 0 mod n. Thus, after we finish this procedure, we return to our starting square and complete the closed Knight's tour.

For the $4 \times n$ algorithm, we can make use of the fact that the column parity changes when wrapping. We can consider the following algorithm:

... | 3 | | | |
| | | 2 | |
| | 4 | | |
| | 1 | | 5 | ...

The algorithm is successful, as it reaches each row exactly once before reaching the

bottom row. However, it is 2 squares away instead of exactly one. However, if column parity switches when wrapping, we can first hit every odd column, then the even ones, and, after the second wrapping, we finish the circuit on the odd square we started on.

Theorem 5

A closed Knight's tour exists on *every* torus.

Proof. Note that we only need to prove the cases $1 \times n$ for all n and $2 \times n, 4 \times n$, with n even, as the cylinder covers all other cases. (The torus is just a cylinder with more options, as a cylinder is a board with more options).

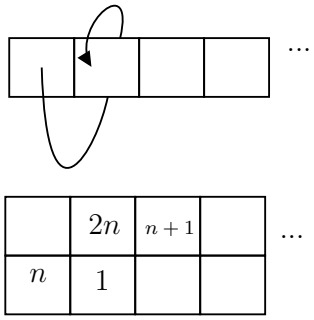
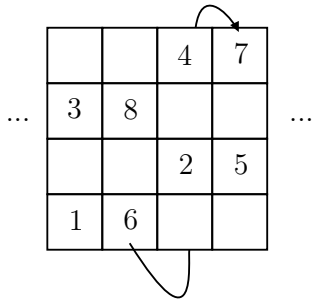


Figure 17: A Knight's Tour on $1 \times n$ and $2 \times n$ tori.

For the $1 \times n$ case, we can move once horizontally and twice wrap vertically, ending on the square directly to the right of the starting square. From here, we can apply this for the whole length of the torus, and we end right back at the starting square.

For the $2 \times n$ case, we can use the result from this. We can work our way around the first row, then jump to the second row and do the same. And since the ending square of the first row is both once to the left of the starting square and a knight's jump away from the starting squrae of the second row, the knight can always jump back to the starting square, completing the knight's tour.

For the $4 \times n$ case, we can construct the following algorithm:



Since the number of columns is even, no iteration will be incomplete. When $n \equiv 2 \mod$

4, the knight simply wraps during the iteration! The ending jump is a knights jump from the starting move, so we have a closed knight's tour. □

5. Conclusion

The Knight's Tour is an interesting puzzle that has fascinated mathematicians for thousands of years. Because of the Knight's unique style of movement, it's nice to model it on a graph and observe its properties. The Knight's Tour is also pretty huge in computer science, with algorithms existing to find these tours on various boards. For further reading, check out George Jelliss's *Knight's Tour Notes*; it has a lot of cool history and variations of the Knight's Tour!

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The Banach–Tarski Paradox

Maliha Aktar

1 Introduction

The Banach–Tarski paradox is a famous dilemma in modern mathematics. It asserts that a solid sphere or any other object that exists in a three-dimensional space can theoretically be divided so that these pieces after being moved around by rotations and translations alone, can be reassembled into *two* spheres identical to the original one.

What makes this difficult to believe is that there is no distortion involved. This seemingly violates not only geometry, but common sense itself. How can an object become two objects of equal size without creating any additional material?



Figure 1: A schematic illustration of the Banach–Tarski paradox. A solid sphere is decomposed into a finite amount of disjoint pieces. Using only rotations and translations it can be reassembled into two spheres congruent to the original.

1.1 Volume and Additivity

1.1.1 Volume Preservation

We consider volume to be the 'space' that an object occupies in the third dimension. It is comparatively similar to 'area' in two-dimensional spaces. An object's volume does not necessarily need to be conserved, evident by the properties of densities, where the same mass can be compacted in different volume amounts.

However, in real world situations, volume behaves predictably. For example, if a log of wood is cut into two pieces, its volume will not change. This is because for volume to change in this regard, it requires distortion through dilation or reduction. If we were to consider an object in a two dimensional space, such as the surface of a pizza, we can cut it as many times as we would like, but we would still have the same total area if we were to reassemble the pieces.

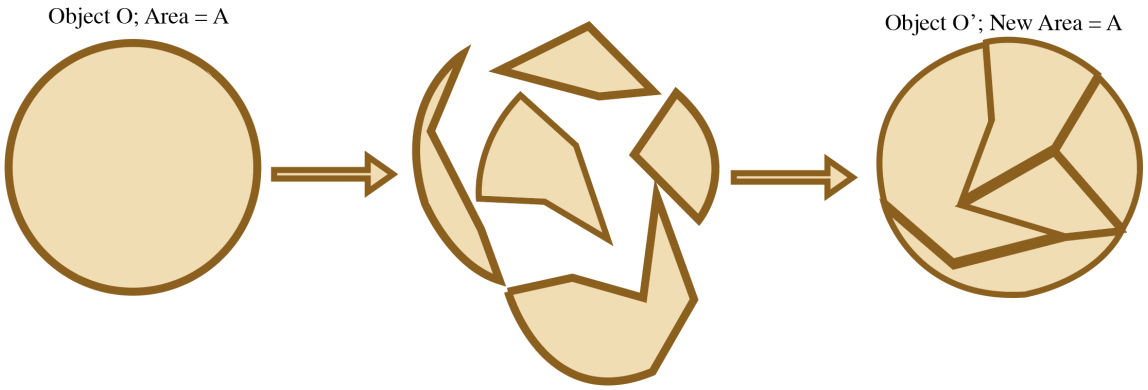


Figure 2: Cutting and reassembling a measurable object (like this 2D shape) preserves its total area. Our physical intuition about volume is rooted in the behavior of such well-behaved sets

We can distort the size of one of these pieces, but then the area of the pizza's surface would no longer be conserved. It does not seem feasible to have two objects with same density *and* volume simply through rotations and translations of pieces from one object.

1.1.2 Defining Volume

We must clarify what exactly we define as 'volume'. The volumes of shapes from Euclidean geometry (balls, cubes, polyhedra, etc.) satisfy the property of *finite additivity*: if two objects are not overlapping, the volume of their union is the sum of their individual volumes.

However, this property assumes *every* subset of three-dimensional space has a well-defined volume. This assumption can be made in regular geometry because typically the only sets we encounter are orderly ones that have regions with clean boundaries. But once we consider arbitrary sets, we can no longer guarantee this assumption. Some sets are so irregular that no consistent notion of volume can be assigned to them at all; we define these sets as *nonmeasurable*.

1.2 Overview

Therefore, the Banach–Tarski paradox does not necessarily have to be a direct contradiction of physics' principle of mass conservation. Rather, it contradicts our intuition that every conceivable set must behave like a physical object. If we assume the Axiom of Choice, then it is possible for three-dimensional space to contain subsets that defy regular geometry.

In the following section, we discuss in more detail the distinction between measurable and nonmeasurable sets. In Section 3 we introduce the Axiom of Choice and its significance in creating non-measurable sets. We will also look at the Vitali Set as a simpler, one-dimensional analogue of the paradox. Section 4 will briefly present the Banach–Tarski theorem itself, and Section 5 will go over concluding remarks and discussion.

2 Measurable and Non-Measurable Sets

2.1 Definition of Measure

When we discuss the notion of *measure*, we mean it as a way to assign a consistent notion "size" to sets. This can come in the form of length, area, or volume.

A measure can be thought of as a function that assigns a nonnegative number to each set in order to represent its "size". For example, on the real number line, the interval $[0, 1]$ has length 1, while $[0, 2]$ has length 2. This is an example of 'measure' for real numbers. However, we would like our definition of measure to behave reasonably for all subsets of $\mathbb{R}$ (as well as higher-dimensional spaces).

For our definition of measure to be useful, a measure should fulfill these particular properties:

- **Non-negativity:** The measure of any set is nonnegative.
- **Countable additivity:** If a set can be expressed as a countable union of disjoint subsets, its measure is equal to the sum of their measures.
- **Translation Invariance** Shifting a set does not change its measure. That is, if A is measurable and t is any real number, then $\mu(A + t) = \mu(A)$. In some cases, this is not considered necessary for a measure, although we will use it for our purposes. The notion of translation may be different for different spaces, such as 1-dimensional or 2-dimensional space.

These properties reflect our physical intuition; cutting or rearranging doesn't change an object's total volume. This is significant because the consistency with physical logic makes the notion of measure meaningful even in higher-dimensional spaces which are difficult to imagine.

2.2 The Lebesgue Measure

The Lebesgue measure was developed in order to handle the measure of more complicated sets. It is mainly based in approximation: for any set $A \subseteq \mathbb{R}$, we can try to cover the set with a collection of intervals whose total length will get as close as possible to the "size" of A.

The smallest total length of a collection of coverings is called the *outer measure* of A, which we define as $\mu^*(A)$. $\mu^*(A)$ represents the least amount of total length needed to completely cover A with disjoint @intervals, regardless of how irregular the set is.

However, not every set can be assigned a measure consistently using only outer measure. Some sets interact with other subsets in ways that make their size ambiguous. Henri Lebesgue addressed this by introducing *Carathéodory's condition* which determines which sets can be considered measurable. A set A is said to be *Lebesgue measurable* if, for every set $E \subseteq \mathbb{R}$,

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A).$$

This condition ensures that A divides any other set E without creating contradictions in how we compute total measure. It basically ensures that if we cut any set by A, we are not interfering with the measurements of the other sets.

The Lebesgue measure, which we denote as μ, is then the outer measure which is restricted to all measurable sets. For intervals, it is aligned with our typical notion of length.

$$\mu([a, b]) = b - a.$$

This definition also extends into more complex sets, such as countable unions of intervals and most regions that one could encounter in analysis and geometry.

The Lebesgue Measure's generality allows us to talk about size across a large range of sets while still preserving the properties that define measurable sets. However, there are some subsets of $\mathbb{R}$ in which no measure satisfying these properties can consistently assign a size. These are also known as *nonmeasurable sets*.

2.3 Conceptual Limits of Measure

The Lebesgue measure seems to be an appropriate and flexible method to assign size to subsets of $\mathbb{R}$. Therefore, it is natural to wonder if the Lebesgue measure can be applied to *every* subset of $\mathbb{R}$.

However, it is not possible to do this. This is not a result of the Lebesgue measure failing. Rather, it is because no function can consistently assign a notion of size to all subsets of $\mathbb{R}$ while also preserving the properties of measure.

Consider the three properties we have previously defined: non-negativity, countable additivity, and translation invariance. Individually, these properties are reasonable and together they are the basis for our natural geometric/physical understanding.

However, these properties cannot all be true for every subset of $\mathbb{R}$. If we try to make *every* subset of $\mathbb{R}$ measurable, either translation invariance or additivity must fail. Conversely, if we try to preserve translation invariance and countable additivity, then some subsets of $\mathbb{R}$ must be non-measurable.

This contradiction occurs because $\mathbb{R}$ contains a large amount of internal symmetry. Translations can shift sets in infinitely many ways. When these shifted sets interact with other sets, additivity imposes rigid constraints on their sizes. Certain subsets can be arranged through translations alone so that they overlap in highly nontrivial ways, creating contradictions if every set were assigned a measure.

Essentially, non-measurable sets exist so that measurable sets, which are defined by their size, additivity, and being translation-additivity, can inform our notion of size consistently. Lebesgue measure does not fail because it excludes some sets, instead it succeeds *because* it does do that.

2.4 Nonmeasurable Sets

Unlike measurable sets, nonmeasurable sets can't be characterized with boundaries or shapes. Rather, they are defined by how their points are *distributed* across space. Nonmeasurable sets are typically constructed by selecting points from collections of equivalence classes (groups of points considered "equivalent" under some rule), rather than by carving out regions of space geometrically.

As a result, nonmeasurable sets are very fragmented. Because they are scattered throughout $\mathbb{R}$, there is no interval of values that can meaningfully capture their size.

Additionally, nonmeasurable sets are very sensitive to translations. Shifting a nonmeasurable set can produce translated copies that interact in a way that conflicts with additivity. If we attempt to assign a consistent measure to the set, translation invariance would require all copies to have the same measure. However, then countable additivity would demand that certain unions of these copies have a measure that can't correspond to the actual amount of space they cover.

Essentially, nonmeasurable sets can't be defined by any explicit rule or formula. Rather, they come from unrestricted selection processes that choose elements from infinitely many overlapping collections without regard to spatial structure. In a sense, nonmeasurable sets are like "weird selections".

To form such sets, one must be able to make infinitely many independent choices without a specific rule. The principle that allows us to do this is known as the *Axiom of Choice*; which is very significant to the existence of nonmeasurable sets.

3 The Axiom of Choice and the Vitali Set

3.1 The Axiom of Choice

The Axiom of Choice (AC) is a principle in set theory that allows us to select elements from arbitrary collections of sets, even if there is no explicit rule given for the selection. This can formally be stated as:

$$\forall \mathcal{C}\,(\emptyset \notin \mathcal{C} \implies \exists f : \mathcal{C} \to \bigcup \mathcal{C} \text{ such that } f(X) \in X \ \forall X \in \mathcal{C}).$$

- $\mathcal{C}$ is defined as any collection of non-empty sets.
- f is defined as a *choice function* that selects one element from each set $X \in \mathcal{C}$.

When $\mathcal{C}$ is finite, a choice function can be constructed explicitly. However, for infinite collections, particularly uncountable ones, AC ensures the existence of a choice function even when there isn't any explicit rule for making the selections.

This principle is important for constructing certain sets that cannot be defined explicitly by a formula, such as the Vitali set. To construct a Vitali set, it is required to make infinitely many independent choices, one from each equivalence class of real numbers modulo $\mathbb{Q}$. The Axiom of Choice is what allows us to make such a selection.

3.2 Equivalence Classes Modulo Rationals

To construct the Vitali set, we first define an equivalence relation on $\mathbb{R}$. For $x, y \in \mathbb{R}$, we define this as:

$$x \sim y \iff x - y \in \mathbb{Q}.$$

This is an equivalence relation because is satisfies the following properties:

- Reflexive: $x - x = 0 \in \mathbb{Q} \implies x \sim x$.
- Symmetric: If $x \sim y$, then $x - y \in \mathbb{Q} \implies y - x = -(x - y) \in \mathbb{Q} \implies y \sim x$.
- Transitive: If $x \sim y$ and $y \sim z$, then $x - z = (x - y) + (y - z) \in \mathbb{Q} \implies x \sim z$.

The set of equivalence classes under this relation is denoted as $\mathbb{R}/\sim$. Each equivalence class is a dense subset of $\mathbb{R}$ consisting of all real numbers differing by a rational number.

Each equivalence class can be thought of as a "thread" running through the real line. No matter how far you move along the line, you will always find points from each class that are spaced by rational distances. For the Vitali Set, we will select exactly one representative from each class.

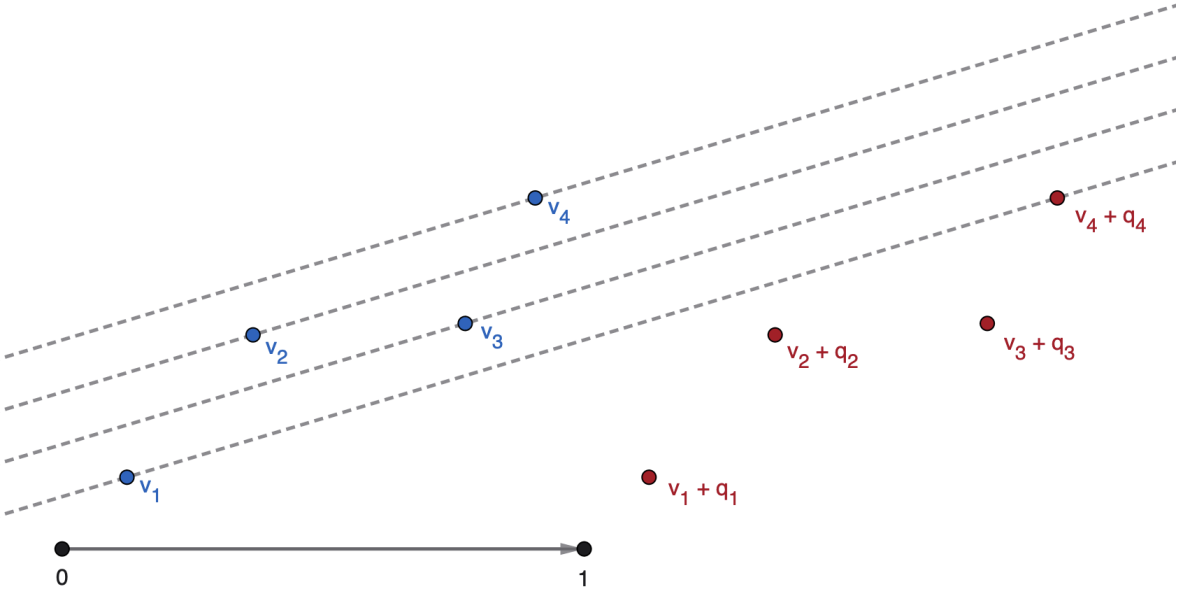


Figure 3: A schematic illustration of equivalence classes of $\mathbb{R}$ modulo $\mathbb{Q}$. Each slanted line represents one equivalence class. Selecting a single point from each class produces the Vitali set V. Horizontal translations of V by rational offsets generate disjoint shifted copies whose union covers an interval, visualizing the source of non-measurability.

3.3 Construction of the Vitali Set

The Vitali set $V \subset [0, 1]$ is constructed by selecting one representation from each equivalence class in $\mathbb{R}/\sim$ that intersects $[0, 1]$. We define this formally using the Axiom of Choice as:

$$V = \{v_\alpha \in \alpha : \alpha \in (\mathbb{R}/\sim) \cap [0, 1]\}.$$

There isn't an explicit formula for the elements of V, so the selection must rely entirely on the Axiom of Choice. This means that the Vitali set can't be constructed explicitly ; it is inherently non-measurable.

3.4 Non-Measurability of the Vitali Set

To see why V is non-measurable, let us consider the set of all rational shifts of V within $[0,1]$:

$$V_r = V + r \quad \text{for} \quad r \in \mathbb{Q} \cap [0,1).$$

These shifted sets are disjoint. If $x \in V_r \cap V_s$ for $r \neq s$, then $x-r, x-s \in V$ and $x-r \sim x-s \implies r-s \in \mathbb{Q}$. This contradicts the construction of V, which includes exactly one representative from each equivalence class.

Furthermore, the union of all such shifts essentially covers $[0,1]$:

$$[0,1] \subseteq \bigcup_{r \in \mathbb{Q} \cap [0,1)} V_r \subseteq [0,2].$$

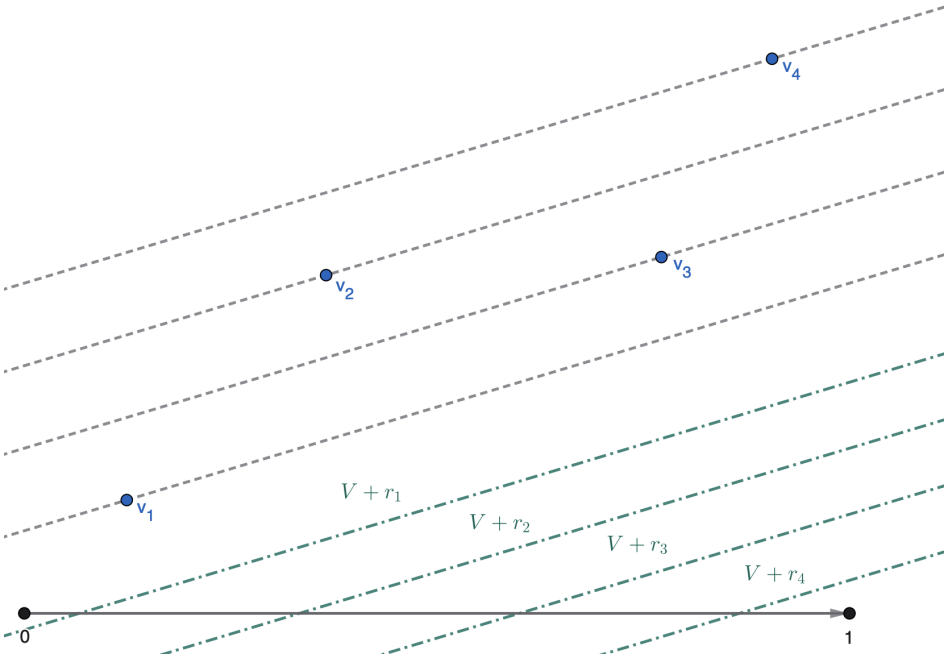


Figure 4: Disjoint rational shifts of the Vitali set. Each copy is non-overlapping, but their union covers $[0,1]$ which shows how it is not feasible to assign a consistent measure.

Hypothetically, let us assume that V is Lebesgue measurable with measure $\mu(V)$. By translation invariance all shifted sets have the same measure: $\mu(V_r) = \mu(V)$. By countable additivity, we have:

$$\mu\Big(\bigcup_{r \in \mathbb{Q} \cap [0,1)} V_r \Big) = \sum_{r \in \mathbb{Q} \cap [0,1)} \mu(V_r) = \sum_{r \in \mathbb{Q} \cap [0,1)} \mu(V).$$

Since $\mathbb{Q} \cap [0,1)$ is countably infinite, the sum is 0 if $\mu(V) = 0$, or it diverges to infinity if $\mu(V) > 0$. Both possibilities contradict the fact that the union is contained in the finite interval $[0,2]$. Therefore $\mu(V)$ can't exist, and V is non-measurable.

4 The Banach–Tarski Theorem

4.1 Statement of the Theorem

Formally, the Banach–Tarski theorem states:

Let B be a solid ball in $\mathbb{R}^3$. There exists a decomposition of B into finitely many disjoint sets $A_1, \ldots, A_n$ such that, using only rotations and translations, these pieces can be reassembled to form two identical copies of B.

Equivalently, there exist rigid motions $T_1, \ldots, T_n$ such that

$$\bigcup_{i=1}^{n} T_i(A_i) = B \cup B',$$

where B' is congruent to B and $A_i \cap A_j = \emptyset$ for $i \neq j$.

We have mentioned previously how this defies physical logic in regards to volume preservation. However, this becomes mathematically possible due to properties of the sets A_i. They are non-measurable, can be infinitely fragmented, and exist due to the Axiom of Choice.

4.2 Group Actions and Orbits

The key to the paradox lies in group theory. Consider the group of rotations in $\mathbb{R}^3$, which we denote as $SO(3)$. An important fact about this group, is that it contains a *free subgroup* F_2 generated by two rotations r_1 and r_2:

$$F_2 = \langle r_1, r_2 \rangle = \{r_1^{n_1} r_2^{m_1} \cdots r_1^{n_k} r_2^{m_k} : n_i, m_i \in \mathbb{Z}\},$$

where no non-trivial combination of these rotations equals the identity.

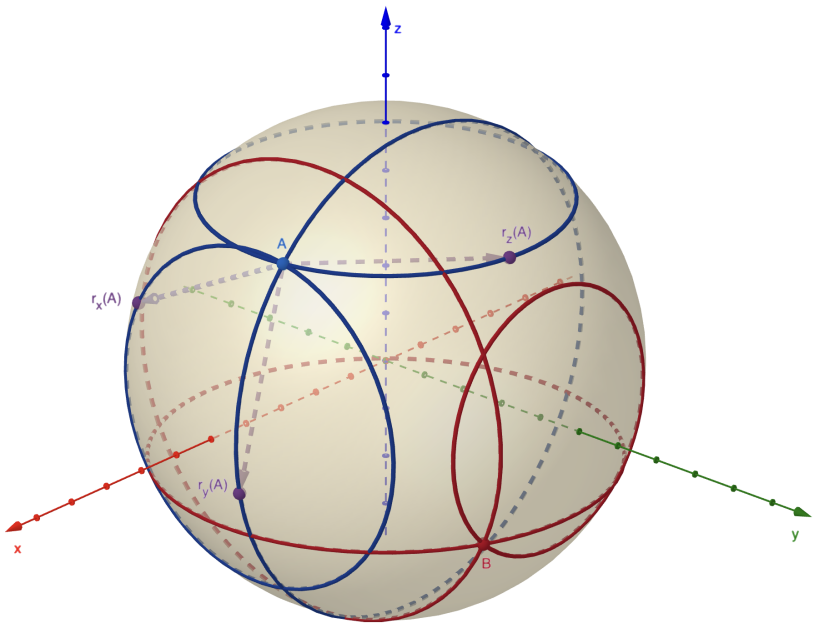


Figure 5: Illustration of orbits of points under rotations. Point A is moved along three rotational axes (x, y, z) to form its orbit. Point B behaves similarly, moving along its own orbit. These orbits are what allows us to select representative points in the Banach–Tarski construction.

Think of each point in the ball as being "moved around" by combinations of the basic rotations. The set of all positions a point can reach with these movements is called its orbit. Almost all points in the ball can be grouped into distinct orbits, with no overlaps between different orbits (with exceptions, like the center).

By picking one point from each orbit, we create a set of representatives that will serve for reconstructing the ball.

For any point $x \in B$, its *orbit* under F_2 is defined as

$$\mathcal{O}(x) = \{g \cdot x : g \in F_2\}.$$

These orbits partition (almost all of) the ball into disjoint equivalence classes under the action of F_2. Using the Axiom of Choice, we select a single representative from each orbit to form a set C. Then the ball can be expressed as

$$B = \bigcup_{g \in F_2} g \cdot C,$$

where the union is disjoint except possibly for a set of measure zero (e.g., the center of the ball).

4.3 Decomposition of the Construction

We can divide the construction into several steps:

1. **Select a free subgroup of rotations.** Let $F_2 = \langle r_1, r_2 \rangle$, generated by two carefully chosen rotations. This subgroup is "paradoxical," meaning that it allows a decomposition similar to the Vitali set construction in 1D.

2. **Partition the ball into orbits.** For each point $x \in B$ outside a negligible set (e.g., the center), its orbit under F_2 defines an equivalence class:

   $$x \sim y \iff \exists g \in F_2 \text{ such that } g \cdot x = y.$$

   The collection of these orbits partitions the ball into infinitely many disjoint classes.

3. **Apply the Axiom of Choice.** Using AC, we pick a single representative from each orbit. This gives us a set C of "choice points."

4. **Form the paradoxical pieces.** By applying group elements $g \in F_2$ to C in certain combinations, we generate disjoint subsets $A_1, \ldots, A_n$ whose union is B and whose rotated/translated copies can form two balls:

   $$B = \bigcup_{i=1}^{n} A_i, \quad B \cup B = \bigcup_{i=1}^{n} T_i(A_i).$$

4.4 Measure-Theoretic Implications

Each piece A_i is non-measurable. If we attempted to assign a Lebesgue measure $\mu(A_i)$, translation invariance and countable additivity would force a contradiction:

- Let $\mu(B) = v$. By translation invariance, each rotated copy $T_i(A_i)$ would have measure $\mu(A_i)$.
- Countable additivity implies the total measure of the union of copies should equal the sum of their measures:

  $$\sum_{i=1}^{n} \mu(A_i) \stackrel{?}{=} \mu\Big(\bigcup_{i=1}^{n} T_i(A_i)\Big) = \mu(B \cup B') = 2v.$$

- But the union of the pieces also equals $B \cup B$, which has a measure of $2v$. This is a contradiction, as the sum $\sum_{i=1}^{n} \mu(A_i)$ cannot equal v and $2v$ under the rules of Lebesgue measure.

So, the paradox does not violate mathematical logic. It can be fully justified because the sets involved here are non-measurable. Measure theory here "fails", but it is in a controlled, understood manner.

5 Conclusions

5.1 Summary

The Banach–Tarski paradox challenges the idea that volume is an intrinsic property of all subsets of space. This intuition for volume is grounded in our real setting in which volume behaves predictably under cutting, rearrangement, and rigid motion. However, once arbitrary subsets are allowed, this assumption no longer holds true.

Using measure theory, we saw that the Lebesgue measure provides a consistent notion of size only by excluding certain pathological sets. The existence of non-measurable sets is not accidental, but rather a consequence of preserving translation invariance and countable additivity. The Axiom of Choice enables the construction of such sets by allowing unrestricted selection from infinitely many equivalence classes, as illustrated by the Vitali set.

In three dimensions, the same ideas manifest through group actions of rotations. The Banach–Tarski Paradox's decomposition relies on the symmetry of $\mathbb{R}^3$, where free subgroups of $SO(3)$ allow space to be partitioned into non-measurable pieces that defy traditional ideas on volume.

5.2 Discussion

Naturally, the paradox is not meant to literally imply physical objects can duplicate, nor does it invalidate conservation of volume as a useful concept. Rather, it shows the limitations in how we apply our definitions. Volume behaves well precisely because it is not defined everywhere for all possible subsets.

Through the Banach–Tarski paradox, it can be seen that by accepting the Axiom of Choice we can gain tools at the cost of admitting objects that resist intuition and visualization. The paradox is both a result that is logically unavoidable and one that is conceptually disturbing to our intuition.

In this way, the paradox serves less as a contradiction and more as a clarification. It reminds us that mathematics is not necessarily constrained to mirror the rules of the physical world, and that the price of generality can often be strangeness.

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Infinite Descent

Zhuo Zhan Chen

1 Introduction

1.1 What is Infinite Descent

Infinite descent, also known as Fermat's method of descent, is a way of proof by contradiction. It uses the fact that there must exist a smallest number satisfying a statement. Proof by infinite descent would show that if the statement holds for a number, then it holds for a smaller one, and so on, meaning there could not be a smallest number. This contradiction allows you to prove that no number satisfies the statement.

1.2 Definitions, Theorems, and Corollaries

Definition 1.1 (Rational Number)**.** Rational numbers are numbers that can be written as the ratio of two integers

Theorem 1.1 (Well-Ordering Principle for natural numbers)**.** *Every non-empty set of natural numbers has a least element*

Theorem 1.2 (Pythagorean Triple)**.** *If primitive ordered triple* (x, y, z) *with* $gcd(x, y) = 1$ *satisfies* $x^2+y^2 = z^2$*, it is possible to write* $x = 2rs, y = r^2 - s^2, z = r^2 + s^2$ *with* r *and* s *being relatively prime integers.(The proof of this is left to the reader) For example,* $(3, 4, 5)$ *is a primitive Pythagorean triple, but* $(6, 8, 10)$ *is a Pythagorean triple, but not primitive.*

Theorem 1.3 (Fermat's Last Theorem)**.** *There is no three positive integers a,b, and c satisfy the equation* $a^n + b^n = c^n$ *for any integer value of n¿2*

2 Some proofs that use infinite descent

2.1 Yeah, it is. So prove it.

$\sqrt{2}$ is irrational; this is true and much more intuitive to believe than to prove. You might have seen this proof that uses infinite descent. It goes as follows:

Proof. Assume that $\sqrt{2}$ is rational, that is, that there exist natural numbers p and q s.t. $\frac{p}{q} = \sqrt{2}$.

By the Well-Ordering Principle, there exists a smallest integer p that satisfies the above equation. (or that there exist p s.t. $\frac{p}{q}$ cannot be further simplified, equivalent to $\gcd(p, q) = 1$)

$\frac{p}{q} = \sqrt{2} \to \frac{p^2}{q^2} = 2 \to p^2 = 2q^2 \to 2 \mid p$

We can rewrite $p = 2p'$ with $p' \in \mathbb{N}$ By substituting $p = 2p'$, through a similar process, we can prove that $2 \mid q$. We will write $q = 2q'$ with $q' \in \mathbb{N}$ This means $\frac{p}{q} = \frac{2p'}{2q'} = \frac{p'}{q'}$ Notice how we can reduce p infinitely, which means the set of possible values of p doesn't have a smallest element. By infinite descent (or contradiction to Well-Ordering Principle), the assumption made is false, $\sqrt{2}$ is irrational.

□

2.2 FERMAT LAST THEOREM?! n=4.

Fermat, in the margin of his copy of the *Arithmetica*, wrote

> It is impossible to separate a cube into two cubes, or a fourth power into two fourth powers, or, in general, any power higher than the second, into two like powers. I have discovered a truly **marvelous** proof of this, which this margin is too narrow to contain.

Due to the limited space of the margin, Fermat's Last Theorem remained unsolved for more than 300 years until Andrew Wiles developed new techniques to prove it. Although this article cannot hope to contain Fermat's "marvelous proof," Fermat did succeed in proving the case where the exponent equals four, which goes as follows:

First, by proving that no ordered triple (x, y, z) satisfies the equation $x^4 + y^4 = z^2$, it would immediately follow that $n = 4$ will not produce any solutions for FLT.

Now, by the Well-Ordering Principle, we can ensure the existence of (x_0, y_0, z_0), with z_0 **being the smallest integer that satisfies the equation**, assuming that there exists at least one solution.

By Theorem 1.2, we can write

$$\begin{cases} x_0^2 = 2rs \\ y_0^2 = r^2 - s^2 \\ z_0 = r^2 + s^2 \end{cases} \tag{1}$$

Claim: s is even

Now we focus on the parity of r and s; by the statement of Theorem 1.2, they cannot both be even or both be odd. Notice that if we make r even and s odd, $y_0^2 \equiv r^2 - s^2 \equiv 0 - 1 \equiv 3 \pmod 4$. And this cannot be true because no perfect square has a remainder of 3 when divided by 4. Therefore, we are left with the case where s is even; our claim is proven.

Using this claim, we can write $s = 2p$, which led us to write $(\frac{x_0}{2})^2 = rp$. This result is especially useful, because since r and s are relatively prime, it is certain that r and p are relatively prime. Therefore, r and p must both be perfect squares, since their product is a perfect square. Thus, we will let $r = R^2$ and $p = P^2$.

By the second column of our definition of y_0^2, we know that $(2P^2)^2 + (y_0)^2 = (R^2)^2$. From here, we can find two relatively prime integers, a and b, such that

$$\begin{cases} 2P^2 = 2ab \\ y_0 = a^2 - b^2 \\ R^2 = a^2 + b^2 \end{cases} \tag{2}$$

Using the result we have found in the previous paragraph, we can prove that a and b are perfect squares because $P^2 = ab$.

Now, we realize that there is a solution $(x, y, z) = (\sqrt{a}, \sqrt{b}, R)$ (why?). It is clear that $R < z_0$, and so an ordered triple with a smaller z can be generated, which contradicts our claim that z_0 is the smallest integer that satisfies the equation, meaning that our assumption that there exist at least one ordered triple (x, y, z) is false.

3 Vieta Jumping

3.1 Notorious Problem Six

3.1.1 Introduction

Ever since 1988, when the first problem using the method of Vieta Jumping appeared as the sixth problem on the IMO, Vieta Jumping has become well-known among problem solvers.

Vieta Jumping is another variant of Infinite Descent that uses Vieta's Formula to generate a smaller solution. The problem that uses this method goes as follows:

"Let a and b be positive integers such that $ab+1$ divides a^2+b^2. Show that $\frac{a^2+b^2}{ab+1}$ is the square of an integer." (Hint: Let $\frac{a^2+b^2}{ab+1} = c$. Think about how to relate a, b, c and how many solutions of a exist, given b and c)

3.2 Group Law on Conics

Theorem 3.1. *For any real coefficient c_1 and c_2, there must be at least one real solution x such that $c_1 \sin x + c_2 \cos x = 0$*

The proof will be left as an exercise for the reader, this lemma will come in handy for the proof of the more important theorem. (Hint: divide $\cos x$ on both sides, do consider the case where $\cos x = 0$ though)

Theorem 3.2. *Every conic section can be represented by an equation of the form $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$, where (A, B, C, D, E, F $\in \mathbb{Q}$). (we will restrict to rational conics) In particular, the mixed term (Bxy) is necessary to describe conics that are rotated with respect to the coordinate axes. Note that a conic with $B \neq 0$ is called an "oblique conic"*

Proof. We begin with an observation on how the coordinates of a point (x_0, y_0) or in its polar form, $(r\cos\alpha, r\sin\alpha)$ can be expressed after rotating by β (counterclockwise, of course), which is clearly $(r\cos(\alpha+\beta), r\sin(\alpha+\beta))$. By angle addition, that would imply that the "new" coordinates, which we denote as x_1 and y_1, will be:

$$\begin{cases} x_1 = r(\cos\alpha\cos\beta - \sin\alpha\sin\beta) \\ y_1 = r(\sin\alpha\cos\beta + \cos\alpha\sin\beta) \end{cases} \tag{3}$$

or even better:

$$\begin{cases} x_1 = x_0\cos\beta - y_0\sin\beta \\ y_1 = y_0\cos\beta + x_0\sin\beta \end{cases} \tag{4}$$

This is extremely useful because we can substitute that into the conic form, and hopefully, we can try to adjust β so that the xy-term is canceled out. We should focus on the x_0y_0 term, which has a degree of two, meaning that in the substitution, only $Ax^2 + Bxy + Cy^2$ can contribute to the coefficient of the x_0y_0. Skipping the computations, after substitution, the coefficient turns out to be $(C-A)\sin 2\beta + B\cos 2\beta$. By Theorem 3.1, this clearly has at least one real β such that the coefficient becomes zero, implying an oblique conic is just a rotation of a "standard" conic. □

Theorem 3.3 (Vieta's Formula for quadratic)**.** *Let x_1 and x_2 be the roots of the quadratic $x^2 + ax + b$, it is true that $x_1 + x_2 = -a$ and $x_1x_2 = b$*

Proof. Clearly $(x - x_1)(x - x_2) = x^2 - (x_1 + x_2)x + x_1x_2 = x^2 + ax + b$. □

Theorem 3.4 (Conjugate Root Theorem for Irrational Roots)**.** *If a polynomial with rational coefficient has an irrational root $A + \sqrt{B}$ then $A - \sqrt{B}$ must also be a root, or that irrational roots must come in pairs. ($A, B \in \mathbb{Q}$*

Theorem 3.5 (Group Law on Conics)**.** *On a non-degenerate conic C, using a fixed rational point N and two other rational points A and B, a new rational point can be found, by drawing a line through N parallel to $\overline{AB}$.*

Proof. The proof is quite elementary, as this can be reworded as follows: The intersection of a non-degenerate conic and a line with rational slope, over a rational point, is rational. Let the coordinates of point N be (x_N, y_N). The fact that A and B need to be rational implies the slope is rational (why?), which means this line can be expressed as $y = m(x - x_N) + y_N$, while the conic can be expressed as $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. Substitution gives you a degree two equation, with one of the roots N being rational and all of its coefficients being rational. By the Conjugate Root Theorem for Irrational Roots, the other intersection must also be a rational point; if not, then there isn't another root that can be its conjugate (A degree two polynomial has at most two distinct roots). □

This is the observation or theorem needed for the intuition of this problem.

3.3 The Solution

There are some other useful observations for this problem listed below:

- With c being fixed, if (a, b) is a solution, then (b, a) is also a solution
- $a = b^3$ generates solutions where c is a perfect square
- With c being fixed, the relation between a and b happens to be a conic

Inspired by the group law of conics, it makes sense for us to first construct a diagram for an arbitrary perfect square c value and start drawing lines between rational points (Note that the problem wants "lattice points" instead of rational points, but the rule also applies similarly). Notice how we can use points A, A', and B to construct more solutions. Focusing on points B and B', it's clear that they have the same y-coordinate. Plugging this y-value as b produces two solutions for a, let's look more closely to it, take $(8, 2)$ as an example. $\frac{8^2+2^2}{8\times 2+1} = 4$.
Let the y-value be replaced with the letter y in this equation $\frac{y^2+8^2}{8y+1} = 4$ which is a quadratic with integer coefficients, with an integer root given, thus the other root must also be an integer.

Proof. Suppose k is not a perfect square, let the lattice point (A, B) lie on the conic produced by $\frac{a^2+b^2}{ab+1} = k$ which can be written as $a^2 - kb(a) + (b^2 - k) = 0$, such that A is minimized. Without loss of generality, we can assume $A > B$.
Clearly, one of the roots of this quadratic is $a = A$. By Vieta's Formula for a quadratic, you can express the other root as $kb - a$ or $\frac{b^2-k}{a}$. Notice how this root has to be a positive integer, that is not zero ($b^2 \neq k$).

In addition, since $A > B$, $B^2 - k < B^2 < A^2$. This other root is also less than A. If k is not a perfect square, there must exist a point (x_2, B) less than (A, B), the minimum element; thus, k must be a perfect square. □

4 Conclusion

In conclusion, infinite descent is a powerful tool for solving Diophantine equations. The fundamental principle that you can disprove the existence of solutions by generating a smaller counterexample amazes me. While studying Vieta jumping, I began to wonder whether other number-theoretic tools could be used to solve some of the same problems, but I got nowhere. This process of asking "what else might work?" proved to be as valuable as learning the methods themselves. As a student, I truly wonder if there are other alternative solutions.

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Boolean Algebra

Yuna Kang

1 Introduction

In 1874, Alexander Graham Bell found the first telephone successful. In 1993, the first smartphone was launched, the IBM Simon, about 8 inches high and weighing 18 ounces. Only about 30 years later, the average size of cell phones is pocket-sized, only about 6.3 inches high, and weighing less than 8 ounces. The abrupt advancement of phones that are used in real life was enabled by computer science and mathematics, specifically Boolean Algebra.

2 Boolean Algebra

Phones and other electronic devices consist of hundreds of circuits that store data. They are formed by a series of microscopic *logic gates*. An input will signal the output to turn off or on, or 0 or 1 respectively. Since it has only two possible outputs, it is called a binary system.

Boolean algebra was invented by George Boole, an English mathematician and philosopher, when he published The Mathematical Analysis of Logic (1847). Boolean algebra simplifies operations with variables that are either true (1) or false (0) by using properties, which results in a faster and smaller device.

2.1 Operations

Operations in Boolean Algebra are logical processes from a certain input that yields a certain output. The common fundamental mathematics have four operations: addition, subtraction, multiplication, and division. While there are more operations such as the NAND, XOR, NOR operations, there are three major operations in Boolean Algebra: the NOT operation, AND operation, and OR operation.

2.1.1 NOT Operation

The NOT Operation, denoted x' or $\neg x$, inverts the boolean digit by using the formula $1 - x$. In terms of mathematical logic, it is known as negation. The operation states that for an input, the output is the inverse of the input.

2.1.2 AND Operation

The AND Operation, denoted $x \cdot y$ or $x \wedge y$, is similar to multiplication $\pmod 2$ or conjunction in mathematical logic. The operation states that only if both inputs are true, the output is true. In other words, if the given statements are "I am a student, and I enjoy school", then both statements "I am a student" and "I enjoy school" would be true.

2.1.3 OR Operation

OR Operation, denoted $x + y$ or $x \vee y$, is similar to addition mod 2 or disjunction in mathematical logic. The operation states that only if both inputs are false, the output is false. For example, if the given statements are "It is snowing, or it is raining", then at least one of the statements "It is snowing" or "It is raining" is true.

However, sometimes there may be a given statement "I am in middle school, or I am in high school." Since a person cannot be both in middle and high school, only one of the statements are true. This is known as the XOR Operation, or the exclusive OR. OR Operations are generally inclusive and are true when both statements are true.

2.2 Truth Tables

The truth table is a way of showing the operation's outcomes by listing out all possible values either true or false for every combination possible. If there are n variables, the number of combinations for the truth table is n^2 since the variables can only be true or false.

Table 1: Boolean Operations

x	x'
0	1
1	0

(a) NOT Operation

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

(b) AND Operation

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

(c) OR Operation

2.3 Logic Gates

Logic gates is a device that does a Boolean operation. The millions of logic circuits inside an electronic device can be mapped out by logic gates.

3 Laws of Boolean Algebra

While Boolean operations relates the inputs to the outputs, there are also laws of Boolean algebra that can simply these logic gates. Simplifying these complex logic gates with the

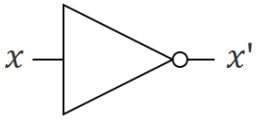


Figure 1: NOT Gate

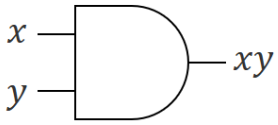


Figure 2: AND Gate

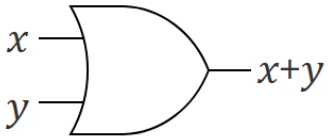


Figure 3: OR Gate

same input and output combination, can help reduce the size of the electronic device while also improving the speed of the device.

Law	AND	OR
Annulment	$a \cdot 0 = 0$	$a + 1 = 1$
Identity	$a \cdot 1 = a$	$a + 0 = a$
Idempotent	$a \cdot a = a$	$a + a = a$
Complement	$a \cdot a' = 0$	$a + a' = 1$
DeMorgan's Theorem	$(ab)' = a' + b'$	$(a + b)' = a'b'$
Commutative	$ab = ba$	$a + b = b + a$
Associative	$(ab)c = a(bc)$	$(a + b) + c = a + (b + c)$
Distributive	$a + bc = (a + b)(a + c)$	$a(b + c) = ab + ac$
Absorption	$a(a + b) = a$	$a + ab = a$

Table 2: Laws of Boolean Algebra

4 The Binary System

It is commonly said that computers "only understand zeros and ones," and indeed information of all kinds from numerical, text, colors, images, colors, letters, to even audio is encoded using these two symbols. In other words, computers rely on the binary number system, a base 2 representation using digits 0 and 1. In contrast, humans generally use the decimal, or base 10, number system, with ten digits ranging from 0 to 9. Despite its minimal alphabet, binary provides a powerful and efficient way to represent and manipulate information electronically.

4.1 Decimal System to Binary System

As mentioned before, computers uses a binary system to calculate numbers, while humans use the decimal system. However, the fundamental for a base 2 system and a base 10 system are very similar.

Take the number 72 in base 10, for example. 72 in the decimal system can be represented as $(7 \times 10^1) + (2 \times 10^0)$. Each numbers from 0 through 9 is being multiplied by 10^n, where n

is an integer. In order to find 72 in the binary system would be to find each number from 0 to 1 being multiplied by 2^n, where n is an integer. 72 can be expressed as $(1 \times 2^6) + (1 \times 2^3)$. So in the binary system, the number 72 is 01001000.

4.2 Binary System Representation

Converting the decimal system into the binary system can be used to represent numbers in the computer.

Computers uses the binary system as well as letters, audio, colors, etc. For example, for the binary codes corresponding to each letter, they use Unicode or ASCII (American Standard Code for Information Interchange). In fact, 01001000 01100101 01101100 01101100 01101111 00100001 in binary code means "Hello!"

All kinds of information are assigned a specific combination of zeros and ones. To prevent a misinterpretation of differentiating 01001000 as the number "72" instead of the letter "H", computers have programs finds if a specific code is referring to a number or a letter using context.

5 Conclusion

Electronics uses only zeros and ones, or the binary system. Boolean algebra also uses a binary system and has been fundamental to the creation and evolution of the digital technologies that make up the necessities of our daily life and will continue to influence the future. While we often use our phones and computers, without noticing the abstract mathematics beneath them, it is worth appreciating how Boolean logic helped make modern computing possible.

Math of Origami Constructions

Sonam Tsephel

1 Introduction

Origami is the art of folding paper to create shapes and objects, originating in Japan. On the surface, origami may seem like another creative hobby, sometimes simple and sometimes very intricate. But when you unfold and look within each crease, the mathematical principles become more evident.

Origami techniques involve all kinds of math. For instance, dividing segments, forming diagonals, and keeping symmetry for constructions are all patterns commonly seen in even the most basic constructions. Furthermore, origami can be used for different kinds of problem solving, such as creating parabolas and even solving cubic equations.

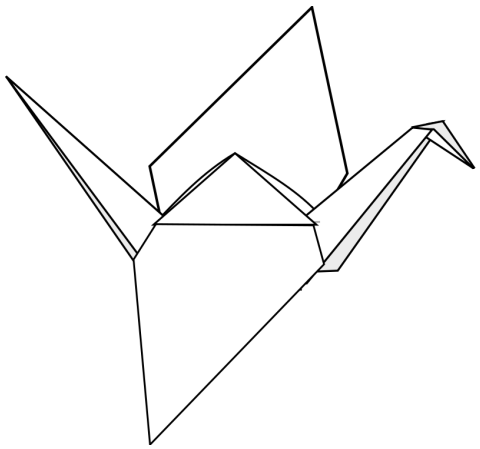


Figure 1: Origami Crane

2 Ruler and Compass Constructions

Before we get into what we can do with origami, we first have to understand lengths formed from rulers and compasses. These numbers are called constructible numbers. But what actually counts as a constructible number? Constructible numbers represent the set of numbers that can also express the length of a segment formed using a straight edge or compass. To put it simply, given a unit length (a construction with the length of one unit), constructible numbers can be formed by drawing segments in between points, drawing circles with a given center and point on its circumference, and finding the intersections of segments and circles.

The use of constructible numbers, not a formalized concept at the time, originated from Ancient Greek geometry. These numbers were fundamental for attempting to solve problems like trisecting an angle, doubling a cube, or squaring a circle. In fact, constructible numbers were implicitly foundational (not algebraically formalized) to Euclid's math, and Euclid would use these concepts to work with ratios of lengths, such as the ratio of a diagonal of a square to its side length.

We can perform basic arithmetic such as adding, multiplying, and even taking the square root of existing constructible numbers to form other constructible numbers. Note that it is assumed we are able to construct a unit length.

1. To copy a line segment, you can use a compass to measure the length of a line segment, and then place the compass on the paper and use a ruler to construct the line segment.

2. You can add line segments by measuring their lengths, marking each distance onto the paper (where the start point of each line segment is the endpoint of the last), and then using a ruler to draw the line segment such that it passes through all of the points of the markings. Subtraction can be done the same way, except the direction of the second segment is opposite to the direction of the first.

3. The product of two constructible numbers, a and b, can be found by drawing them to form an acute angle, drawing a line from the endpoint of segment a to b and drawing another line from the endpoint of b to the extension of a, parallel to the previous line, and taking the length from the shared endpoint to the endpoint of the second constructed line.

4. To find the quotient of two constructible numbers, with the same setup of segments of lengths a and b, construct a line from the endpoint of b to a and then construct a line parallel to the previous one where the endpoint on line b is 1 unit away to the shared endpoint of a and b. The length from the shared endpoint to the second line, at a closer distance to the shared endpoint than the first, represents $\frac{a}{b}$.

5. To divide a line segment into n parts, you can construct a line that makes an acute angle with our original line and use a compass to make n markings on the construction. Then construct a line from the last marking to the endpoint of our original line and construct all the other lines from the each marking to the original line so that all the constructions are parallel.

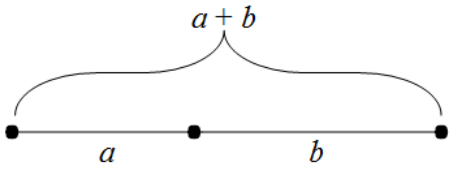


Figure 2: Adding Constructible Lengths

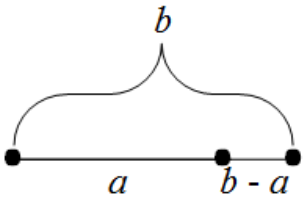


Figure 3: Subtracting Constructible Lengths

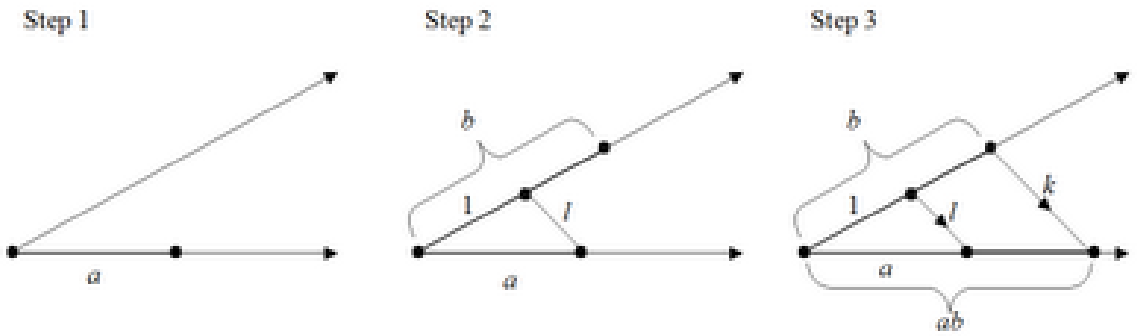


Figure 4: Product of Two Constructible Lengths

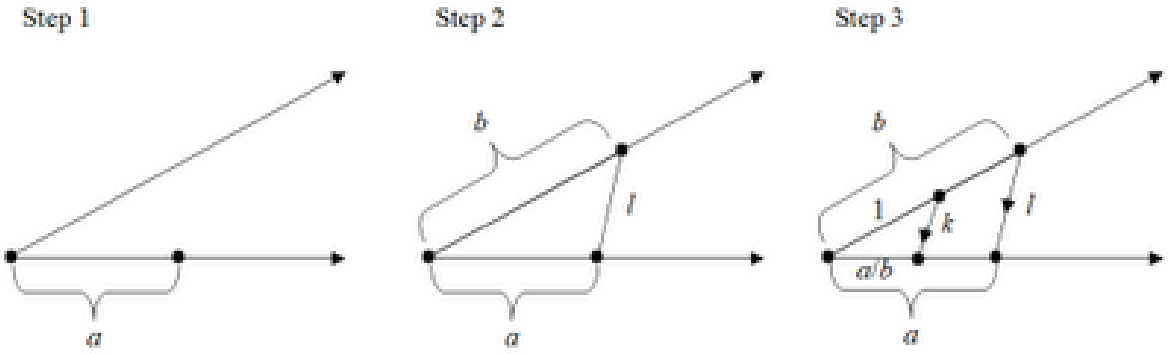


Figure 5: Quotient of Two Constructible Length

3 The Field of Constructible Numbers and Impossible Constructions

Our previous claim that we can perform the basic operations to constructible numbers to form other constructible numbers leads to the question of knowing whether or not a value is constructible. For example, is $\sqrt[3]{2}$ a constructible number? Let's first start with $\sqrt{2}$.

If we construct 2 unit lengths that are perpendicular and share an endpoint, the construction of the diagonal would have a length of $\sqrt{2}$ units because it is the hypotenuse of a right triangle.

In a similar manner, it is possible to construct the square root of any constructible number. In fact, since we are always adding lines and circles when constructing lengths, new constructible numbers are always just solutions to quadratics with coefficients of other constructible numbers. In other words, constructible numbers can always be represented as combinations of the basic 4 operations and square roots applied to rational numbers.

Put another way, the constructible numbers are the smallest extension of Q such that the square root of every constructible number is constructible.. However, you cannot form through the intersections of constructible numbers, the values of cube roots; therefore constructing a line of length $\sqrt[3]{2}$ with only a compass and ruler is impossible.

The impossibility of constructing $\sqrt[3]{2}$ was not proven until the 19th century, along with many other lengths as a constructible number such as π and e. This is simply because no combination of the four basic operations or square roots of any combinations of the four operations is equivalent to a cube root (aside from the cube roots of 0 and 1), and that is how a constructible number is defined. This is also the reason the Ancient Greeks were not able to construct the length to double the volume of a cube using a ruler or compass, one of their major classical problems. (We now know it's impossible, as well as the other problems: trisecting an angle and squaring a circle).

The problem of doubling the cube is the following: given an isometric drawing of a cube, construct an isometric drawing of a cube with double the volume. If s is the side length of the original cube, then clearly, the new cube should have a side length of $s\sqrt[3]{2}$. Therefore, doubling the cube is equivalent to constructing the length $\sqrt[3]{2}$, which is impossible as we just discussed.

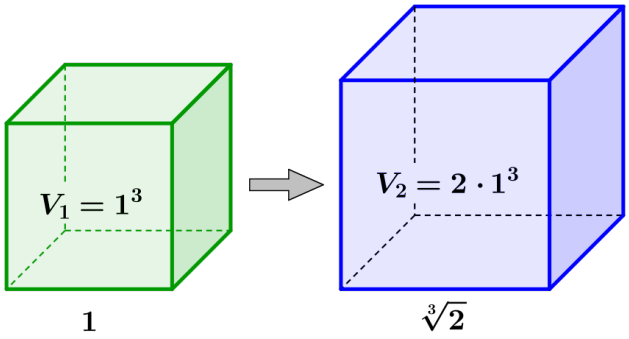


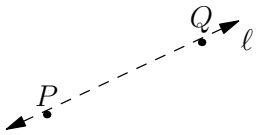
Figure 6: Doubling the Volume of a Cube

4 Huzita-Justin Axioms

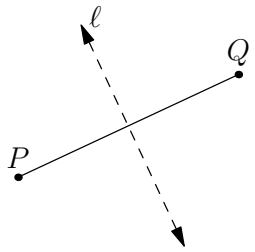
As we shift to constructions by Origami, it is important to note that all ruler and compass constructions can be performed with origami. Origami constructions don't have a strictly defined field like the constructible numbers, but it can be thought of as an extension to the field of constructible numbers that includes values of cube roots, which we will be examined in the following sections. But before that, we introduce the Huzita Justin Axioms.

The Huzita-Justin axioms are a set of origami operations that provide the fundamentals for folds and creases. There are seven axioms in total, where two are special cases, and most of them can be understood intuitively. Note that the construction of lines in these axioms represent the physical folding of paper

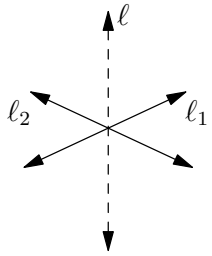
1. Given two points, P and Q, we can construct a line that contains both points.



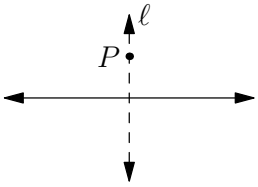
2. Given two points, P and Q, we can construct a line such that P and Q are reflections over that line.



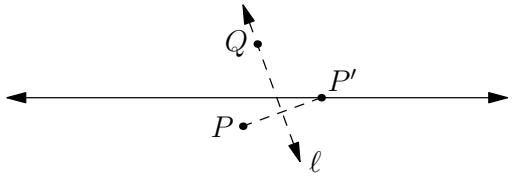
3. Given two lines, ℓ_1 and ℓ_2, we can construct a line such that ℓ_1 and ℓ_2 are reflections over this line.



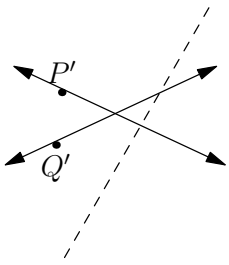
4. Given a point P and a line ℓ, we can fold a line that contains P and is perpendicular to ℓ.



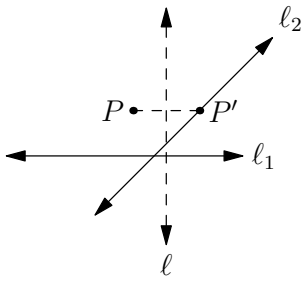
5. Given a line ℓ, and two points, P and Q, not on line ℓ, we can fold a line that places P onto ℓ such that the crease passes through Q.



6. Given two points, P and Q, and two lines, ℓ_1 and ℓ_2, we can fold a line that places P onto ℓ_1 and Q onto ℓ_2



7. Given a point P and two lines, ℓ_1 and ℓ_2, we can fold a line perpendicular to ℓ_1 that places P onto ℓ_2.



The origami axioms are very similar to constructions in geometry, where a crease is equivalent to a line.

5 Constructing a Parabola

One way to define a parabola is with a focus F and a directrix d, which is a line. A parabola is the set of all points that are equidistant from the focus and directrix. A parabola can be constructed through paper folding with if you draw a line to represent the directrix d and a point, preferably centered and somewhat near the directrix, to represent the focus. It is best to use origami paper or thin, transparent paper.

Given a line ℓ and a point P, and you know the distance between ℓ and P. Folding P onto line ℓ near one end of line will form a crease. If you adjust your pull on the flap of paper to place point P further along line ℓ, until you reach the other edge of the line, you will see the creases form a parabola shape that surround point P. This reveals a parabola, where each of the crease lines are the tangents to the parabola.

More simply, you can draw many distinct points on line ℓ and fold the lines that place point P onto each of the points you drew on line ℓ. If you trace the curve formed from all the creases of the tangents, you will see the formation of a U shaped curve: the parabola. The focus of the parabola is point P and the directrix is line ℓ. For parabolas with end behaviors facing up, the parabola is above the directrix, and for parabolas with end behaviors facing down, the parabola is below the directrix.

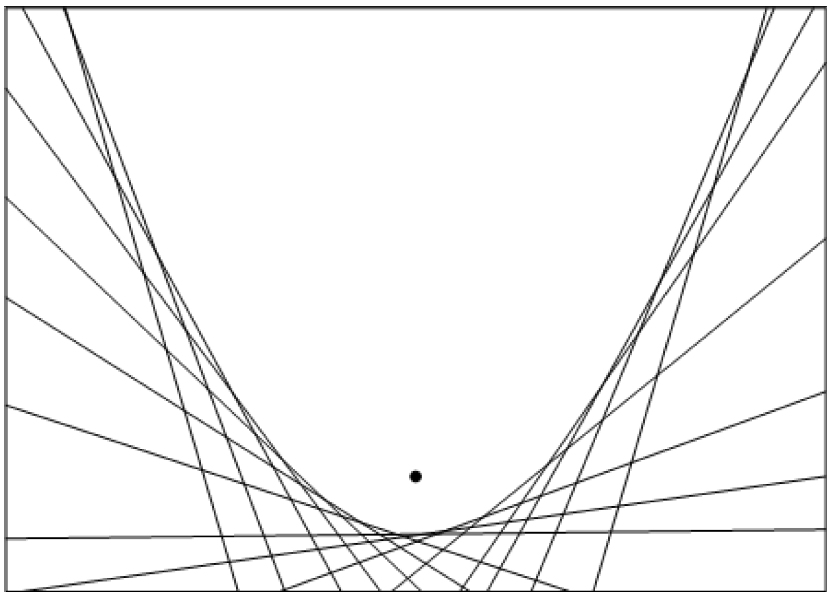


Figure 7: Parabola Formed from Paper Folding

6 Beloch's Big Idea

Margherita Piazzola Beloch is credited for discovering the method to solve any cubic equation using origami. The 6th origami axiom is also known as Beloch's fold, which is key to solving the cubic.

First let's understand how origami can help us solve a quadratic. When we fold a point P to a line m, it will form a tangent to the parabola formed with focus P and directrix m. The crease formed is the perpendicular bisector of the segment connecting P to m, and it is equivalent to finding the intersection of that tangent to the parabola. It can basically be thought of as solving a quadratic.

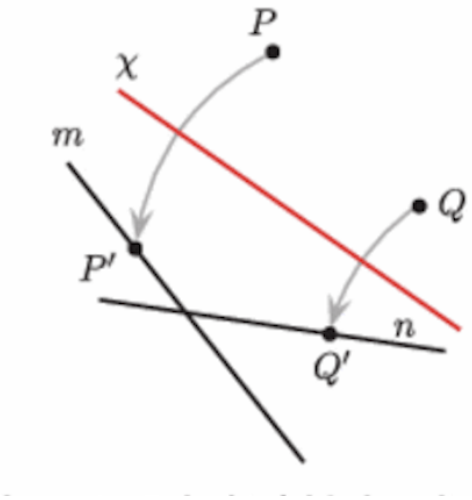


Figure 8: Beloch's Fold

Beloch's fold is important because it forms the crease that is tangent to more than one parabola. Let's say we have one parabola with focus P, directrix m, and a second with focus Q, directrix n. We can use Beloch's fold to fold the line that places P onto m and Q onto n, and the crease made from folding a focus to the directrix of a parabola represents a tangent line to the parabola. The other two tangents aren't as clear to obtain but can be found by folding P to m, and adjusting the fold to be tangent to the second parabola, and by folding Q to n, and adjusting the fold to be tangent to the first parabola. Technically, the ability to find all three tangents of the two parabolas is equivalent to solving a cubic equation.

Extending this concept to solve cubic equations is a little more complex. In order to do so, we need to find the tangent lines of two parabolas. Mathematically, two parabolas can have up to three common tangents.

Suppose we have the parabolas $y = x^2$ and $x = y^2$, and we would like to find the equations of the tangents between them. To find a line $y = mx + b$ to be tangent to the first parabola, we must set them equal to each other and the discriminant should be equal to zero. Therefore, we have

$$\begin{aligned} mx + b &= x^2 \\ x^2 - mx - b &= 0 \\ \Delta = m^2 + 4b &= 0, \end{aligned}$$

so $b = \frac{-m^2}{4}$. We can do the same for the second parabola:

$$\begin{aligned}(mx+b)^2 &= x\\ m^2x^2 + 2mbx + b^2 &= x\\ m^2x^2 + (2mb-1)x + b^2 &= 0\\ \Delta = (2mb-1)^2 - 4m^2b^2 &= 0\\ 4m^2b^2 - 4mb + 1 - 4m^2b^2 &= 0\\ -4mb + 1 &= 0.\end{aligned}$$

Therefore, we have $b = \frac{1}{4m}$. Equating the two expressions we derived for b, we see that

$$\frac{1}{4m} = \frac{-m^2}{4} \implies m^3 + 1 = 0.$$

Thus, finding tangent lines to two parabolas reduces to solving a cubic (in the slope of the tangent line). One can change the two parabolas to change the resulting cubic, although we won't show this in full generality here.

Beloch's big idea was able to relate finding the tangents of two parabolas to solving a cubic equation. The solutions of the cubic gives the slopes of the tangent lines, and if we plug them into either of the general tangent equations you can find the y-intercept, and then substituting both values back into $y = mx + b$ form can give you the equation to the tangent line. For our cubic $m^3 + 1 = 0$, there is only one real solution, $m = -1$, and if we substitute -1 into $b = \frac{1}{4m}$, we get $b = \frac{-1}{4}$. Therefore, $y = x^2$ and $x = y^2$ have only one real tangent line $y = -x - \frac{1}{4}$ (and two non real ones).



Figure 9: The Three Tangent Lines of Two Parabolas

7 Constructing a Cubic Length

Here is a more concrete example of constructing the solution to cubic, known as the Beloch Square.

Given two points, A and B, and two lines, r and s, we can construct a square $WXYZ$ where X lies on line r, and Y lies on line s, and sides WX, ZY, or their extensions, pass through A and B.

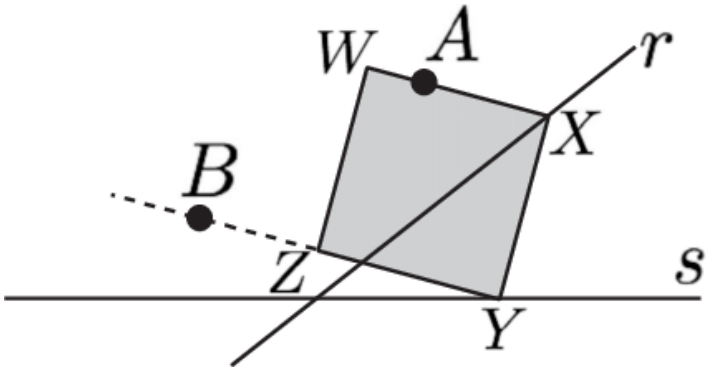


Figure 10: The Beloch Square

To construct this square, we can first construct a line r' parallel to r and twice the distance from point A than r, and do the same thing to construct a line s', using point B and line s. We can then use the Beloch fold to place A onto r' and B onto s', producing points A' and B'. The crease formed from the fold forms a perpendicular bisector to lines AA' and BB', and the intersection form points X and Y. To form the rest of the square, we just have to copy the length XY onto the lines AA' and BB'.

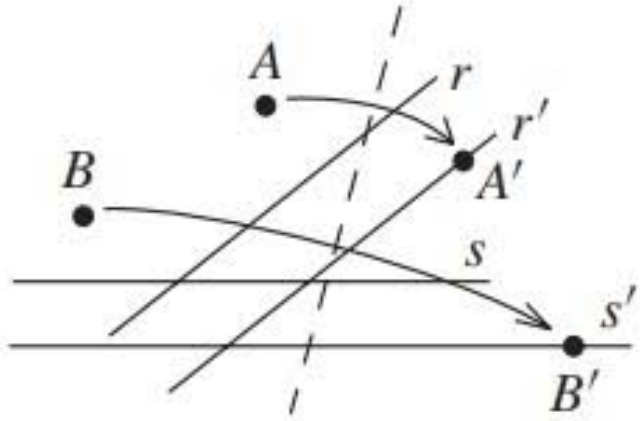


Figure 11: Constructing the Beloch Square

Beloch demonstrated her discovery by using origami to form the length for $\sqrt[3]{2}$. If we place Beloch's square on a coordinate plane, where r is the y-axis, s is the x-axis, O is the origin, A is the point $(-1, 0)$, B is the point $(0, -2)$.

1. $\frac{OX}{OA} = \frac{OY}{OX} = \frac{OB}{OY}$ because the points form similar right triangles.
2. We know $OA = 1$ and $OB = 2$.
3. This leads to the equation $|OX|^3 = 2$.
4. Therefore $OX = 2^{1/3}$ and $X = (0, \sqrt[3]{2})$.

This discovery also found the solution to the classic Greek problem of doubling the volume of cube that required to find $\sqrt[3]{2}$

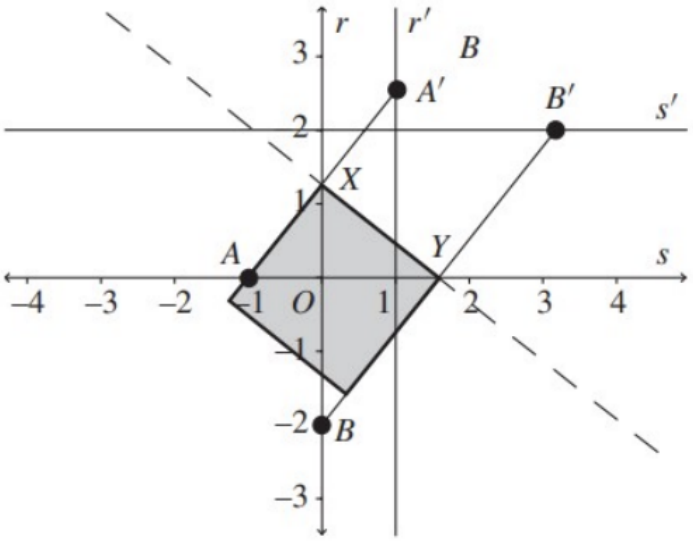


Figure 12: The Beloch Square on a Coordinate Plane

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Gamma Function for Non-Integer Factorials

1 Factorial recurrence

For a natural number n, $n! = 1 \cdot 2 \cdot 3 \cdot ... \cdot n$. We also define $0! = 1$, and with this convention factorials satisfy a distinct recurrence property of the next factorial obtained from the previous by multiplying by the next integer: $(n+1)! = (n+1) \cdot n!$

A natural question is: how could factorial be extended to quantities like $\left(\frac{1}{2}\right)!$ or $1.7!$? The problem of extending factorials was studied by Daniel Bernoulli and Leonhard Euler. Adrien-Marie Legendre introduced the modern symbol $\Gamma(x)$ for the standard extension, the Gamma function. This notation embeds a shift:

$$\Gamma(n+1) = n! \text{ for } n = 1, 2, 3, \ldots \text{ so more precisely, } n! = \Gamma(n+1).$$

This extension of factorials represents a smooth function that satisfies recurrence $\Gamma(x+1) = x\Gamma(x)$, or equivalently, using the exclamation symbol, $x! = x \cdot (x-1)!$.

2 The Gamma Function

Definition

$$\text{For } x > 0,\ \Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt.$$

This integral represents an area under a curve. If this area is finite, then the quantity is a real number. If the area is infinite, then the formula would not define an ordinary finite value.

That is why the convergence of the integral is important. Before using the function defined with the help of the integral, we will check that the area it describes is finite. In two ways, this is an improper integral. First, because the interval begins at $t = 0$ where t^{x-1} is not defined for *every* $x > 0$, and second, because is extends to infinity.

Convergence of the integral

First, we will prove that the integral defining $\Gamma(x)$ converges for every $x > 0$:

$$\int_0^\infty t^{x-1} e^{-t} dt = \int_0^1 t^{x-1} e^{-t} dt + \int_1^\infty t^{x-1} e^{-t} dt.$$

For $t \in (0, 1]$, $e^{-t} \leq 1$, and $0 \leq t^{x-1} e^{-t} \leq t^{x-1}$ holds for the integrand. Since $x > 0$, integral $\int_0^1 t^{x-1} dt = \frac{t^x}{x}\Big|_0^1 = \frac{1}{x}$, which is finite for any given $x > 0$. Thus, the integral will converge near 0.

For the tail integral, we will use the fact that the exponential decay dominates polynomial growth as t approaches infinity, for which reason $t^{x-1} e^{-\frac{t}{2}}$ is bounded on $[1, \infty)$. Denoting the bound C we have for $t \geq 1$:

$$0 \leq t^{x-1} e^{-t} = (t^{x-1} e^{-\frac{t}{2}}) \cdot e^{-\frac{t}{2}} \leq C \cdot e^{-\frac{t}{2}}.$$

Since integral $\int_1^\infty e^{-t/2} dt$ converges, the Gamma function integral also converges on $[1, \infty)$, and the Gamma function is defined; see Figure 1 for its graph and the value of $\Gamma(4)$. Notice that the graph passes through factorial values shifted by one unit: $\Gamma(1) = 0!$, $\Gamma(2) = 1!$, $\Gamma(3) = 2!$, and so on.

Scaling identity



Figure 1: Desmos graph of the Gamma function with marked point (4,6). Values for negative x are shown for illustration only.

For the Beta function, the following equality is important.

$$\text{If } x > 0 \text{ and } s > 0, \text{ then } \int_0^\infty t^{x-1}e^{-st}dt = \frac{\Gamma(x)}{s^x}. \text{ Equivalently, } \Gamma(x) = s^x \int_0^\infty t^{x-1}e^{-st}dt.$$

The one-line proof uses the substitution $u = st$. Then $t = \frac{u}{s}$ and $dt = \frac{du}{s}$, and since $s > 0$, the limits remain 0 and ∞: when $t = 0$, $u = 0$, and when t approaches infinity, u also approaches infinity. Therefore

$$\int_0^\infty t^{x-1}e^{-st}dt = \int_0^\infty \left(\frac{u}{s}\right)^{x-1} e^{-u}\frac{du}{s} = \frac{1}{s^x}\int_0^\infty u^{x-1}e^{-u}du = \frac{\Gamma(x)}{s^x}.$$

The functional equation

The Gamma function is important because it satisfies the relationship similar to factorial recurrence. Using integration by parts, we will prove the functional equation for all $x > 0$

$$\Gamma(x+1) = x\Gamma(x).$$

Let $u(t) = t^x, dv = e^{-t}dt$, and we get $du = xt^{x-1}dt$ with $v(t) = -e^{-t}$.

$$\Gamma(x+1) = \int_0^\infty t^x e^{-t}dt = \int_0^\infty udv = uv\Big|_0^\infty - \int_0^\infty vdu = \left(-t^x e^{-t}\right)\Big|_0^\infty + x\int_0^\infty t^{x-1}e^{-t}dt = 0 + x\Gamma(x).$$

The term $\left(-t^x e^{-t}\right)\Big|_0^\infty = 0$ since for $x > 0$, $\lim\limits_{t\to 0+} t^x e^{-t} = 0$ as $0 < e^{-t} \leq 1$, and $\lim\limits_{t\to\infty} t^x e^{-t} = 0$.

Extension of factorial

Using above functional equation, we can now prove that for all integers $n \geq 0$,

$$\Gamma(n+1) = n!$$

First, $\Gamma(1) = \int_0^\infty e^{-t}dt = 1 = 0!$ Then, by repeatedly using $\Gamma(x+1) = x\Gamma(x)$, we get

$$\begin{aligned}\Gamma(n+1) &= n\Gamma(n) = \\ &= n(n-1)\Gamma(n-1) = \\ &= \cdots = \\ &= n(n-1)\cdot ... \cdot 1 \cdot \Gamma(1) = \\ &= n!\end{aligned}$$

For example, $5! = \Gamma(6) = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot \Gamma(1) = 120$.

The factorial of one-half

To determine the value discussed in the beginning, we will use the functional equation, and first compute the value of $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}} e^{-t}dt = \sqrt{\pi}$. Using the substitution

$$t = u^2, dt = 2udu, \text{ and since } u > 0,\ t^{-\frac{1}{2}} = \frac{1}{u}.$$

since t goes from 0 to infinity, u also goes from 0 to infinity. Thus we have

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty \frac{1}{u} e^{-u^2} 2udu = 2\int_0^\infty e^{-u^2}du = \int_{-\infty}^\infty e^{-u^2}du.$$

This is the Gaussian integral, and we can use Fubini's theorem and polar coordinates with substitution $r^2 = x^2 + y^2, dA = rdrd\theta$:

$$\int_{-\infty}^\infty e^{-u^2}du = \sqrt{\left(\int_{-\infty}^\infty e^{-u^2}du\right)^2} = \sqrt{\left(\int_{-\infty}^\infty e^{-x^2}dx\right)\left(\int_{-\infty}^\infty e^{-y^2}dy\right)} = \sqrt{\int\int_{\mathbb{R}^2} e^{-(x^2+y^2)}dA} =$$

$$= \sqrt{\int_0^{2\pi}\int_0^\infty e^{-r^2} rdrd\theta} = \sqrt{\int_0^{2\pi} d\theta \int_0^\infty e^{-r^2} rdrd\theta}$$

The last substitution is $z = r^2$ with $\frac{1}{2}dz = rdr$:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\int_0^{2\pi} d\theta \int_0^\infty e^{-r^2} rdrd\theta} = \sqrt{2\pi \cdot \frac{1}{2}\int_0^\infty e^{-z}dz} = \sqrt{\pi \cdot 1} = \sqrt{\pi}.$$

Using the functional equation,

$$\left(\frac{1}{2}\right)! = \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

By extension, $\left(-\frac{1}{2}\right)! = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, $\left(\frac{3}{2}\right)! = \Gamma\left(\frac{5}{2}\right) = \frac{3\sqrt{\pi}}{4}$, and $\left(\frac{5}{2}\right)! = \Gamma\left(\frac{7}{2}\right) = \frac{15\sqrt{\pi}}{8}$.

In general, half-integer factorials for $n = 0, 1, 2, ...,$ are

$$\boxed{\left(n - \frac{1}{2}\right)! = \Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)!}{4^n \cdot n!}\sqrt{\pi}.}$$

3 Stirling's Formula for Factorials

Stirling's approximation

The asymptotic approximation for factorials named after James Stirling is

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

So, the ratio of both sides approaches one as n approaches infinity:

$$\lim_{n\to\infty} \frac{n!}{\sqrt{2\pi n}\cdot(n/e)^n} = 1.$$

This approximation is powerful because factorials grow extremely quickly, and Striling's formula provides a simpler expression close to $n!$ for sufficiently large n.

For the proof below, we will represent the formula as follows:
for each n $\exists$ ω_n with $|\omega_n| \leq 1$ such that $n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n}\left(1+\omega_n\sqrt{\frac{2}{\pi n}}\right)$.
This formulation provides the main term and also indicates that the relative error is at most of the order of $\frac{1}{\sqrt{n}}$.
To understand the motivation, we will first notice that for $n! = \Gamma(n+1)$, the integrand $t^n e^{-t}$ has a maximum at $t = n$, because

$$\frac{d}{dt}(t^n e^{-t}) = nt^{n-1}e^{-t} - t^n e^{-t} = (n-t)x^{n-1}e^{-t}, \text{ which has a zero at } t = n.$$

After factoring out the maximum,

$$t^n e^{-t} = \left(\frac{n}{e}\right)^n \left(\frac{t}{n}\right)^n e^{n-t}, \text{ and}$$

$$n! = \Gamma(n+1) = \left(\frac{n}{e}\right)^n \int_0^\infty \left(\frac{t}{n}\right)^n e^{n-t}dt.$$

We choose new variable s so that $\left(\frac{t}{n}\right)^n e^{n-t} = e^{-s^2}$, and taking natural logarithm of both sides gives

$$-s^2 = n\cdot ln\left(\frac{t}{n}\right) + n - t, \text{ or } s^2 = t - n - n\cdot ln\left(\frac{t}{n}\right).$$

Selecting $s < 0$ when $t < n$ and $s > 0$ when $t > n$, with this sign conversion, $t \in (0,\infty)$ corresponds to $s \in (-\infty,\infty)$.

Differentiating $-s^2 = n\cdot ln\left(\frac{t}{n}\right) + n - t$, we get $-2s\frac{ds}{dt} = \frac{n}{t} - 1 = \frac{n-t}{t}$, hence $\frac{dt}{ds} = 2s\frac{t}{t-n}$. Taylor's theorem for logarithm at 1 gives $ln(1+u) = u - \frac{u^2}{2(1+\theta u)^2}$ for some $\theta \in (0,1)$. With $u = \frac{t-n}{n}$, we have:

$$s^2 = t - n - n\cdot ln\left(\frac{t}{n}\right) = \frac{n}{2}\frac{u^2}{(1+\theta u)^2}, \text{ or } s = \frac{\sqrt{n/2}}{\frac{n}{t-n}+\theta}, \text{ and } \frac{t}{t-n} = \frac{n}{t-n} + 1 = \frac{\sqrt{n/2}}{s} + 1 - \theta.$$

Now we apply $\frac{t}{t-n} = \frac{\sqrt{n/2}}{s} + 1 - \theta$ to $\frac{dt}{ds} = 2s\frac{t}{t-n}$:
$\frac{dt}{ds} = 2s\frac{t}{t-n} = \sqrt{2n} + 2s(1-\theta)$ with θ depending on t and therefore s is between 0 and 1.
Returning to integration, we obtain

$$n! = \Gamma(n+1) = \left(\frac{n}{e}\right)^n \int_{-\infty}^{\infty} e^{-s^2}\left[\sqrt{2n} + 2s(1-\theta(s))\right]ds =$$

$$= \left(\frac{n}{e}\right)^n \left(\sqrt{2n}\int_{-\infty}^{\infty} e^{-s^2}ds + \int_{-\infty}^{\infty} e^{-s^2}2s(1-\theta(s))ds\right)$$

This equals to $\left(\frac{n}{e}\right)^n \left(\sqrt{2n\pi} + \int_{-\infty}^{\infty} e^{-s^2} 2s(1-\theta(s))ds\right)$. However, since $0 < \theta(s) < 1$, the absolute value of the remaining integral is less than $\int_{-\infty}^{\infty} e^{-s^2} 2|s|ds = 2$ which means that $\exists \omega_n : |\omega_n| \leq 1$ such that

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \left(1 + \omega_n \sqrt{\frac{2}{\pi n}}\right).$$

Example

For example, using Stirling's approximation, $10! \approx \sqrt{20\pi} \left(\frac{10}{e}\right)^{10} \approx 3.60 \times 10^6$, while the exact value is $3,628,800$.

4 Applications to Geometry, Combinatorics, and Probability

Geometry

The volume of unit sphere in n-dimensional space is $V_n = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)}$.

The unit sphere is the set of points in n dimensions such that:

$$x_1^2 + x_2^2 + ... + x_n^2 \leq 1.$$

For $n = 2$, this is the unit circle area equaling $\frac{\pi^{2/2}}{\Gamma\left(\frac{2}{2}+1\right)} = \pi$.

For $n = 3$, unit sphere volume is $\frac{\pi^{3/2}}{\Gamma\left(\frac{3}{2}+1\right)} = \frac{\pi^{3/2}}{\frac{3\sqrt{\pi}}{4}} = \frac{4\pi}{3}$.

For $n = 4$, $V_4 = \frac{\pi^2}{\Gamma(3)} = \frac{\pi^2}{2}$.

Combinatorics

Striling's formula helps estimate $2n$ choose n for large n:

$$\boxed{\binom{2n}{n} = \frac{(2n)!}{(n!)^2} \sim \frac{4^n}{\sqrt{\pi n}}}.$$

This is because:

$$\frac{(2n)!}{(n!)^2} \sim \frac{\sqrt{4\pi n}\left(\frac{2n}{e}\right)^{2n}}{\left(\sqrt{2\pi n}\left(\frac{n}{e}\right)^n\right)^2} = \frac{4^n}{\sqrt{\pi n}}$$

Probability and Statistics: the Beta function

Beta distribution is an important distribution in statistics and probability theory. It is used, for example, to model online advertisement click-through rates. Beta function is used to normalize this distribution's density function so that area under the density function curve is exactly 1, since total probability cannot exceed 1.

For $\alpha > 0$ and $\beta > 0$ the function is defined as $B(\alpha, \beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx$.

Equivalently, $B(\alpha, \beta) = \int_0^\infty \frac{\tau^{\alpha-1}}{(1+\tau)^{\alpha+\beta}}d\tau$.

The most important connection between the Beta function and the Gamma function is the Beta-Gamma identity

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

This leads to the recurrence of the Beta function: $B(\alpha+1, \beta+1) = \frac{\alpha\beta}{(\alpha+\beta)(\alpha+\beta+1)}B(\alpha, \beta)$.

Example

For example, $B(3,4) = \frac{2 \cdot 3}{5 \cdot 6} \cdot B(2,3) = \frac{1}{5} \cdot \frac{\Gamma(2)\Gamma(3)}{\Gamma(5)} = \frac{1}{5} \cdot \frac{1! \cdot 2!}{4!} = \frac{1 \cdot 2}{5 \cdot 24} = \frac{1}{60}$, which first applies the recurrence and then the definition of the Beta-Gamma identity.

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The Principle of Inclusion and Exclusion: A Deeper Connection to Pascal's Triangle

Sophia DeGregorio

1 Introduction

Whether we realize it or not, we have all used the Principle of Inclusion and Exclusion (PIE) before. In its simplest form, it tells us that the number of items in 2 groups is the sum of the number of items in each group minus the number of items in both groups. For example, we would use it to find the number of students in at least 1 of a math or science class, given that there are 34 students taking math, 23 taking science, and 16 taking both: there are $34 + 23 - 16 = 41$ students in at least 1 of the classes. More formally, this formula can be written as

$$|A \cup B| = |A| + |B| - |A \cap B|$$

for 2 sets A and B, where $|X|$ denotes the cardinality of a set X, or the number of elements of X.

But what if we have more than 2 sets? Or what if we want to know the number of elements in, say, at least 2 of the sets? At least 3? This first question is answered by a widely known formula for PIE for n sets. The answer to the others, however, is less well-known, and it turns out it is closely related to Pascal's Triangle.

2 PIE for Counting the Number of Items in the Union of n Sets

For n sets $A_1, A_2, \ldots, A_n$, PIE states that

$$\begin{aligned}
&|A_1 \cup A_2 \cup \cdots \cup A_n| \\
=&|A_1| + |A_2| + \cdots + |A_n| \\
&- (|A_1 \cap A_2| + |A_1 \cap A_3| + \cdots + |A_{n-1} \cap A_n|) \\
&+ |A_1 \cap A_2 \cap A_3| + \cdots + |A_{n-2} \cap A_{n-1} \cap A_n| \\
&- (|A_1 \cap A_2 \cap A_3 \cap A_4| + \cdots + |A_{n-3} \cap A_{n-2} \cap A_{n-1} \cap A_n|) \\
&\vdots \\
&+ (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|.
\end{aligned}$$

We can write this more succinctly by letting

$$S_k = \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} |A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}|.$$

PIE then states that for $A_1, A_2, \ldots, A_n$,

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{i=1}^{n} (-1)^{i-1} S_i.$$

2.1 Proof of PIE

To prove that this formula accurately counts the cardinality of the union of the n sets $A_1, A_2, \ldots, A_n$, we must ensure that each element in this union is counted exactly once. Take an arbitrary element x that is in m of the n sets for some $1 \leq m \leq n$. We then consider the value of each of the lines in the multiline expression for PIE:

1. x is in $\binom{m}{1}$ of the sets.
2. x is in $\binom{m}{2}$ of the intersections of 2 sets.
3. x is in $\binom{m}{3}$ of the intersections of 3 sets.

   $\vdots$

m. x is in $\binom{m}{m}$ of the intersections of m sets, namely the intersection of the m sets it is in, on the mth line. After this line, x is counted 0 times because it does not appear in any of the subsequent intersections.

We thus aim to show

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots + (-1)^{m-1}\binom{m}{m} = 1.$$

Notice that the left-hand side of this equation is equal to

$$-\binom{m}{0} + \binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots + (-1)^{m-1}\binom{m}{m} + 1,$$

and we have by the Binomial Theorem that

$$\binom{m}{0} - \binom{m}{1} + \binom{m}{2} - \cdots + (-1)^{m}\binom{m}{m} = (1-1)^m = 0,$$

so the left-hand side of the equation equals 1, as desired.

3 Generalizing PIE for Counting Elements in More than 1 Set

Suppose we want to count the number of elements in at least k of n sets $A_1, A_2, \ldots, A_n$. The generalized form of PIE states that this count equals

$$\sum_{i=k}^{n} (-1)^{i-k} \binom{i-1}{k-1} S_i.$$

3.1 Understanding the Formula

To understand this formula, let's investigate an example. Say we have a group of students taking math, science, literature, and history classes. There are 60 students taking math and science; 52 taking math and literature; 49 taking math and history; 30 taking science and literature; 23 taking science and history; 21 taking literature and history; 25 taking math, science, and literature; 14 taking math, science, and history; 12 taking math, literature, and history; 11 taking science, literature, and history; and 9 taking all 4. How would we find the number of students taking at least 2 of the classes?

Starting without using the formula, we can begin writing an expression to count these students. We see that there are $60 + 52 + 49 + 30 + 23 + 21 = 235$ students taking 2 of the classes. However, students in 3 of the classes have been counted $\binom{3}{2} = 3$ times, as they are counted for each 2-element subset of the set of the 3 classes they are taking. We want to count each student exactly once, so we subtract 2 times the number of students in 3 of the classes, which is $25 + 14 + 12 + 11 = 62$, to get $235 - 2(62) = 111$. We have now overcounted the students taking all 4 classes; they have been counted $\binom{4}{2} - 2\binom{4}{3} = -2$ times, so we add back 3 times the number of students in all 4 classes, which is 9, to get $111 + 3(9) = 138$ students taking at least 2 of the classes.

Applying the formula directly yields a count of $\binom{1}{1}(235) - \binom{2}{1}(62) + \binom{3}{1}(9) = 138$ students in at least 2 of the classes, which matches our previous answer. But how do we know this formula always counts all elements correctly?

3.2 Proof of the Generalization of PIE

We must show that each element in at least k of the n sets $A_1, A_2, \ldots, A_n$ is counted exactly once. Consider some element x in r of the n sets, where $k \leq r \leq n$. Since x is counted $\binom{r}{i}$ times by S_i, we want to show

$$\binom{k-1}{k-1}\binom{r}{k} - \binom{k}{k-1}\binom{r}{k+1} + \binom{k+1}{k-1}\binom{r}{k+2} - \cdots + (-1)^{r-k}\binom{r-1}{k-1}\binom{r}{r} = 1.$$

We have omitted the subsequent terms because for $i > r$, $\binom{r}{i} = 0$ (alternatively, x is counted 0 times because it cannot be in the intersection of more than r sets). We now proceed by induction on k.

We have already proven the base case for $k = 1$; it is simply the formula from PIE for calculating $|A_1 \cup A_2 \cup \cdots \cup A_n|$. Then, suppose

$$\binom{j-1}{j-1}\binom{r}{j} - \binom{j}{j-1}\binom{r}{j+1} + \binom{j+1}{j-1}\binom{r}{j+2} - \cdots + (-1)^{r-j}\binom{r-1}{j-1}\binom{r}{r} = 1 \quad (1)$$

for some $j \leq r \leq n$. We aim to show

$$\binom{j}{j}\binom{r}{j+1} - \binom{j+1}{j}\binom{r}{j+2} + \binom{j+2}{j}\binom{r}{j+3} - \cdots + (-1)^{r-j-1}\binom{r-1}{j}\binom{r}{r} = 1. \quad (2)$$

By subtracting equation (2) from equation (1), we see that equation (2) holds if and only if

$$\binom{j-1}{j-1}\binom{r}{j} - \binom{j+1}{j}\binom{r}{j+1} + \binom{j+2}{j}\binom{r}{j+2} - \cdots + (-1)^{r-j}\binom{r}{j}\binom{r}{r} = 0,$$

using $\binom{a}{j-1} + \binom{a}{j} = \binom{a+1}{j}$ by Pascal's Identity for $j \le a \le n-1$. Note that

$$\begin{aligned}
&\binom{a}{b}\binom{b}{c} = \frac{a!}{b!(a-b)!} \cdot \frac{b!}{c!(b-c)!} = \frac{a!(a-c)!}{c!(a-c)!(a-b)!(b-c)!} = \binom{a}{c}\binom{a-c}{b-c} \\
\implies &\binom{j+s}{j}\binom{r}{j+s} = \binom{r}{j}\binom{r-j}{s},
\end{aligned}$$

which we can apply for $1 \le s \le r-j$ to get

$$\begin{aligned}
&\binom{j-1}{j-1}\binom{r}{j} - \binom{j+1}{j}\binom{r}{j+1} + \binom{j+2}{j}\binom{r}{j+2} - \cdots + (-1)^{r-j}\binom{r}{j}\binom{r}{r} \\
=&\binom{j-1}{j-1}\binom{r}{j} - \binom{r}{j}\binom{r-j}{1} + \binom{r}{j}\binom{r-j}{2} - \cdots + (-1)^{r-j}\binom{r}{j}\binom{r-j}{r-j} \\
=&\binom{r}{j}\left(\binom{j-1}{j-1} - \binom{r-j}{1} + \binom{r-j}{2} - \cdots + (-1)^{r-j}\binom{r-j}{r-j}\right) \\
=&\binom{r}{j}\left(\binom{r-j}{0} - \binom{r-j}{1} + \binom{r-j}{2} - \cdots + (-1)^{r-j}\binom{r-j}{r-j}\right) \\
=&\binom{r}{j}(0) \qquad \text{(Binomial Theorem)} \\
=&\,0,
\end{aligned}$$

as desired.

4 Conclusion

We have seen that the number of elements in at least k of n sets $A_1, A_2, \ldots, A_n$ is $\displaystyle\sum_{i=k}^{n} (-1)^{i-k}\binom{i-1}{k-1} S_i$. Notice that this sum takes the expression from PIE for finding $|A_1 \cup A_2 \cup \cdots \cup A_n|$ and multiplies each term by $\binom{i-1}{k-1}$ for $k \le i \le n$. It follows that the coefficient of these terms come from the kth diagonal of Pascal's Triangle, using 1-based indexing. This means that whenever we use PIE to count items with at least k of some n properties, we can find the coefficients by simply taking the terms of the kth diagonal of Pascal's triangle with alternating signs, starting with a positive coefficient.

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Fast Fourier Transform

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Math Survey 2025-26

§1 Fourier Series and Fourier Transforms

The idea of a Fourier Series is to be able to write a function (usually periodic) as an infinite sum of sine and cosine terms, similar to how a Power Series is an infinite sum of polynomial terms. Because sin and cos can be expressed as complex exponentials, particularly:

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$
$$\sin x = \frac{e^{ix} - e^{-ix}}{2i},$$

for our purposes, we will treat a Fourier Series of a function f as an infinite sum of e^{inx} terms for integers n. In other words,

$$f(x) = \sum_{n=-\infty}^{\infty} a_n e^{inx}$$

would be a Fourier series for $f(x)$. The following theorem gives a way to actually calculate the Fourier Coefficients a_n.

Theorem 1.1

For a real number $L > 0$ and a continuous function $f : [-L, L] \to \mathbb{C}$, if there is a Fourier Series for f, i.e.

$$f(x) = \sum_{n=-\infty}^{\infty} a_n e^{in\pi x/L}$$

for some choice of $a_n \in \mathbb{C}$, then for $n \in \mathbb{Z}$,

$$a_n = \frac{1}{2L} \int_{-L}^{L} f(t) e^{-in\pi t/L}\, dt.$$

Proof. Suppose we can write $f(x)$ in the desired form and converges over all of $[-L, L]$. Notice that for $m \neq n$, we have

$$\int_{-L}^{L} e^{i(m-n)t/L}\, dt = \frac{1}{i(m-n)/L} e^{i(m-n)t/L}\Big|_{-L}^{L} = 0.$$

Now consider

$$f(t) = \sum_{n=-\infty}^{\infty} a_n e^{in\pi x/L}.$$

To extract the coefficient a_n, we can multiply both sides by $e^{-int/L}$ and integrate from $-L$ to L. Every term with index not equal to n vanishes due to the above integral and we are left with

$$\int_{-L}^{L} f(t)e^{-in\pi t/L}\,dt = \int_{-L}^{L} a_n\,dt = 2La_n.$$

Dividing by $2L$ gives the desired conclusion. □

Based on this technique we used to extract coefficients for a Fourier Series, we define the Fourier Transform as follows:

Definition 1.2 — Given a complex-valued function $f(x)$ on the real line, the complex-valued function $\hat{f}(\xi)$, the Fourier transform of $f(x)$, is defined as

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2i\pi\xi x}\,dx.$$

Roughly speaking, the Fourier Transform can detect when a certain frequency is present in a function. For example, if $\hat{f}(2)$ is very big (in magnitude) compared to the values of $\hat{f}$ around 2, then there is probably a large similarity of f to $e^{4\pi i}$, which would indicate a frequency of 2. Sometimes we may also denote $\hat{f}$ by $\mathcal{F}(f)$.

§2 Discrete Fourier Transform

Now we define the Discrete Fourier Transform (DFT), which is a discrete analogue of the continuous Fourier Transform, given in Definition 1.2. In the continuous case, we integrate a function f times an exponential factor over the entire real line. For the discrete case, we turn the integral into a sum, and the values of the exponential term we use end up being the nth roots of unity.

Definition 2.1 — Suppose f is a complex-valued function on the set $\{0, 1, \ldots, n-1\}$ for some positive integer n. Then, for integers k, define the *Discrete Fourier Transform* of f as

$$\hat{f}(k) = \sum_{j=0}^{n-1} f(j)e^{-2\pi ijk/n}.$$

Notice in the definition that f is defined on $\{0, 1, \ldots, n-1\}$. A better way to phrase is that f is defined on all integers but its value only depends on its input mod n. In other words, f is defined on the integers mod n, which we denote as $\mathbb{Z}/n\mathbb{Z}$. Note that the same is true of $\hat{f}$ since plugging since the exponential term in $\hat{f}$ cycles as k cycles mod n. We can write the definition for $\hat{f}$ more concisely by using roots of unity. In particular, let $\omega = e^{-2\pi i/n}$, which is a primitive nth root of unity. Then, we have

$$\hat{f}(k) = \sum_{j=0}^{n-1} f(j)\omega^{jk}.$$

When we are using the Discrete Fourier Transform, we are often considering all the values of f at once and trying to compute the values of $\hat{f}$. Thus, it is useful to think of the Discrete Fourier Transform as a matrix whose input is a vector of the values of f and whose output is the a vector of the values of $\hat{f}$. The above formula then tells us that the coefficient in the jth row and kth column

of the matrix is exactly ω^{jk}. In other words, we have the following matrix equation:

$$\begin{pmatrix} \hat{f}(0) \\ \hat{f}(1) \\ \hat{f}(2) \\ \vdots \\ \hat{f}(n-1) \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \dots & \omega^{n-2} \\ 1 & \omega^3 & \omega^6 & \omega^9 & \dots & \omega^{n-3} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{n-2} & \omega^{n-3} & \dots & \omega \end{pmatrix} \begin{pmatrix} f(0) \\ f(1) \\ f(2) \\ \vdots \\ f(n-1) \end{pmatrix}$$

Notice that we can reduce the powers of ω in the matrix mod n since ω is an nth root of unity.

§2.1 Inverse DFT

Given the DFT of a function $f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{C}$, we can apply a simple formula to obtain the original values for f that almost looks like applying the DFT again. In fact, we will prove the following:

Theorem 2.2 (Inverse DFT)

For a function $f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{C}$ and $k \in \mathbb{Z}/n\mathbb{Z}$, we have

$$f(k) = \frac{1}{n} \sum_{j=0}^{n-1} \hat{f}(j)\omega^{-jk}.$$

In other words, this means that $f = \frac{1}{n}\hat{\hat{f}}$, so taking an inverse FFT is essentially equivalent to taking a standard FFT.

Proof. We will consider all k at once and think of the above as a matrix equation. The left hand side is just a vector containing all the values of f. The right hand side is

$$\begin{aligned} &\frac{1}{n} \begin{pmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega^{-1} & \omega^{-2} & \omega^{-3} & \dots & \omega \\ 1 & \omega^{-2} & \omega^{-4} & \omega^{-6} & \dots & \omega^2 \\ 1 & \omega^{-3} & \omega^{-6} & \omega^{-9} & \dots & \omega^3 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{-1} \end{pmatrix} \begin{pmatrix} \hat{f}(0) \\ \hat{f}(1) \\ \hat{f}(2) \\ \vdots \\ \hat{f}(n-1) \end{pmatrix} \\ = &\frac{1}{n} \begin{pmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega^{-1} & \omega^{-2} & \omega^{-3} & \dots & \omega \\ 1 & \omega^{-2} & \omega^{-4} & \omega^{-6} & \dots & \omega^2 \\ 1 & \omega^{-3} & \omega^{-6} & \omega^{-9} & \dots & \omega^3 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{-1} \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \dots & \omega^{n-2} \\ 1 & \omega^3 & \omega^6 & \omega^9 & \dots & \omega^{n-3} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{n-2} & \omega^{n-3} & \dots & \omega \end{pmatrix} \begin{pmatrix} f(0) \\ f(1) \\ f(2) \\ \vdots \\ f(n-1) \end{pmatrix} \end{aligned}$$

We want to show that $\frac{1}{n}$ times the two matrices in the middle turn out to be the identity matrix I. Recall that if we have vectors A and B, the entry in the ith row and jth column of AB is the dot product of the ith row of A and the jth column of B. We have to show that this dot product is 0 for $i \neq j$ and n for $i = j$ (for this particular case of matrices). Note we will be zero-indexing our rows and columns for the proceeding calculations, since everything else is zero-indexed.

For $i \neq j$, the ith row of the left matrix is $(\omega^{i\ell})_\ell$, and the jth column of the right matrix is $(\omega^{-j\ell})_\ell$. The dot product of these is

$$\sum_{\ell=0}^{n-1} \omega^{i\ell}\omega^{-j\ell} = \sum_{\ell=0}^{n-1} \omega^{(i-j)\ell}.$$

Since $i - j \neq 0$, ω^{i-j} is a root of unity other than 1, so when summing all its powers, we get 0. Now we have the other case when $i = j$. Then, the above dot product we computed is just

$$\sum_{\ell=0}^{n-1} 1 = n$$

as desired. □

§3 Fast Fourier Transform

Computing the DFT for an input of size n naively requires $O(n^2)$ operations since each entry of the DFT matrix needs to get multiplied by some value in the input once. However, this matrix has a lot of symmetry which we aren't using. The Fast Fourier Transform is an algorithm that leverages this symmetry to compute the DFT in a more efficient time complexity of $O(n \log n)$.

§3.1 The Cooley-Tukey Algorithm

The key idea of FFT is to reduce the computation of an N-point FFT into two computations of $\frac{N}{2}$-point FFT's. For now, we will assume that N is even for convenience. Then, if we want to take the FFT of a function f which is defined on $\mathbb{Z}/N\mathbb{Z}$, notice that we can split up the even and odd terms in the definition of $\hat{f}$ as follows:

$$\begin{aligned}\hat{f}(k) &= \sum_{j=0}^{N-1} f(j)\omega_N^{jk} \\ &= \sum_{j=0}^{\frac{N}{2}-1} f(2j)\omega_N^{2jk} + \sum_{j=0}^{\frac{N}{2}-1} f(2j+1)\omega_N^{(2j+1)k}.\end{aligned}$$

Then, notice that $\omega_N^2 = \omega_{N/2}$, so this is equal to

$$\sum_{j=0}^{\frac{N}{2}-1} f(2j)\omega_{N/2}^{jk} + \omega_N^k \sum_{j=0}^{\frac{N}{2}-1} f(2j+1)\omega_{N/2}^{jk}.$$

These two sums are essentially $\frac{N}{2}$-point FFT's–one of them for the vector $[f(2j)]$ and the other for the vector $[f(2j+1)]$. This algorithm works by recursively computing these smaller FFT's using the same subdividing strategy. In the simplest case, we have that N is a power of 2. We would like to analyze the efficiency of this algorithm in this case. We will assume that all additions and multiplications take the same amount of time.

Suppose that $N = 2^n$ and let a_n be the number of operations necessary for a 2^n-point FFT. To compute this FFT, we must compute two 2^{n-1}-point FFT's as well as the values of $1, \omega_N, \ldots, \omega_N^{N-1}$. Then, to compute each value $\hat{f}(k)$, we just have to add a pre-computed value of the FFT of $[f(2j)]$,

a pre-computed value of ω_N^k, and a pre-computed value of the FFT of $[f(2j+1)]$. Overall, the number of operations used is $2a_{n-1} + cN = 2a_{n-1} + c \cdot 2^n$ where c is some constant.

We thus have the recurrence $a_n = 2a_{n-1} + c \cdot 2^n$. We can actually solve this recurrence. Let $A(x)$ be the generating function for the sequence a_n. In other words,

$$A(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Taking out the first term, we can apply the recurrence to get

$$\begin{aligned} A(x) &= a_0 + \sum_{n=1}^{\infty} (2a_{n-1}x^n + c \cdot 2^n x^n) \\ &= a_0 + x\sum_{n=0}^{\infty} 2a_n x^n + c\sum_{n=1}^{\infty} (2x)^n \\ &= a_0 + 2xA(x) + \frac{2cx}{1-2x}. \end{aligned}$$

We can solve for $A(x)$ to get

$$A(x) = \frac{a_0}{1-2x} + \frac{2cx}{(1-2x)^2}.$$

From this formula, we can extract the value of a_n by expanding out the geometric series. The contribution of the term $\frac{a_0}{1-2x}$ to the coefficient of x^n in $A(x)$ is exactly $a_0 \cdot 2^n$, and it is not too hard to check that the contribution of the second term is $c(n+1) \cdot 2^n$. Therefore, we have $a_n = c(n+1) \cdot 2^n + a_0 \cdot 2^n$. In terms of N, which we are measuring the time complexity based off, this is equal to $cN(\log_2 N + 1) + a_0 N$. Thus, the time complexity of the Cooley-Tukey Algorithm is on the order of $O(N \log N)$. This means that multiplying by the DFT matrix can be done much more efficiently than the naive $O(n^2)$ solution. We will more thoroughly discuss the implications of this algorithm and its applications in the modern-day in later sections.

§4 Non-powers of 2

One limitation of the Cooley-Tukey Algorithm is that it only works for powers of 2. The algorithm can also be used to decompose a N-point FFT into N_1 N_2-point FFT's where $N = N_1 N_2$. However, for this decomposition to be effective, it is best for N to have only small prime factors. This way, one can decompose the N-point FFT into small pieces eventually. However, we would like to come up with an FFT algorithm that works for any N, especially large primes N. We will accomplish this by using Bluestein's Algorithm and the Convolution Theorem.

§4.1 Convolution Theorem

One tool we will need is the Convolution Theorem. This is an identity we can use for both continuous and discrete Fourier Transforms, so we introduce both cases. Suppose we have two sufficiently nice functions $u, v : \mathbb{R} \to \mathbb{C}$ and define their convolution r as the function

$$r(x) = \int_{-\infty}^{\infty} u(t)v(x-t)\, dt.$$

Notice that if we switch the order and u and v, r stays the same, which we can prove by making a substitution in the integral. We will also use the notation $r = u * v$.

Theorem 4.1 (Convolution Theorem)
Given the definitions earlier, we have $\hat{r}(\xi) = \hat{u}(\xi)\hat{v}(\xi)$ for all $\xi \in \mathbb{R}$. In other words, $\widehat{u * v} = \hat{u} \cdot \hat{v}$.

Proof. We can just carry out the computation:

$$\begin{aligned}
\hat{r}(\xi) &= \int_{-\infty}^{\infty} r(t) e^{-2\pi i \xi t}\, dt \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(q) v(t-q)\, dq\, e^{-2\pi i \xi t}\, dt \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(q) v(t-q) e^{-2\pi i \xi q} e^{-2\pi i \xi (t-q)}\, dt\, dq \\
&= \int_{-\infty}^{\infty} u(q) e^{-2\pi i \xi q} \left(\int_{-\infty}^{\infty} v(t-q) e^{-\pi i \xi (t-q)}\, dt \right) dq \\
&= \int_{-\infty}^{\infty} u(q) e^{-2\pi i \xi q} \hat{v}(\xi)\, dq \\
&= \hat{u}(\xi)\hat{v}(\xi).
\end{aligned}$$

□

This can be generalized to convolutions of discrete functions using the Discrete Fourier Transform. In particular, let f, g be functions defined on $\mathbb{Z}/n\mathbb{Z}$, and define their convolution h as

$$h(k) = \sum_{j=0}^{n-1} f(j) g(k-j).$$

Notice that the argument of g will sometimes be negative, but that is fine since f and g are just defined mod n. The discrete version of the Convolution Theorem states that $H(k) = F(k)G(k)$ where F, G, and H are the DFT's of f, g, and h respectively. This is not too hard to prove, and is quite similar to the continuous case:

$$\begin{aligned}
H(k) &= \sum_{j=0}^{n-1} h(j) \omega^{jk} \\
&= \sum_{j=0}^{n-1} \sum_{\ell=0}^{n-1} f(\ell) g(j-\ell) \omega^{jk} \\
&= \sum_{\ell=0}^{n-1} f(\ell) \omega^{\ell k} \sum_{j=0}^{n-1} g(j-\ell) \omega^{(j-\ell)k} \\
&= \sum_{\ell=0}^{n-1} f(\ell) \omega^{\ell k} G(k) \\
&= F(k)G(k).
\end{aligned}$$

A useful way to write the convolution theorem for computational purposes is:

Theorem 4.2

Let $f, g : \mathbb{Z}/n\mathbb{Z} \to \mathbb{C}$ be functions defined mod n. Then, $f * g = \mathcal{F}^{-1}(\mathcal{F}(f)\mathcal{F}(g))$.

Therefore, rather than computing convolutions directly using the definition, we can just perform two FFT's and one inverse FFT.

§4.2 Bluestein's Algorithm

One limitation to the Cooley-Tukey Theorem is that it only works for composite values of N, and in the case above, we only considered it when N is a power of 2. The worst case is when N is a large prime number, and we can't factor it and simplify the computation of the FFT using the Cooley-Tukey Theorem. Bluestein's Algorithm gets around this by reducing general FFT's to FFT's of power-of-2 order. The main idea is to write the DFT as a convolution of other sequences and then proceed to use Theorem 4.2 and the Cooley-Tukey Algorithm to finish.

In particular, suppose we are given a function $f : \mathbb{Z}/N\mathbb{Z} \to \mathbb{C}$ and let ω be an Nth root of unity. We are trying to compute

$$\hat{f}(k) = \sum_{j=0}^{n-1} f(j)\omega^{jk}.$$

Now observe that

$$jk = \frac{-(k-j)^2}{2} + \frac{j^2}{2} + \frac{k^2}{2}.$$

This allows us to write the desired DFT as

$$\begin{aligned}\hat{f}(k) &= \sum_{j=0}^{n-1} f(j)\omega^{jk} \\ &= \omega^{\frac{k^2}{2}} \sum_{j=0}^{n-1} \left(f(j)\omega^{\frac{j^2}{2}} \right) \omega^{-\frac{(k-j)^2}{2}}.\end{aligned}$$

Notice that we have now written $\hat{f}(k)$ as a convolution of two sequences, which we will define as

$$a_m = f(m)\omega^{\frac{m^2}{2}}$$

and

$$b_m = \omega^{-\frac{m^2}{2}}.$$

In other words, we have

$$\hat{f}(k) = \omega^{\frac{k^2}{2}} \sum_{j=0}^{n-1} a_j b_{k-j} = \omega^{\frac{k^2}{2}} (a * b)_j.$$

where $a * b$ is the convolution of a and b. Thus, if we can efficiently compute the sequence $a * b$, we can efficiently compute the Fourier transform of f. We can use Theorem 4.2 to do this, turning the operation of convolution into a few FFT's. However, the problem is that these FFT's don't operate on a vector with a power-of-2 number of elements. We can solve this problem by zero-padding. As an example, suppose we were trying to find the convolution of two sequences a_k and b_k which both only have 5 terms.

a_1	a_2	a_3	a_4	a_5
b_1	b_2	b_3	b_4	b_5

In the diagram above, we have the sequences a_k on top of b_k and the terms which are color coded will be multiplied together in the calculation of $(a * b)_1$. Note that for example $a_1 b_5$ is included because we add the indices mod 5. We can pad both sequences with 0's in the following way to have the same calculation:

a_1	a_2	a_3	a_4	a_5	0	0	0
0	0	0	b_1	b_2	b_3	b_4	b_5

Now we are calculating $(a * b)_1$ where we consider a and b to be the 8-term sequences as above. Note that now $a_1 b_5$ is still included in the calculation since a_1 is the 1st term of a and b_5 is now the 8th term of b, and $1 + 8 \equiv 1 \pmod 9$. Because of the 0's we added, we always get the same pairs of terms as we did in the 5-term case, even if we were to compute a different term of $a * b$. Thus, if we want to compute the convolution of two sequences of arbitrary length, it suffices to do this zero-padding to a power-of-two length, and then apply Theorem 4.2 with the Cooley-Tukey Algorithm to efficiently perform FFT's. Going back, this allows us to perform arbitrary-length DFT's in $O(n \log n)$ time complexity, albeit with a considerably larger constant factor than for powers of 2.

§5 Implementation

In this section, we will show an implementation of the FFT and related algorithms we have discussed. Our input to the FFT will be an array of numbers; for the Cooley-Tukey Algorithm, we must assume that the length of the array is a power of 2. In the Cooley-Tukey Algorithm, we split up these values into the even and odd indices, perform the FFT on both parts, and then take a linear combination of the results for our final answer. This is essentially the structure that our code will take.

```
import numpy as np
def FFT(x):
    N = len(x)

    if N == 1:
        return x
    else:
        even = FFT(x[::2])
        odd = FFT(x[1::2])
        factor = np.exp(-2j*np.pi*np.arange(N)/ N)

        X = np.concatenate(
            [even+factor[:int(N/2)]*odd,
             even+factor[int(N/2):]*odd])
        return X

print(FFT([1, 2, 1, 1]))
```

Clearly, this can be expanded to an implementation of Bluestein's algorithm for arbitrary length FFT's through the use of convolutions we explained in the previous section.

In python, the FFT can be used easily using the `numpy` package. In particular, `numpy` allows for easy use of many math-related commands, using special `numpy` arrays. To use the FFT, you must use the `numpy.fft.fft` command on an array constructed using the `np.array` function. Here is an example:

```python
import numpy as np
x = np.array([1, -1, 2, 6, 4])
print(np.fft.fft(x))

# output: [12.+0.j, -4.54508497+7.10642359j,
# 1.04508497-0.8652998j, 1.04508497+0.8652998j,
# -4.54508497-7.10642359j]
```

`numpy.fft` also includes other useful commands such as `numpy.fft.ifft` (Inverse FFT), `numpy.fft.fft2` (2-dimensional FFT), and `numpy.fft.fftn` (n-dimensional FFT). Furthermore, it is important to remember that in python, such as in the output provided in the above example code, the imaginary unit i is displayed using the letter `j`. Regarding the actual output, we can see that the first element of FFT output is the sum of all the elements in `x` since it is really the sum of each element of the array times $\omega^0 = 1$, where $\omega = e^{-\frac{2\pi i}{5}}$.

§6 Multiplication

§6.1 Polynomial Multiplication

Suppose we have a polynomial $A(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$. We will denote its coefficient vector, containing the a_i's as just A. If $\omega = e^{-\frac{2\pi i}{n}}$ as a usual, we have $\hat{A}(0) = a_n\omega^0 + a_{n-1}\omega^0 + \cdots + a_0 = A(1) = A(\omega^0)$. In fact, we have that $\hat{A}(k) = A(\omega^k)$ in general. Now, suppose we have another polynomial $B(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_0$. Consider taking the FFT of the product $A(x)B(x)$. The result will consist of values $A(\omega^k)B(\omega^k)$ for a root of unity ω. However, we can find all the values of $A(\omega^k)$ and $B(\omega^k)$ fast by FFT'ing them first. From there, we can take the IFFT of the resulting vector.

One thing we must do first is to zero-pad the coefficients for A and B. If A has degree n and B has degree m, then we will zero-pad both vectors so they have a total length L where L is a power of 2 greater than or equal to $m + n$. We want L to be a power of 2 since FFT's are fastest when the length of the input is a power of 2 (using Cooley-Tukey). Furthermore, we want $L \geq m + n$ since the length of the coefficient vector AB will be $m + n$. Then, $\mathcal{F}(A)$ is a vector consisting of the values of $A(\omega^k)$ where $\omega = e^{-\frac{2\pi i}{L}}$, and similarly for $\mathcal{F}(B)$. Then, notice that $\mathcal{F}(AB)$ consists of the values of $A(\omega^k)B(\omega^k)$, so $\mathcal{F}(AB) = \mathcal{F}(A) \cdot \mathcal{F}(B)$, where the dot represents the pairwise products of two vectors. Finally, taking the IFFT, we have $AB = \mathcal{F}^{-1}(\mathcal{F}(A) \cdot \mathcal{F}(B))$. All these operations can be computed in $O(n \log n)$ time, so we have an algorithm for polynomial multiplication that is $O(n \log n)$. Note that the naive algorithm with distributing all the coefficients is $O(n^2)$, so this is a significant improvement.

§6.2 Schönhage-Strassen Algorithm

The Fast Fourier Transform can be used to construct an efficient algorithm for multiplication of large numbers, known as the Schönhage-Strassen Algorithm. First of all, note that the ordinary algorithm for multiplying two n-digit numbers (using column multiplication) has time complexity $O(n^2)$ because for each digit in the first number, we must multiply it by each digit in the second number. The first algorithm with a faster time complexity for this problem was Karatsuba's Algorithm, which divides the multiplication of two n-digit numbers into 3 multiplications of two $n/2$-digit numbers (rather than 4). This achieves a time complexity of $O(n^{\log_2 3}) \approx O(n^{1.58})$ multiplications of 1-digit numbers.

The Toom-Cook Algorithm uses a similar approach and for example, can reach a time complexity of $O(n^{\log_3 5}) \approx O(n^{1.46})$.

On the other hand, the Schönhage-Strassen Algorithm uses the FFT to achieve a time complexity of $O(n \log n \log\log n)$. In this algorithm, we calculate the product of two numbers a and b mod a number of the form $2^N + 1$.

In particular, we want to choose some large N where $N = 2^k \cdot m$. Then, we split a and b into $D = 2^k$ blocks of m bits each. These D blocks for a and b can be interpreted as vectors of length D, and multiplying a and b essentially comprises of taking a convolution of these two vectors, so we would like to use the Convolution Theorem to do so. Computing the FFT of these two vectors will require a Dth primitive root. It is not too hard to see that $g = 2^{2m}$ is a Dth primitive root of unity mod $2^N + 1$, so we will use this for the FFT used in the Convolution Theorem, which will give us the final product.

First of all, for this setup to work, we must have $ab < 2^N + 1$. From there, the FFT's used for the Convolution Theorem are of length 2^k, so the time complexity should be $O(k \cdot 2^k)$. However, we must remember that the FFT requires us to multiply many m-digit numbers, so the time complexity of multiplication can't be treated as constant (after all, we are describing a multiplication algorithm). While we won't go over the details here, by finding the optimal value of k and m, this algorithm works with time complexity $O(n \log n \log\log n)$ where n is the number of digits.

The Schönhage-Strassen only starts to get faster than other previously mentioned algorithms like Tom-Cook for significantly large numbers, but they are not unfathomably large (we start seeing major improvements with 100000+ digits). In fact, this algorithm has been used in the Great Internet Mersenne Prime Search and calculating approximations of π. There is another theoretical algorithm for multiplication with a time complexity of $O(n \log n)$ known as the Harvey-Hoeven Algorithm

§7 Signal Processing

The Discrete Fourier Transform, as the name suggests, is just a discrete version of the continuous Fourier Transform. This makes it perfect for working with digital data. Since running a Fourier Transform on a wave can give you information about its component frequencies, the DFT can do this for digital data. One particular example of this is essential for digital communications.

§7.1 Orthogonal Frequency-Division Multiplexing

Orthogonal Frequency-Division Multiplexing (OFDM) is a crucial technology that allows for wireless communications as we know it. When sending a wireless signal, one must transmit a wave or collection of waves through the air, known as carrier waves. Suppose we are trying to send a series of bits in this way. One way we can do this is by fixing some frequency for the wave, and for every 1 in the signal, the wave has a certain phase, and for every 0, the wave has a certain different phase. This method is not very favorable because every bit of information has to be sent separately in time, and there is no way to send them simultaneously.

One solution to this is to send data that is encoded by the frequency of carrier waves, leading us to Frequency-Divison Multiplexing. Suppose we wanted to send a sine wave given by $\sin(2x)$. This is a linear combination of e^{2ix} and e^{-2ix}, so the Fourier Transform will have peaks at $2i$ and $-2i$. In other words, these are the frequencies inherent to the sine function.

However, in reality, we would only be sending a sine wave for a limited amount of time, and when applying the Fourier Transform on this signal, the most apparent frequencies are still $\pm 2i$, but there are still nonzero contributions from other frequencies.

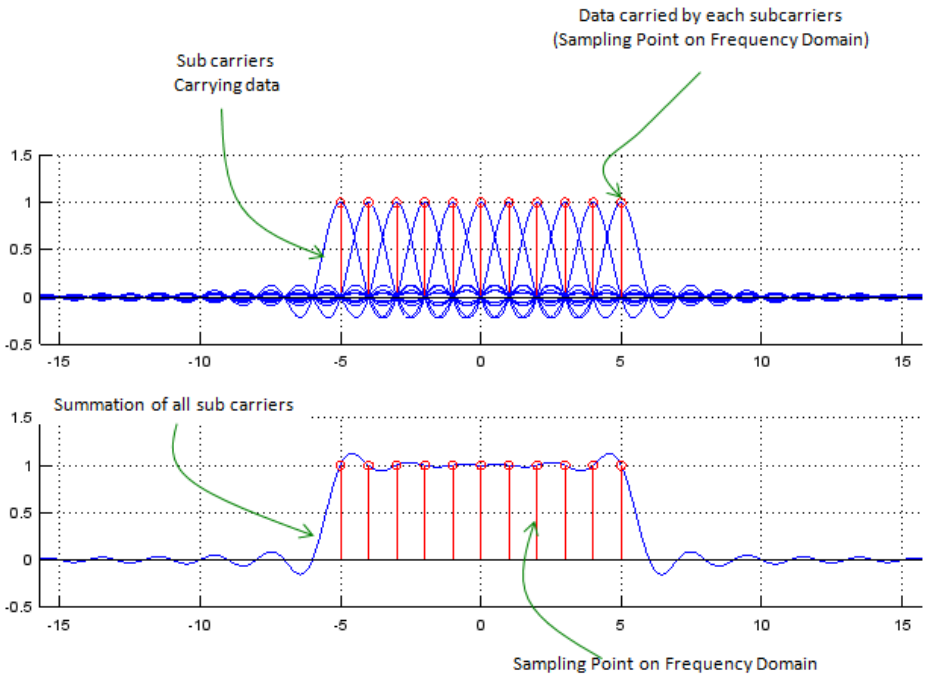


Figure 1: The range of frequencies used by different OFDM subcarriers

Now suppose we want to transmit data using different frequencies. For the sender, this would just consist of adding waves of the particular frequencies. The receiver of the signal can then use the Fourier Transform to extract the frequencies themselves. In practice, this will be done using the FFT since all digital waves are received as discrete signals. However, as mentioned earlier, the Fourier transforms of the transmitted waves are not just singular peaks, so the presence of one wave could possibly interfere with the presence of another if they share frequencies in the Fourier Transform.

As a result, the distinct frequencies used in this type of signal transmission have to be spaced out further apart so that they don't interfere with one another. This increases the total bandwidth, the overall range of frequencies that a device can transmit. Bandwidth is costly, so we would like to have a way to decrease it.

Figure 7.1 shows how the frequencies used in OFDM are arranged. The top graph shows different frequencies that can be transmitted (each by a different subcarrier), and they overlap. However, when one frequency is at its peak, all others are 0, so in the transmitted signal, each frequency can still be detected independently from the others. Furthermore, keep in mind that when the transmitted frequencies are selected, they have to be converted from the frequency domain to the time domain, which requires an inverse FFT. At the receiving end, the signal just needs to go through a regular FFT.

Compared to other types of digital signals, OFDM is signals are more resistant from changes due to the environment (such as bouncing off objects) since such environmental factors typically don't change the frequencies of waves. This has caused OFDM to be crucial for technologies like Wi-Fi and 5G, which we use every day.

§8 Data Compression

§8.1 Multi-Dimensional FFT

So far, we have been considering FFT's of vectors of complex numbers. First, we will generalize the DFT to multiple dimensions. For example, let us consider a 2-dimensional DFT. In this case, we consider a 2-variable function $f(x, y) : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \to \mathbb{C}$, and we have

$$\hat{f}(k_1, k_2) = \sum_{i \in \mathbb{Z}/n\mathbb{Z}} \sum_{j \in \mathbb{Z}/m\mathbb{Z}} f(i, j)\omega_n^{ik_1}\omega_m^{jk_2}$$

where $\omega_n = e^{-2\pi i/n}$ and $\omega_m = e^{-2\pi i/m}$ as usual. The general multi-dimensional case is analogous to this, but with a lot more summations.

Definition 8.1 — Let $f(x_1, \ldots, x_\ell) : \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_\ell\mathbb{Z} \to \mathbb{C}$ be a function. We define its Disrete Fourier Transform as the function

$$\hat{f}(k_1, \ldots, m_\ell) = \sum_{i_1 \in \mathbb{Z}/n_1\mathbb{Z}} \cdots \sum_{i_\ell \in \mathbb{Z}/n_\ell\mathbb{Z}} f(i_1, \ldots, i_\ell)\omega_{n_1}^{i_1k_1} \ldots \omega_{n_\ell}^{i_\ell k_\ell}.$$

It turns out that performing a 2-dimensional DFT is the same as performing a DFT of the columns first and then on the rows, as stated in the following theorem. An analogous statement is true for n-dimensional DFT's

Theorem 8.2

Consider a function $f(x, y) : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \to \mathbb{C}$. Then, the 2D array of values given by $[\hat{f}(k_1, k_2)]$ is the same as the result of applying a DFT to every row of $[f(x, y)]$ and then to every column, or vice versa.

§8.2 Image Compression

The JPEG Image Compression algorithm can typically compress the size of an image by 10 times with virtually no loss to the image quality. The simple idea of how this compression is by using a 2-dimensional FFT on the image, with a couple of nuances.

First of all, the JPEG algorithm actually uses the DCT, or the Discrete Cosine Transform, which is similar to the DFT as both convert the information of the pixels into the frequency domain. From here, the JPEG algorithm essentially throws away high frequency information in the data, which don't really effect our eyes' perception of the image. For example, high frequency data might include subtle changes in the color of the background whereas lower frequencies might account for the boundaries of different objects. This information is discarded by being stored less precisely the lower frequency information.

JPEG compression divides an image into 8×8 squares of pixels and for each square, performs the DCT on the Y, C_b, and C_r components. Roughly speaking, the Y component is the greyscale version of the image while the C_b and C_r components are the blue and red chrominance components, as shown in Figure 8.2.

Figure 8.2 shows the 64 fundamental frequencies that make up 8×8 images. Each cell of the grid represents a different pair of frequencies, one for each dimension of the grid. The frequencies on the top and left sides are more significant as they carry less noise, so the JPEG algorithm prioritizes these more.

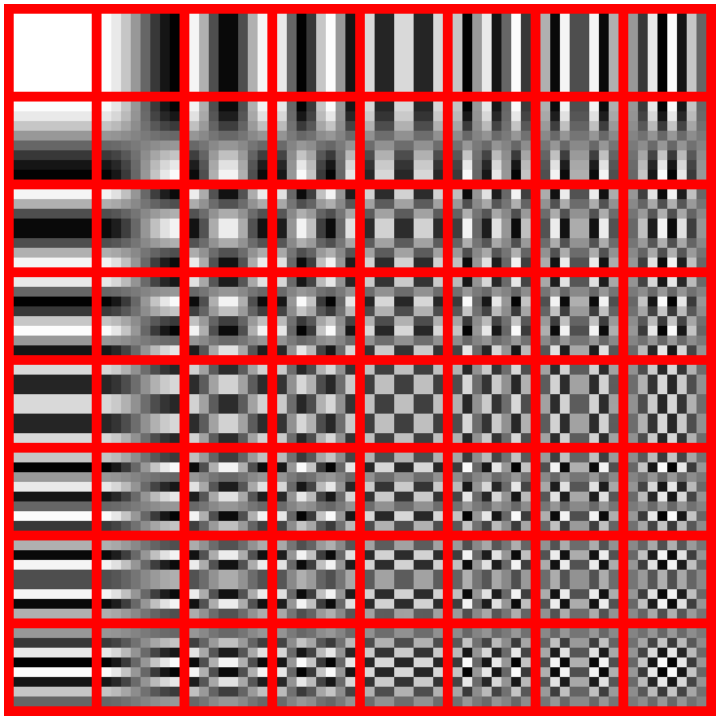


Figure 2: The 64 fundamental frequencies on an 8×8 grid of pixels

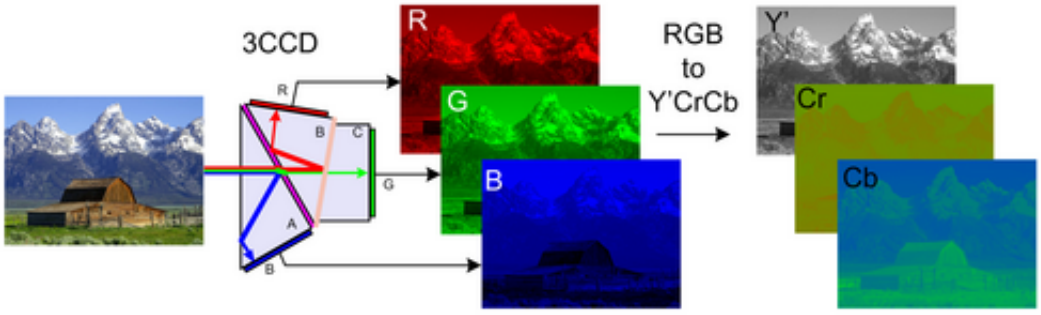


Figure 3: An image split into its Y, C_b, and C_r components

§9 Conclusion

Overall, the FFT takes advantage of the symmetry present in roots of unity and the calculation of a DFT, allowing for a very large speed improvement from $O(n^2)$ to $O(n \log n)$. Because the DFT is such an essential operation in signal processing, so many modern technologies can only function because of the speedup provided by the FFT. If we were stuck with a $O(n^2)$ complexity, we wouldn't have the luxury of fast wireless communications or near-flawless data compression. As Gilbert Strang, a mathematics professor at MIT said, the FFT "is the most important numerical algorithm of our lifetime."

Who's Who and What's What?

Steven Breger



Figure 1: Isn't Gemini amazing?

1 The digital ID problem

How does a police officer know who you are? Well, that's easy. You show them an ID. In fact, it's how you prove your identity to almost anyone, whether that's a bank, a bar, or an airport.

But, when your computer connects to a server, how does it truly know who that server is? ID cards prevent people from getting away with lying about their identity, but how does your computer verify the identity of the server without just taking its word? Moreover, how does the server verify your computers identity without taking its word?

Just like police don't want to arrest the wrong people, banks don't want to give others access to your money, bars don't want underage people drinking, and airports want to verify who you are, servers and clients need to know that the connection they have established is legitimate and is with whatever entity they thought it was. If these measures don't exist, a hacker could just intercept a connection and steal sensitive data.

So, now that we see the benefits of creating a digital version of identity verification, the question becomes how. How can this be achieved? Well, your first thought might be to generate a unique ID card for each person. We could generate a random 2048-bit string and have that be an ID (something like 10111011101000110101111110000111...). Although this does work, there's a major problem. Unlike a physical ID card, which is unique and has anti-counterfeit protections, as soon as you give this binary string to someone else, they could pretend to be you by copy and pasting this string to someone else. That's why for digital identity verification we have to be a bit more clever. Take a moment to make a rough guess for how we could implement digital ID cards before moving on.

2 More Intuition

Note that from here on out, we will switch to using the word "key" instead of "ID" as that is the more common term within computer science, but that keys can essentially be thought of as ID cards. Okay, have you thought about it? Well, the answer is that we need a private key. Something that only you have and that you can't give to anyone else. This, however, gives rise to another problem. How can anyone verify your identity if you can't share your private key with others? To solve this problem, we would generate a corresponding public key that goes along with the private key. Once you share your public key with someone, they would then hopefully somehow be able to verify that you have the corresponding private key. Think of the public key as saying "I claim my identity to be X," after which whoever is verifying your identity can test that you really are X by verifying that you have the corresponding private key.

The careful reader may have caught a few problems with the above proposal. Namely,

1. Whoever is verifying your identity has to know that the public key you provided is actually the key corresponding to you. Otherwise, a hacker could just generate their own public-private key pair, claim that their public key corresponds to you, and steal your identity.

2. It must be impossible to obtain the private key from the public key. Moreover, it must be impossible to obtain the private key while verifying that you have the private key. So, in essence, you have to prove that you have the private key without actually letting anyone get any information about it.

The first of these problems will be tackled later on and the second is solved by the nature of the math that we will employ to implement this algorithm.

3 The Math Under the Hood

Here's the key mathematical idea that we will use to design our algorithm: the discrete root problem is really hard to solve. In other words, given a, r, and n, it is very difficult to find a g (assuming that such a g exists), such that $g^a \equiv r \pmod{n}$.

It's pretty clear how we can begin to encrypt data with this. Given data d, we can encrypt it with public key (a, n) by computing $d' \equiv d^a \pmod{n}$. Someone seeing d' wouldn't easily be able to compute d by taking the ath root. It's computationally infeasible: taking an ath is harder than raising something to the ath power for positive integers a.

Okay, now that the original data is impossible to retrieve from the encrypted data with the public key, let's try to find a private key that can decrypt our data. It seems like the most logical way to do this is to take another exponent. So, we can try to compute a b such that

$$\begin{aligned} & d'^b \equiv d \pmod{n} \\ \iff & d^{ab} \equiv d^1 \pmod{n} \end{aligned}$$

If we can guarantee that $\varphi(n) \mid ab - 1$, then the above equation will be true by Euler's totient theorem. Looking at this condition in light of modular arithmetic, it is sufficient that the private key involves the modular inverse of a modulo $\varphi(n)$. So, our private key becomes (b, n) where $b \equiv a^{-1} \pmod{\varphi(n)}$.

Aside from the issue of everything having to be relatively prime and "nice" for these modular inverses and exponentials to work, we also have another problem. If someone has the public key a and $\varphi(n)$, then they would be able to compute b by using the Euclidean algorithm to compute a modular inverse. So, it follows that we must make $\varphi(n)$ impossible to obtain from n (since n must be in the public key). If $n = p$ for a prime p, obtaining $\varphi(n)$ is easy, just compute $p - 1$. However, if we simply let $n = pq$ for primes p and q, $\varphi(n) = (p-1)(q-1) = pq - p - q + 1 = n - p - q + 1$. Aha! Although this is easy for us to compute if we generate p and q first, in order to find $\varphi(n)$ an attacker must first find p and q just from n. In other words, an attacker would have to factor n. But, for large n this is incredibly slow and computationally infeasible!

Notice that our decryption relied on the fact that $d^{ab} \equiv d^1 \pmod{n}$ when $\varphi(n) \mid ab - 1$. Usually, we only consider these types of statements for d relatively prime to n. However, it turns out that because this

equation has a factor of d on both sides, it is actually true for any value of d when n is square-free). To see this, let $n = p_1 p_2 \cdots p_k$ for primes p_i. By Chinese remainder theorem, our equation is true if and only if

$$d^{ab} \equiv d^1 \pmod{p_i}$$

for each i. Since $p_i - 1 \mid \varphi(n)$ and $\varphi(n) \mid ab - 1$, we can let $ab - 1 = t(p_i - 1)$ for some integer t. Since d is not relatively prime to n, there are two cases for each i. Either $p_i \mid d$, which makes the above equation obviously true, or $p_1 \nmid d$, in which case Fermat's little theorem tells us that

$$d^{ab} \equiv d^{ab-1} d \equiv d^{t(p_i-1)} d \equiv d \pmod{p_i}$$

As mentioned before, by the Chinese remainder theorem, since our congress is true modulo each p_i, it is true modulo n.

Of course, when d is not relatively prime to n, the set of powers of d modulo n is much smaller and thus the encryption will be slightly less secure. This doesn't happen to often though and there is still a large enough set of possibilities, so it's safe to ignore.

Last, we must solve the problem of a having to be relatively prime to $\varphi(n)$. To solve this, we can just always set $a = 2^{16} + 1$. Then, when generating n, if on the slim chance a is not relatively prime to $\varphi(n)$, we can just regenerate n until it is. Notice that it doesn't matter if everyone in the world chooses the same value of a for their public key, because the real security comes from the large random value of n and the fact that it is impossible to find b, the inverse of a, modulo a specific n.

4 The RSA Algorithm

To sum everything up, here is the condensed version of what we created above. This algorithm is actually known as RSA, one of the most famous and useful cyber security algorithms that we have invented.

Key Generation

1. Generate two large random prime numbers p and q. Note: to see how to generate large primes, look at the appendix.
2. Compute $n = pq$ and $\varphi(n) = (p-1)(q-1)$.
3. Pick a to be a somewhat small prime number such as $2^{16} + 1$. If a is not relatively prime to n, repeat steps 1 and 2.
4. Compute b, the inverse of a modulo $\varphi(n)$. This can be done using the Euclidean algorithm since we have both of these values.
5. Share your public key in the form (a, n) and keep your private key in the form (b, n).

Encryption

1. Convert your data into an integer, d. If your data is really large, you can split it up into sections and encode each section as an integer. For example, ABZ might become the integer 010226 if we encode each letter based on its position in the alphabet.
2. Use the public key, (a, n), to compute $d' \equiv d^a \pmod{n}$. d' is the encrypted data.

Decryption

1. Given encrypted data d', use the private key (b, n) to compute $d \equiv d'^b \pmod{n}$. d is the encoded form of the decrypted data.
2. decode d into it's original form. Using the example from before, $d = 010226$ might become ABZ.

Identity Verification

1. Person A and person B each generate a public-private key pair. They then share their public keys with each other.

2. Person A encrypts a bunch of random data with person B's public key. They send this encrypted data to person A and person B then decrypts it and sends the decrypted data back. Person A then verifies that person B was able to successfully decrypt all the data. Person A can then conclude that person B did in fact have the corresponding private key and has thus verified the identity of person B.

3. Person B now does the same for person A. In other words, repeat step 2, but with A and B swapped.

5 How Do We Verify Public Keys?

There's still one last problem with RSA. When you receive someone's public key, how do you know that that key truly corresponds to their identity? Well, in truth, there isn't an amazing answer. However, there are still many common techniques that are used in the real world:

1. The trust on first use (TOFU) model. When you connect to someone for the first time, you take it as a fact that the public key you get from them is correct and you store it for your own records. As long as a hacker didn't breach the first connection, the first key truly corresponds to whoever you are connecting to. Then, whenever you reconnect you can verify that the key is still the same as what you have stored. If it isn't, that means that the connection has been compromised.

2. Get keys in person. In order to avoid having to do all this digitally, you can just talk to whoever you want to connect to in person to be sure that they are giving you their key and that a digital hacker won't be able to intercept your conversation.

3. Use a certificate authority (CA). This is more practical for large numbers of connections. It would be impossible for each amazon customer to go talk to someone in the company to obtain a key. Instead, amazon has well known key that is verified by a trusted third party. Using a reliable third party verifier can be another way to ensure you have the correct public key for a specific connection.

6 Session Keys and the Diffie-Hellman Algorithm

It turns out that RSA encryption is actually rather slow. It works fine for a few probabilistic identity tests, but it will be noticeably slow for encrypted large files or messages. There is in fact a much faster encryption algorithm out there called AES. AES is a symmetric algorithm, meaning that it uses the same key for both encryption and decryption (of course, RSA is asymmetric). This type of key should be unique for every time you establish a connection, which is why it's called a session key. This is needed for added security so that if a hacker obtains one key, they can only decrypt one connection worth of messages.

Covering how AES works is not in the scope of this article and frankly, the algorithm just shifts data around in many convoluted ways to make it impossible to get the original data from the encrypted data. However, what is within the scope of this article is how two people can secretly share such a symmetric encryption key to use AES with.

It is tempting to turn to the obvious solution here: after verifying someones identity with RSA, you can simply decide on a random session key, encrypt it, and have the other person decrypt it. Although this works, it creates a problem: if a hacker can crack or find a flaw in our RSA algorithm, they can obtain our session keys as well. We want to separate out our layers of security whenever possible and not have them rely on each other since a chain is only as strong as its weakest link.

So, we have a new problem: If two people are communicating with each other without any prior communication, how do they generate a shared secret key such that if the third person hears everything they say, they will still not be able to decern the shared secret key.

The solution lies in the discrete logarithm problem: given g, r, and n, it is very difficult to find an a (assuming that such an a exists), such that $g^a \equiv r \pmod{n}$. Note that despite being similar, it is not the

same as the discrete root problem. Regardless, both are quite difficult to solve and sufficient for encryption. Here are the steps to follow:

1. Person A sends person B a prime number p and another positive integer g that is relatively prime to p.
2. Person A generates a random exponent a and person B generates a random exponent b.
3. Person A sends g^a to person B and person B sends g^b to person A.
4. Both person A and person B can now compute a shared secret key g^{ab} since $g^{ab} = (g^a)^b = (g^b)^a$.

Even if a third person listened to everything that person A and B said to each other, they would only obtain p, g, g^a, and g^b. In order to compute g^{ab}, this third person must either find a or b, but this is impossible because the discrete logarithm problem is infeasible to solve. The above algorithm is called the Diffie-Hellman key exchange algorithm and just like RSA, it is still commonly used today.

As always, there is one final catch to deal with. If we pick a bad value of g, there will be very few distinct powers of g modulo p. For example, if $p = 101$ and we pick $g = 10$, there will only be 4 distinct powers of g modulo p since the order of 10 modulo 101 is 4. To solve this problem, we have to pick p to be a safe prime. A safe prime is of the form $p = 2q + 1$ for another prime q. Thus, $\varphi(p) = 2q$. It follows that the order of any element is either 1, 2, q, or $2q$ (all the divisors of $\varphi(p)$). Then, as long as we don't pick any value of g with order 1 or 2, we can guarantee that g has at least q distinct powers modulo p. Eliminating the trivially bad values of g for safe primes p, we can make sure we pick a secure enough set of powers modulo p to share our secret key with.

It is important to note that safe primes are not easy to generate and quite rare. Thus, it is okay to use known safe primes which can be found in various pieces of documentation. This is because the modulus for this key exchange algorithm is not the secret. This is in contrast to RSA where the security lies behind knowing the prime factors of the modulus. In fact, always picking the same value of g is not a problem either since all the security lies in the randomness of a and b.

7 Conclusion

Cryptography is one of the coolest fields within computer science. It combines some of the most difficult concepts from number theory with simple questions about verifying information. At the same time, while designing all of these algorithms and thinking about all this math, we have to keep in mind the constantly changing landscape of computing and all the real world limitations that are present. Isn't it wild that this whole article, and so much more academic literature on this subject all came from the idea of sharing secrets and looking at ID cards? I, for one, think it is.

8 Appendix: Generating Large Primes and the Miller-Rabin Test

In order to implement RSA we must be able to generate large prime numbers. To prevent modern computational power, our primes must be anywhere from 1024 to 4096 bits in size. Common algorithms such as the sieve of Eratosthenes will not make even a dent in finding prime numbers of this size. Let's think of another way to test for primes.

Suppose n is a prime. Then, $a^{n-1} \equiv 1 \pmod{n}$ for any positive integer a less than n. However, the converse is not necessarily true. Moreover, the converse because even less likely to be true if we only check a few values of a. Just because this does end up being true for some a, it doesn't mean that n is prime.

However, we can hope that if we strengthen this test and apply it to a bunch of a, we will pretty accurately be able to determine whether n is a prime. Here is how we can strengthen the test: Let $n - 1 = 2^s * t$, where s and t are integers, and t is odd. Since the only square roots of 1 modulo a prime are ± 1, if n were prime, one of the following $s + 1$ conditions must be true

$$a^{2^i t} \equiv -1 \pmod{n} \quad \text{for some } 0 \leq i < s$$
$$a^t \equiv 1 \pmod{n}$$

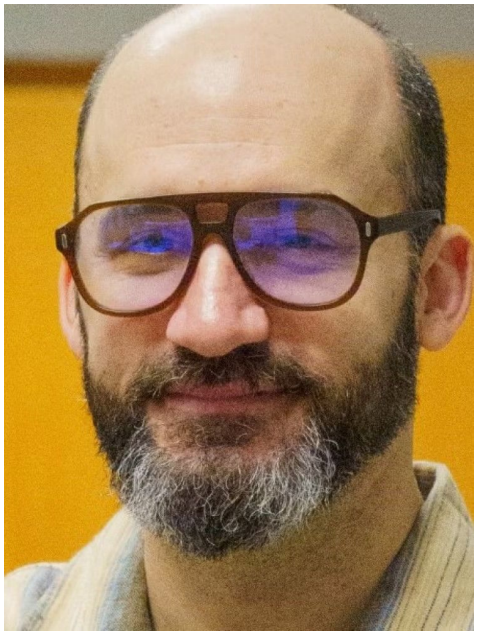
This comes from taking the square root of the equation $a^{n-1} \equiv 1 \pmod{n}$ until the right hand side is no longer 1 or until the exponent of a is no longer even.

It turns out that when n is not prime, on average $\frac{1}{4}$ of the values of a less than n still satisfy these conditions. To be clear, whenever n is prime, this test will of course pass since one of those conditions will always be true. But, when n is not prime, this test may fail. That means that if we run this test for a bunch of randomly chosen values of a, say 100 values of a, the chance that we falsely identify a composite number as prime is $\frac{1}{4^{100}}$, which is essentially 0. There's a higher chance that a random bit flips and breaks all our algorithms anyway or that our computer systems simply shut down and break.

Isn't this crazy? The best way we know of generating large primes involves picking a random number, probabilistically checking if it's prime, and then incrementing it until it actually is prime. For all that we know about mathematics and about our world, it's rather beautiful that generating something as simple as primes takes an algorithm that's so clever and yet barbaric at the same time.

What's the point?

Steven Breger



"Believe in the universe." –Mr. Scheiman

1 Introduction

Mathematics is infamously difficult. It starts off with simple counting and operations, and quickly begins to blow up. Often, even the most simple looking of questions require very long, difficult, and non-elementary solutions. We don't need to look further than the counting numbers (the things that kindergartners know about!): simple-looking questions like the Collatz conjecture or the twin prime conjecture are still not solved, despite being explainable to a well-versed elementary schooler.

Why then, you may ask, is it even worth studying mathematics? Doesn't it feel hopeless and pointless? Besides, most of us won't even need something as simple as high-school algebra throughout our lives, let alone basic calculus, rigorous analysis, or the countless branches of essentially inapplicable pure math. Even if you want to be an engineer, requiring an understanding of algebra, calculus, and physics, it seems pointless to go deeper into pure math. After all, when would an engineer need to explore the ultra theoretical spaces studied within topology or know what the Riemann hypothesis implies about prime asymptotics? Shouldn't we just leave all of this crazy stuff to the few hyper-geniuses of the world that want to lock themselves in a room all day and learn all this theory for it's own sake?

Well, according to Mr. Scheiman, who teaches calculus and differential equations at Stuyvesant, doing so would be a grave mistake. Seeing how Mr. Scheiman's teaching journey has influenced his perspective on mathematics, we can get a sense of why he does math, and why we all should engage with it too.

2 Connection

Besides his teaching at Stuyvesant, Mr. Scheiman simultaneously teaches at Rutgers University, where he earned his mathematics degree. As you can imagine, teaching at Stuy until around 3, commuting for an hour, and then teaching afternoon classes at the university. When asked why he does this, Mr. Scheiman said "Of course I need to pay the rent, but I've always loved it. I think it's about being in front of a group of people in a lecture and helping them out... I get an excitement from it." Mr Scheiman added that he especially loves the daily tight-knit environment in the Stuy high-school classrooms: "Here I can say a witty joke, something sarcastic, and they will understand. You might not get that in college where there's large lecture halls and students are taking classes just because they need to, unlike here where students are taking math classes as electives and there's a general liking to it."

Surprisingly, human connection is one of the biggest benefits of doing math. We stereotypically see math as a solitary art, but it doesn't have to be. As Mr. Scheiman points out, a big part of teaching it is that human connection with your students. Importantly though, you don't even need to be a teacher or have a student to connect over math. Anyone can do it. Working on a problem set with your peers, talking about a problem with someone, or explaining something to someone who is stuck, is a shockingly effective way to meet new people and create new friendships. In fact, one of Mr. Scheiman's favorite ways to bond with his students is to make jokes about "difficult–I mean differential equations" or to start off class by saying "GooooOOOOOOOOOOOOD afternoon everyone!" He points out that it's important to know the students, read the energy level in the room, and adjust to create the best relationship possible with the people you do mathematics with. That way, both you and they can get the most out of it.

3 Thinking is Power

Aside from the fact that math lets you make so many new friends, it improves your problem solving and ability to understand complex ideas. Whether you are training for high-school math competitions, explaining algebra to your brother, or reading a topology textbook, there is some part of you that is thinking, gaining an intuition for approaching difficult problems, and learning how to explain what you see. Mr. Scheiman noted how these problem-solving skills are always useful, saying that "Math is the great equalizer. People won't take advantage of you. Car salesmen, accountants, things like that. The more math you know, the better you are in the world. I say that first and foremost."

These abilities of critical thinking are useful regardless of what you do. Sadly, this viewpoint is not commonly expressed because people often think that math is just about calculation, about solving some crazy long equation with a set procedure that was taught to you by someone else. But in reality, math is much more like an art form. Calculation is merely a means of obtaining data, intuition, and deductions in order to find a new proof, make a discovery, and think more deeply about a problem. If you view math in this way it, the skills you gain from math quickly become applicable to a wide range of disciplines, even if the exact math itself isn't. Don't believe it? See for yourself:

- If you read literature, learning about logic, mathematical proof, and deduction can help you draw more interesting conclusions, perhaps giving you the capacity to connect and analyze more pieces of evidence at once.
- If you are a construction worker, well each construction site is essentially its own giant problem. There's a lot of problem solving that goes into what parts to use where, how to adjust something without tearing down an existing wall, and the best way to budget. It's the same kind of thinking that goes on in a hard math problem.
- If you are a chef there's countless times when you have to figure out the best way to create a certain flavor, texture, or style. Going through all the possibilities and having an intuition for what works is the same skill that used in many math problems.

The list goes on. Mr. Scheiman put it best when he said that "if the world is more math literate, the world would be a better place."

4 Conclusion

So, to answer the question, the point is that no one, and I really do mean no one, can ever know enough math. Learning more math is always a great thing to do so long as you are a living breathing human. Math satisfies natural instincts of curiosity, creates a surprising amount of socialization, and fosters critical thinking, useful in any part of life. Yes, it's useful even if you don't need to solve daily difficult–I mean differential equations.

